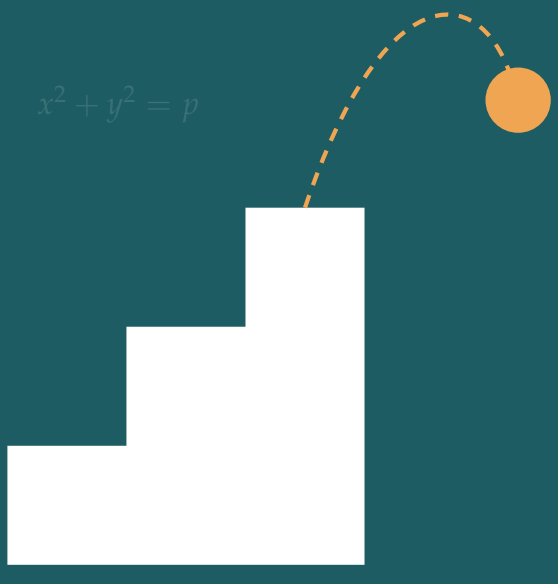
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2005 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2005 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2005 Worked Solutions

Worked solutions to all 42 questions of the 2005 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2005, Question 1

Pure • Counting Digit Sums

> **STEP I 2005, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/05-s1-q1

The key idea

The largest possible digit sum of a five-digit number is $9 \times 5 = 45$, and both target sums are close to it. So instead of counting the digits themselves, count how far each digit falls *short* of 9. The shortfalls are small non-negative numbers with a small fixed total, and counting those is easy, either by listing cases or by the "stars and bars" trick.

Solution

Write the number as $d_1d_2d_3d_4d_5$, where $d_1 \in \{1, \ldots, 9\}$ (no leading zero) and $d_2, \ldots, d_5 \in \{0, \ldots, 9\}$. Put

$$e_i = 9 - d_i \qquad (i = 1, \ldots, 5),$$

the *shortfall* of the ith digit. Each e_i is a whole number with $0 \leq e_i \leq 9$ (and $e_1 \leq 8$, since $d_1 \geq 1$), and

$$d_1 + d_2 + d_3 + d_4 + d_5 = 45 - (e_1 + e_2 + e_3 + e_4 + e_5).$$

Different numbers give different lists $(e_1, \ldots, e_5)$ and vice versa, so counting numbers with digit sum S is the same as counting ordered lists of shortfalls with total $45 - S$.

Part (i)

Digit sum 43 means the shortfalls total $45 - 43 = 2$. There are only two ways to split 2 into non-negative parts:

- **One digit is short by 2, the rest by 0.** One digit is 7 and the other four are 9. The 7 can be in any of the 5 positions: 5 numbers $(79999, 97999, \ldots, 99997)$.
- **Two digits are short by 1 each, the rest by 0.** Two digits are 8 and the other three are 9. We choose which 2 of the 5 positions hold the 8s: $\binom{5}{2} = 10$ numbers $(88999, 89899, \ldots, 99988)$.

Every digit is at least 7, so the leading digit is never 0 and all of these really are five-digit numbers. The two cases don't overlap and cover every possibility, so there are

$$5 + 10 = 15$$

five-digit numbers whose digits sum to 43, as required.

Part (ii)

Now the shortfalls must total $45 - 39 = 6$. Since $6 \leq 8$, no single shortfall can exceed its limit: $e_1 \leq 6 \leq 8$ keeps $d_1 \geq 3$, and $e_i \leq 6 \leq 9$ keeps the other digits between 3 and 9. So we simply need the number of ordered solutions of

$$e_1 + e_2 + e_3 + e_4 + e_5 = 6, \qquad e_i \geq 0 \text{ whole numbers.}$$

Ask yourself How can we count these without listing them all?

Picture 6 stars in a row, separated into 5 groups by 4 bars:

$$\star\star \,|\, |\, \star \,|\, \star\star\star \,| \quad \text{stands for} \quad (e_1, e_2, e_3, e_4, e_5) = (2, 0, 1, 3, 0).$$

Every arrangement of 6 stars and 4 bars gives exactly one solution, and every solution comes from exactly one arrangement. An arrangement is fixed by choosing which 4 of the 10 places hold bars, so the number of solutions is

$$\binom{10}{4} = 210.$$

Hence there are **210** five-digit numbers whose digits sum to 39.

If you'd rather count by cases, as in part (i), list the ways to split 6 into at most five positive parts and count arrangements of each:

shortfalls	numbers	shortfalls	numbers
6	5	$3,2,1$	$5 \cdot 4 \cdot 3 = 60$
$5,1$	$5 \cdot 4 = 20$	$3,1,1,1$	$5 \cdot 4 = 20$
$4,2$	$5 \cdot 4 = 20$	$2,2,2$	$\binom{5}{2} = 10$
$4,1,1$	$5 \cdot \binom{4}{2} = 30$	$2,2,1,1$	$\binom{5}{2}\binom{3}{2} = 30$
$3,3$	$\binom{5}{2} = 10$	$2,1,1,1,1$	5

The total is $5 + 20 + 20 + 30 + 10 + 60 + 20 + 10 + 30 + 5 = 210$, agreeing with stars and bars. (The partition $1,1,1,1,1,1$ of 6 needs six digits, so it doesn't occur.)

Checks

- Stars and bars reproduces part (i): shortfalls totalling 2 give $\binom{6}{4} = 15$.

- The extreme cases work too: total shortfall 0 gives $\binom{4}{4} = 1$ number (99999), and total shortfall 1 gives $\binom{5}{4} = 5$.
- A computer search through all 90 000 five-digit numbers finds exactly 15 with digit sum 43 and 210 with digit sum 39, with the case breakdown in the table above.

Common slips

- Counting $\{8, 8, 9, 9, 9\}$ as one number, or as 5: the order of the digits matters, and there are $\binom{5}{2} = 10$ arrangements.
- Forgetting to check the leading digit. Here it can't be 0 because every digit is at least 3; for a smaller digit sum (say 20) stars and bars would overcount and the cases with $d_1 = 0$, or digits above 9, would have to be removed.

Beyond STEP

The number of ways to write n as an ordered sum of k non-negative whole numbers is $\binom{n+k-1}{k-1}$, and it is the coefficient of x^n in $(1 + x + x^2 + \cdots)^k$. Restricting each digit to at most 9 replaces this with $(1 + x + \cdots + x^9)^k$, and the cut-off is handled by the inclusion–exclusion principle. These generating-function methods are a staple of first-year combinatorics; Siklos, *Advanced Problems in Mathematics*, and Davenport, *The Higher Arithmetic*, are good places to keep exploring digit and counting problems.

WORKED SOLUTION

STEP I 2005, Question 2

Pure • Parabola Tangents and Normals

STEP I 2005, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q2

The key idea

Parametrising the parabola $y^2 = 4x$ by $(t^2, 2t)$ makes every tangent and normal a short linear equation in x and y, so finding intersections is just solving two simultaneous linear equations. The condition that $(1, 0)$, the focus, lies on PQ turns into a single neat relation between p and q, and that relation is exactly what makes the tangents at P and Q perpendicular. Once you have that, the rectangle almost draws itself.

Solution

The tangents and the point R

At the point $(t^2, 2t)$ we have $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 2t$ and $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 2$, so the gradient of C there is $\dfrac{2}{2t} = \dfrac{1}{t}$ (fine, since $t \neq 0$). The tangent at P is therefore

$$y - 2p = \frac{1}{p}(x - p^2) \quad \Longleftrightarrow \quad py = x + p^2,$$

and likewise the tangent at Q is $qy = x + q^2$. Subtracting,

$$(p - q)\, y = p^2 - q^2 = (p - q)(p + q),$$

and since $p \neq q$ we may divide by $p - q$ to get $y = p + q$. Then $x = py - p^2 = p(p + q) - p^2 = pq$. So

$$R = (pq,\ p + q),$$

as required.

The normals and the point S

The normal at $(t^2, 2t)$ is perpendicular to the tangent, so it has gradient $-t$. The normal at P is

$$y - 2p = -p\,(x - p^2) \quad \Longleftrightarrow \quad px + y = 2p + p^3,$$

and the normal at Q is $qx + y = 2q + q^3$. Subtracting,

$$(p-q)\,x = 2(p-q) + (p^3 - q^3) = (p-q)(2 + p^2 + pq + q^2),$$

so, dividing by $p - q \neq 0$ again,

$$x = p^2 + pq + q^2 + 2, \qquad y = 2p + p^3 - px = -p^2q - pq^2 = -pq\,(p+q).$$

These hold for any p and q; we now use the extra condition.

Using the condition that $(1,0)$ lies on PQ

The chord PQ has gradient

$$\frac{2p - 2q}{p^2 - q^2} = \frac{2}{p+q}$$

(when $p + q \neq 0$), so its equation is $y - 2p = \dfrac{2}{p+q}(x - p^2)$, which simplifies to

$$(p+q)\,y = 2x + 2pq.$$

This form also covers $p + q = 0$, when it gives the vertical line $x = -pq = p^2$ through P and Q. Putting $(x, y) = (1, 0)$ gives $0 = 2 + 2pq$, so

$$pq = -1.$$

Substituting into the coordinates of S:

$$S = \big(p^2 + q^2 + 2 + pq,\ -pq(p+q)\big) = \big(p^2 + q^2 + 1,\ p+q\big),$$

as required. Notice also that now $R = (-1,\ p+q)$, which lies on the line $x = -1$, the directrix.

$PSQR$ is a rectangle

> **Ask yourself** What is the cleanest way to prove that four given points form a rectangle?

A good route is: show it is a parallelogram (the diagonals bisect each other), then show one angle is a right angle.

Parallelogram. In $PSQR$ the diagonals are PQ and SR. Using $pq = -1$,

$$R + S = \big(pq + p^2 + q^2 + 1,\ 2(p+q)\big) = \big(p^2 + q^2,\ 2p + 2q\big) = P + Q,$$

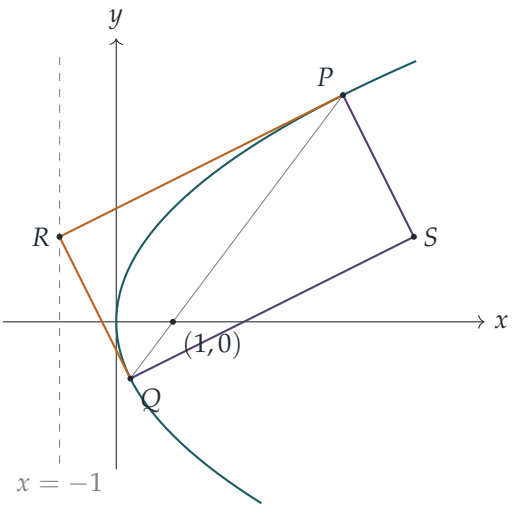
so the midpoint of RS equals the midpoint of PQ. The diagonals bisect each other, so $PSQR$ is a parallelogram. It is a genuine one, not a degenerate one: P, R and Q are not collinear, because R lies on both tangents and the tangents at two different points of a parabola meet only once and neither passes through the other point of contact.

A right angle. The sides RP and RQ lie along the tangents at P and Q, with gradients $\frac{1}{p}$ and $\frac{1}{q}$. Their product is

$$\frac{1}{p}\cdot\frac{1}{q}=\frac{1}{pq}=-1,$$

so the two tangents are perpendicular and the angle PRQ is a right angle.

A parallelogram with one right angle is a rectangle, so $PSQR$ is a rectangle. (Equivalently: the angles at P and Q are right angles because each normal is perpendicular to its own tangent, and with the right angle at R the fourth angle must be one too.)



The figure shows the case $p=2$, $q=-\frac{1}{2}$: tangents in amber, normals in plum, and the focal chord PQ in grey.

Checks

Take $p=2$, $q=-\frac{1}{2}$, so $pq=-1$. Then $P=(4,4)$ and $Q=(\frac{1}{4},-1)$, and the line through them, $y-4=\frac{4}{3}(x-4)$, does pass through $(1,0)$. The formulae give $R=(-1,\frac{3}{2})$ and $S=(\frac{21}{4},\frac{3}{2})$. Check directly: $\overrightarrow{RP}=(5,\frac{5}{2})$ and $\overrightarrow{QS}=(5,\frac{5}{2})$ are equal, and $\overrightarrow{RQ}=(\frac{5}{4},-\frac{5}{2})$ has dot product $\frac{25}{4}-\frac{25}{4}=0$ with $\overrightarrow{RP}$. So $PSQR$ is indeed a rectangle. We also checked all the coordinate formulae with exact fractions for about two thousand random choices of p and q.

Common slips

- Dividing by $p-q$ without saying why it is allowed; the question gives $p\neq q$ precisely so you can.
- Showing only that adjacent sides are perpendicular at one corner, which does not

by itself make a rectangle: you need either a parallelogram plus one right angle, or right angles at three corners.

- Forgetting to use $pq = -1$ in S: the general normal intersection is $\big(p^2 + pq + q^2 + 2,\ -pq(p+q)\big)$, and it only simplifies to the given form on the focal chord.

Beyond STEP

A chord through the focus is a *focal chord*, and this question proves two classical facts about it: the tangents at its ends are perpendicular and meet on the directrix. Both are special cases of the theory of poles and polars for conics, in which the directrix is the polar of the focus; projective geometry makes such results almost automatic. A good place to explore parametric methods for conics in this spirit is Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2005, Question 3

Pure • Rational Equations

STEP I 2005, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q3

The key idea

Multiply through by $(x-a)(x-b)$ to turn each equation into a polynomial equation, then count its real roots. The point that needs care is that clearing denominators could, in principle, add "solutions" $x=a$ or $x=b$ that the original equation doesn't allow, so we must check that this never happens. In part (ii) the quadratic has a repeated root exactly when its discriminant is zero, and that gives the condition.

Solution

Part (i)

For $x \neq a, b$ we may multiply by $(x-a)(x-b)$, and the equation is equivalent to

$$\begin{aligned} x(x-b)+x(x-a) &= (x-a)(x-b) \\ 2x^2-(a+b)x &= x^2-(a+b)x+ab \\ x^2 &= ab. \end{aligned}$$

Since a and b have the same sign, $ab>0$, so this has the two distinct real roots $x=\sqrt{ab}$ and $x=-\sqrt{ab}$.

We still need these to be allowed values. If $x=a$ were a root, then $a^2=ab$, and since $a\neq 0$ this gives $a=b$, which is false. In the same way $x=b$ is not a root. So both roots are genuine solutions, and the equation has exactly two distinct real solutions, $x=\pm\sqrt{ab}$.

Part (ii)

Again, for $x \neq a, b$ multiply by $(x-a)(x-b)$:

$$2x^2-(a+b)x = (1+c)\big(x^2-(a+b)x+ab\big),$$

which rearranges to

$$q(x) \equiv (1-c)x^2 + c(a+b)x - (1+c)ab = 0.$$

Because $c \neq 1$, the leading coefficient is non-zero, so this really is a quadratic.

Ask yourself Could one of the roots of q be a or b, so that it has to be thrown away?

Substitute and see:

$$q(a) = (1-c)a^2 + c(a+b)a - (1+c)ab = a^2 - ab = a(a-b),$$
$$q(b) = b^2 - ab = b(b-a)$$

(the c terms cancel). Both are non-zero because a, b are non-zero and distinct. So a and b are never roots of q. This means every real root of q is a solution of the original equation, and the reverse is also true.

The discriminant of q is

$$\begin{aligned}\Delta &= c^2(a+b)^2 + 4(1-c)(1+c)ab \\ &= c^2\left[(a+b)^2 - 4ab\right] + 4ab \\ &= c^2(a-b)^2 + 4ab.\end{aligned}$$

If $c^2 = -\dfrac{4ab}{(a-b)^2}$ (and $a \neq b$, so we may divide by $(a-b)^2$), then $\Delta = 0$. The quadratic then has exactly one (repeated) real root,

$$x = -\frac{c(a+b)}{2(1-c)},$$

and by the check above this root is not a or b. So the equation has exactly one real solution, as required. (In fact the converse holds as well. If $\Delta > 0$ there are two solutions and if $\Delta < 0$ there are none, so the condition is also necessary.)

Rewriting the condition. Over a common denominator,

$$1 - \left(\frac{a+b}{a-b}\right)^2 = \frac{(a-b)^2 - (a+b)^2}{(a-b)^2} = \frac{-4ab}{(a-b)^2},$$

so the condition is $c^2 = 1 - \left(\dfrac{a+b}{a-b}\right)^2$, as required.

Deducing $0 < c^2 \leq 1$. The square $\left(\frac{a+b}{a-b}\right)^2$ is non-negative, so the second form gives $c^2 \leq 1$. The first form gives $c^2 = -4ab/(a-b)^2$. This is non-zero because $ab \neq 0$, and $c^2 \geq 0$ for real c, so $c^2 > 0$. (This also tells us that $ab < 0$: a and b have opposite signs, which fits with part (i).) Hence the condition can only hold if $0 < c^2 \leq 1$.

The upper value is attained. $c^2 = 1$ exactly when $a + b = 0$, and then $c = -1$ (since $c \neq 1$).

Checks

- Take $a = 4$, $b = -1$. Then $c^2 = 16/25$; choose $c = \frac{4}{5}$. The repeated root is $x = -\frac{4}{5} \cdot 3/(2 \cdot \frac{1}{5}) = -6$, and indeed $\frac{-6}{-10} + \frac{-6}{-5} = 0.6 + 1.2 = 1.8 = 1 + c$.
- The extreme case $a = 1$, $b = -1$, $c = -1$ gives $\frac{x}{x-1} + \frac{x}{x+1} = 0$, i.e. $2x^2 = 0$: one solution, $x = 0$.
- In part (i), $a = 1$, $b = 4$ gives $x = \pm 2$: $\frac{2}{1} + \frac{2}{-2} = 1$ and $\frac{-2}{-3} + \frac{-2}{-6} = 1$.

Common slips

- Forgetting to check that the roots are not a or b after clearing denominators. Here it turns out to be harmless, but it has to be shown.
- Expanding the discriminant without grouping the c^2 terms. Writing $(a+b)^2 - 4ab = (a-b)^2$ is what makes the condition fall out neatly.
- Claiming $c^2 \geq 0$ instead of $c^2 > 0$. The strict inequality comes from $ab \neq 0$.

Beyond STEP

The left-hand side is a rational function, and "how many times does its graph meet the horizontal line $y = 1 + c$?" is a question about its range. The condition found here picks out the turning-point values of that function, the heights where a horizontal line just touches the graph. Asking when a polynomial has a repeated root leads to the discriminant and, more generally, to the resultant, which appear in first-year algebra. For more problems in this style, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2005, Question 4

Pure • Multiple Angle Formulae

STEP I 2005, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q4

The key idea

Both parts are about multiple angles. In (i) the only real decision is the sign of $\sin\theta$, which the given range settles; after that the double- and triple-angle formulae do the work. In (ii) we build $\tan 3\theta$ as $\tan(2\theta+\theta)$, and then the "hence" turns into a cubic in $t=\tan\theta$. A cubic in a STEP question nearly always has a root you can spot, and the range for θ tells us which root to keep.

Solution

Part (i)

Since $\cos^2\theta+\sin^2\theta=1$,

$$\sin^2\theta = 1-\tfrac{9}{25} = \tfrac{16}{25}, \qquad \text{so} \qquad \sin\theta = \pm\tfrac{4}{5}.$$

For $\frac{3\pi}{2}\le\theta\le 2\pi$, θ lies in the fourth quadrant, where $\sin\theta\le 0$. So $\sin\theta=-\frac{4}{5}$, and

$$\sin 2\theta = 2\sin\theta\cos\theta = 2\left(-\tfrac{4}{5}\right)\left(\tfrac{3}{5}\right) = -\frac{24}{25},$$

as required.

For $\cos 3\theta$ we need the triple-angle formula. Writing $c=\cos\theta$ and $s=\sin\theta$,

$$\begin{aligned}\cos 3\theta=\cos(2\theta+\theta) &= \cos 2\theta\cos\theta-\sin 2\theta\sin\theta\\ &= (2c^2-1)c-2s^2c\\ &= (2c^2-1)c-2(1-c^2)c\\ &= 4c^3-3c.\end{aligned}$$

With $c=\frac{3}{5}$,

$$\cos 3\theta = 4\cdot\frac{27}{125}-\frac{9}{5} = \frac{108-225}{125} = -\frac{117}{125}.$$

(Here the sign of $\sin\theta$ doesn't matter, because $\cos 3\theta$ depends only on $\cos\theta$.)

Part (ii)

The identity. Write $t = \tan\theta$. By the double-angle formula, $\tan 2\theta = \dfrac{2t}{1-t^2}$, and by the addition formula,

$$\tan 3\theta = \tan(2\theta+\theta) = \frac{\tan 2\theta + \tan\theta}{1 - \tan 2\theta \tan\theta} = \frac{\dfrac{2t}{1-t^2} + t}{1 - \dfrac{2t^2}{1-t^2}}.$$

Multiplying the top and bottom by $1-t^2$,

$$\tan 3\theta = \frac{2t + t(1-t^2)}{(1-t^2) - 2t^2} = \frac{3t - t^3}{1 - 3t^2},$$

which is the identity. The derivation needs $\tan 2\theta$ to exist, so $t^2 \neq 1$; when $t = \pm 1$ we can check directly: for $\theta = \frac{\pi}{4}$, say, the right-hand side is $\frac{3-1}{1-3} = -1 = \tan\frac{3\pi}{4}$, and $t = -1$ is the same with signs flipped.

Solving $\tan 3\theta = \frac{11}{2}$. By the identity,

$$\frac{3t - t^3}{1 - 3t^2} = \frac{11}{2} \quad\Longrightarrow\quad 6t - 2t^3 = 11 - 33t^2 \quad\Longrightarrow\quad 2t^3 - 33t^2 - 6t + 11 = 0.$$

Multiplying out is safe because $1 - 3t^2 \neq 0$ here: if it were zero, $\tan 3\theta$ would be undefined, not $\frac{11}{2}$.

Ask yourself How do we find a root of $2t^3 - 33t^2 - 6t + 11$?

Any rational root p/q in lowest terms has p dividing 11 and q dividing 2, so we only need to try $\pm 1, \pm 11, \pm\frac{1}{2}, \pm\frac{11}{2}$. Trying $t = \frac{1}{2}$:

$$2\cdot\tfrac{1}{8} - 33\cdot\tfrac{1}{4} - 3 + 11 = \tfrac{1}{4} - \tfrac{33}{4} + 8 = 0.$$

So $2t - 1$ is a factor, and comparing coefficients gives

$$2t^3 - 33t^2 - 6t + 11 = (2t-1)(t^2 - 16t - 11).$$

The quadratic has roots $t = 8 \pm \sqrt{64+11} = 8 \pm 5\sqrt{3}$. So the three candidates are

$$t = \tfrac{1}{2}, \qquad t = 8 + 5\sqrt{3} \approx 16.66, \qquad t = 8 - 5\sqrt{3} \approx -0.66.$$

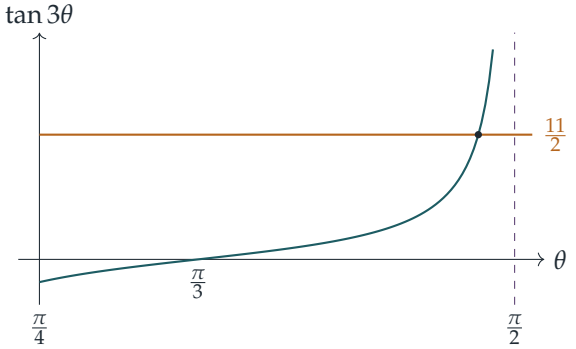
These correspond to the three angles in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ whose triple has tangent $\frac{11}{2}$.

Now use the range. Since tan is increasing on $\left[\frac{\pi}{4}, \frac{\pi}{2}\right)$ and $\tan\frac{\pi}{4} = 1$, we need $t \geq 1$ (and $\theta = \frac{\pi}{2}$ itself is excluded, as $\tan\theta$ is undefined there). Only one candidate qualifies:

$$\tan\theta = 8 + 5\sqrt{3}.$$

The other two are genuine roots of the cubic but belong to angles outside the range: $\frac{1}{2} < 1$ and $8 - 5\sqrt{3} < 0$.

The graph confirms there is exactly one solution: as θ runs from $\frac{\pi}{4}$ to $\frac{\pi}{2}$, 3θ runs from $\frac{3\pi}{4}$ to $\frac{3\pi}{2}$, and on that interval $\tan 3\theta$ increases continuously from -1 to $+\infty$, so it crosses $\frac{11}{2}$ exactly once.



Checks

- In (i), $\theta = 2\pi - \arccos\frac{3}{5} \approx 5.3559$, and a calculator gives $\sin 2\theta = -0.96$ and $\cos 3\theta = -0.936 = -\frac{117}{125}$.
- In (ii), $\theta = \arctan(8 + 5\sqrt{3}) \approx 1.5108$, which is between $\frac{\pi}{4} \approx 0.785$ and $\frac{\pi}{2} \approx 1.571$, and $\tan 3\theta \approx 5.5000$.
- A quick test of the identity: at $\theta = \frac{\pi}{12}$, $t = 2 - \sqrt{3}$, and the right-hand side simplifies to $1 = \tan\frac{\pi}{4}$.

Common slips

- Taking $\sin\theta = +\frac{4}{5}$ in (i), which gives the wrong sign for $\sin 2\theta$.
- Stopping at $t = \frac{1}{2}$ in (ii) because it is the root you spotted. It is a root of the cubic, but $\arctan\frac{1}{2} < \frac{\pi}{4}$.

Beyond STEP

The cubic had three roots because $\tan 3\theta = \frac{11}{2}$ has three solutions for θ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, spaced $\frac{\pi}{3}$ apart. The same mechanism runs in reverse in the classical trigonometric solution of cubics, and more generally de Moivre's theorem gives $\tan n\theta$ as a ratio of polynomials in $\tan\theta$ whose roots are the tangents of equally spaced angles. Needham's *Visual Complex Analysis* shows why multiple-angle formulae come out of $(\cos\theta + \mathrm{i}\sin\theta)^n$ so naturally.

WORKED SOLUTION

STEP I 2005, Question 5

Pure • Integrating Powers

STEP I 2005, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q5

The key idea

Powers of $x+1$ integrate by the reverse chain rule, $\int (x+1)^n \,\mathrm{d}x = \frac{(x+1)^{n+1}}{n+1}$, with one exception: when $n=-1$ the answer is a logarithm. So every case split here comes from asking "when does the power become -1?". In (ii) the extra factor x is removed by writing $x=(x+1)-1$, which leaves two plain powers and so *two* exceptional values of m.

Solution

Part (i)

If $k \neq 0$, the power $k-1$ is not -1, so

$$\int_0^1 (x+1)^{k-1}\,\mathrm{d}x = \left[\frac{(x+1)^k}{k}\right]_0^1 = \frac{2^k-1}{k}.$$

If $k=0$, the integrand is $\frac{1}{x+1}$, and since $x+1>0$ on $[0,1]$,

$$\int_0^1 \frac{1}{x+1}\,\mathrm{d}x = \Big[\ln(x+1)\Big]_0^1 = \ln 2.$$

Ask yourself Why should the answer for $k \neq 0$ have anything to do with the answer for $k=0$?

Because the *integrand* barely changes when k moves a little away from 0. For $0 \le x \le 1$,

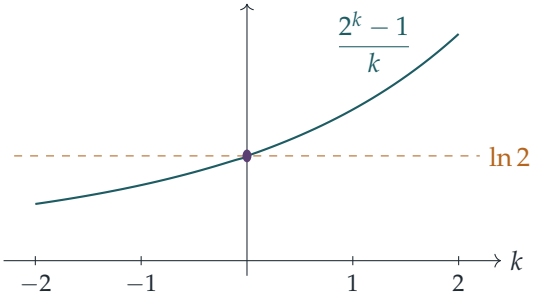
$$\left|(x+1)^{k-1} - (x+1)^{-1}\right| = \frac{\left|(x+1)^k - 1\right|}{x+1} \le \left|2^k - 1\right|,$$

since $1 \le x+1 \le 2$ means $(x+1)^k$ lies between 1 and 2^k. Integrating over an interval of length 1, the two integrals differ by at most $|2^k-1|$, which tends to 0 as $k \to 0$. Hence,

for $k \approx 0$ (and $k \neq 0$),

$$\frac{2^k - 1}{k} = \int_0^1 (x+1)^{k-1}\,\mathrm{d}x \approx \int_0^1 (x+1)^{-1}\,\mathrm{d}x = \ln 2,$$

as required. In other words, the formula for $k \neq 0$ has a "hole" at $k = 0$, and the $k = 0$ integral fills it in exactly.



Part (ii)

Write $x = (x+1) - 1$, so that

$$x(x+1)^m = (x+1)^{m+1} - (x+1)^m.$$

Each term is a power of $x+1$, and integrating it produces a logarithm exactly when its power is -1. That happens for $m+1 = -1$ or $m = -1$, so the cases are $m = -1$, $m = -2$, and everything else.

General case, $m \neq -1, -2$.

$$\int_0^1 x(x+1)^m\,\mathrm{d}x = \left[\frac{(x+1)^{m+2}}{m+2} - \frac{(x+1)^{m+1}}{m+1}\right]_0^1 = \frac{2^{m+2}-1}{m+2} - \frac{2^{m+1}-1}{m+1}.$$

Over a common denominator the numerator is

$$(m+1)(2^{m+2}-1) - (m+2)(2^{m+1}-1) = 2^{m+1}\big(2(m+1) - (m+2)\big) + 1 = m\,2^{m+1} + 1,$$

so

$$\int_0^1 x(x+1)^m\,\mathrm{d}x = \frac{m\,2^{m+1}+1}{(m+1)(m+2)}, \qquad m \neq -1, -2.$$

Case $m = -1$.

$$\int_0^1 \left(1 - \frac{1}{x+1}\right)\mathrm{d}x = \Big[x - \ln(x+1)\Big]_0^1 = 1 - \ln 2.$$

Case $m = -2$.

$$\int_0^1 \left(\frac{1}{x+1} - \frac{1}{(x+1)^2}\right)\mathrm{d}x = \left[\ln(x+1) + \frac{1}{x+1}\right]_0^1 = \ln 2 + \tfrac{1}{2} - 1 = \ln 2 - \tfrac{1}{2}.$$

Ask yourself The general formula has $0/0$ at $m = -1$ and $m = -2$. Is that a contradiction?

No: it is part (i) again. At $m = -1$ the numerator is $-1 \cdot 2^0 + 1 = 0$, and at $m = -2$ it is $-2 \cdot 2^{-1} + 1 = 0$, so the formula is undefined there rather than wrong, and as in part (i) its limiting values are the special-case answers ($1 - \ln 2 \approx 0.307$ and $\ln 2 - \frac{1}{2} \approx 0.193$).

Checks

- $m = 0$: $\int_0^1 x \, dx = \frac{1}{2}$, and the formula gives $\frac{0+1}{1 \cdot 2} = \frac{1}{2}$. For $m = 1$: $\int_0^1 (x^2 + x) \, dx = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$, and the formula gives $\frac{4+1}{6}$.
- Numerically, $k = 10^{-4}$ gives $\frac{2^k - 1}{k} \approx 0.693171$, against $\ln 2 \approx 0.693147$.
- Simpson's rule on all the integrals, for many values of k and m (including $m = -1, -2$), agrees with the formulas to 10^{-9}. Putting $m = -1 \pm 10^{-5}$ into the general formula gives $0.30685\ldots$, matching $1 - \ln 2$.

Common slips

- Writing $\frac{(x+1)^0}{0}$ for $k = 0$, or missing that (ii) has *two* exceptional values of m, not one.
- Integrating by parts in (ii) and still dividing by $m + 1$ without noticing that this fails when $m = -1$.
- In the $m = -2$ case, getting the sign of $\int (x+1)^{-2} \, dx = -(x+1)^{-1}$ wrong.

Beyond STEP

The deduction in (i) says $\lim_{k \to 0} \frac{2^k - 1}{k} = \ln 2$, which is the derivative of 2^k at $k = 0$. The general fact used is that if integrands converge *uniformly*, their integrals converge too. That is a central theorem of first-year analysis, and it fails without uniformity. The same idea explains why $\int x^n \, dx$ "becomes" $\ln x$ as $n \to -1$. Abbott's *Understanding Analysis* treats uniform convergence and integration carefully.

WORKED SOLUTION

STEP I 2005, Question 6

Pure • Circles of Apollonius

STEP I 2005, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q6

The key idea

A condition like $AP = 2BP$ involves square roots, so square it first: distances squared are polynomials, and the $x^2 + y^2$ terms come with the same coefficient on both sides, which is why the path is a circle. In part (ii) we do the same for a general pair of points on the x-axis, and then two equations of the form $x^2 + y^2 + px + q = 0$ describe the same circle exactly when p and q agree. Comparing coefficients gives two expressions for k^2, and eliminating k gives both results.

Solution

Part (i)

Both lengths are non-negative, so $AP = 2BP$ is equivalent to $AP^2 = 4BP^2$:

$$(x-5)^2 + (y-16)^2 = 4\big[(x+4)^2 + (y-4)^2\big].$$

Expanding,

$$x^2 - 10x + y^2 - 32y + 281 = 4x^2 + 32x + 4y^2 - 32y + 128.$$

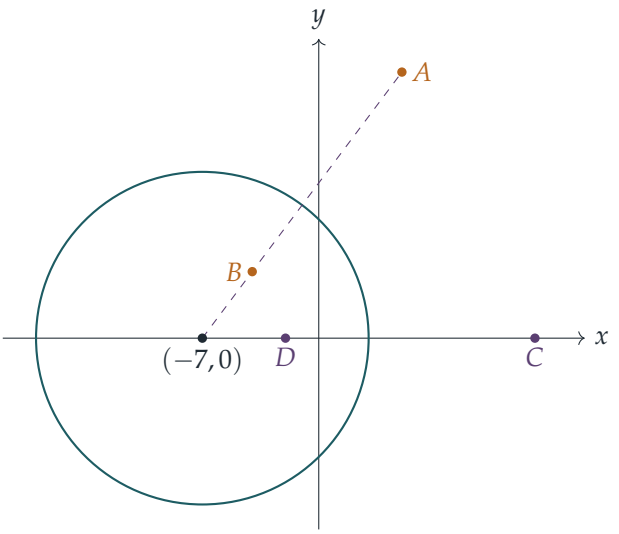
The y terms cancel, and collecting everything on the right gives

$$3x^2 + 42x + 3y^2 - 153 = 0, \qquad \text{that is,} \qquad x^2 + 14x + y^2 - 51 = 0.$$

Completing the square, $x^2 + 14x = (x+7)^2 - 49$, so

$$(x+7)^2 + y^2 = 100,$$

as required. The path is the circle with centre $(-7, 0)$ and radius 10.



(The points $C = (13, 0)$ and $D = (-2, 0)$ are one possible pair for part (ii), with $k = 2$. Notice that A, B and the centre are collinear too.)

Part (ii)

As before, $QC = kQD$ is equivalent to $QC^2 = k^2QD^2$. With $Q = (x, y)$,

$$(x-a)^2 + y^2 = k^2\left[(x-b)^2 + y^2\right].$$

Collecting terms on the right,

$$(k^2-1)(x^2+y^2) - 2(k^2b - a)x + (k^2b^2 - a^2) = 0.$$

Since $k > 1$, we have $k^2 - 1 \neq 0$ and we may divide by it:

$$x^2 + y^2 - \frac{2(k^2b-a)}{k^2-1}\,x + \frac{k^2b^2 - a^2}{k^2-1} = 0.$$

Ask yourself Why are we allowed to compare coefficients with $x^2 + 14x + y^2 - 51 = 0$?

An equation $x^2 + y^2 + px + q = 0$ can be rewritten as $(x + \frac{1}{2}p)^2 + y^2 = \frac{1}{4}p^2 - q$. If it is the circle with centre $(-7, 0)$ and radius 10, then the centre forces $p = 14$ and the radius forces $\frac{1}{4}p^2 - q = 100$, so $q = -51$. A circle has only one centre and one radius, so the two equations must have the same p and the same q.

Comparing the coefficients of x:

$$\frac{k^2b - a}{k^2 - 1} = -7 \implies k^2b - a = -7k^2 + 7 \implies k^2(b+7) = a + 7.$$

Comparing the constant terms:

$$\frac{k^2b^2-a^2}{k^2-1}=-51 \implies k^2b^2-a^2=-51k^2+51 \implies k^2(b^2+51)=a^2+51.$$

Now $b^2+51>0$, and $b+7\neq 0$: if $b=-7$, the first equation would give $a=-7=b$, which is ruled out. So we can divide by both to get two expressions for k^2:

$$\frac{a+7}{b+7}=k^2=\frac{a^2+51}{b^2+51},$$

which is the first result.

For the second, cross-multiply:

$$(a+7)(b^2+51)=(b+7)(a^2+51).$$

Expanding, the constant 357 cancels from both sides and we are left with

$$ab^2+51a+7b^2=a^2b+51b+7a^2.$$

Group the terms so that $a-b$ appears in each:

$$\begin{aligned}0&=(a^2b-ab^2)-51(a-b)+7(a^2-b^2)\\&=(a-b)\big[ab-51+7(a+b)\big].\end{aligned}$$

Since $a\neq b$, we can divide by $a-b$, so

$$ab+7a+7b=51, \quad \text{so} \quad ab+7a+7b+49=100, \quad \text{that is,} \quad (a+7)(b+7)=100,$$

as required.

Checks

- Try $b=-2$: then $b+7=5$, so $a+7=20$ and $a=13$, giving $k^2=20/5=4$. The point $(3,0)$ is on the circle, and $QC=10=2\times 5=2\,QD$. The point $(-17,0)$ is on it too, with $QC=30=2\times 15$.
- In part (i), the point $(3,0)$ gives $AP=\sqrt{4+256}=\sqrt{260}$ and $BP=\sqrt{49+16}=\sqrt{65}$, and indeed $\sqrt{260}=2\sqrt{65}$.
- We also checked numerically that random points on the circle satisfy $QC=kQD$ for many pairs with $(a+7)(b+7)=100$, and that every pair with $a\neq b$ satisfying the first result also satisfies the second.

Common slips

- Comparing coefficients before dividing by k^2-1: the two equations only need to agree up to an overall factor.
- Dividing by $b+7$ without checking it is non-zero.
- Losing the factor $a-b$: it is exactly the trivial solution $a=b$ that the question excludes.

Beyond STEP

The locus of points whose distances from two fixed points are in a fixed ratio $k \neq 1$ is a *circle of Apollonius*. The result $(a+7)(b+7) = 100$ says that C and D are on the same side of the centre with $MC \cdot MD = r^2$, where M is the centre and r the radius: C and D are *inverse points* in the circle. (So are A and B: $MA = 20$, $MB = 5$.) Inversion in a circle is a powerful tool in geometry and in complex analysis, where it is the map $z \mapsto 1/\bar{z}$; Needham's *Visual Complex Analysis* has a beautiful account.

WORKED SOLUTION

STEP I 2005, Question 7

Pure • Telescoping Products

STEP I 2005, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q7

The key idea

All three products *telescope*: write each factor as a ratio $g(r+1)/g(r)$, or as a product of such ratios, and almost everything cancels, leaving only the first and last values. Parts (i) and (ii) are warm-ups that show the mechanism. In part (iii) the work is to spot that the trigonometric factor is secretly such a ratio. Putting it over the common denominator $\sin\frac{(2r-1)\pi}{n}$ turns the numerator into a compound-angle sine.

Solution

Part (i)

Writing the product out,

$$\prod_{r=1}^{n}\frac{r+1}{r}=\frac{2}{1}\times\frac{3}{2}\times\frac{4}{3}\times\cdots\times\frac{n+1}{n}.$$

Each numerator cancels the denominator of the next factor, so only the last numerator and the first denominator survive:

$$\prod_{r=1}^{n}\frac{r+1}{r}=\frac{n+1}{1}=n+1.$$

Part (ii)

Factorise the numerator, $r^2-1=(r-1)(r+1)$, and split each factor into two:

$$\frac{r^2-1}{r^2}=\frac{r-1}{r}\times\frac{r+1}{r}.$$

Since the product of products is the product of the factors, we can treat the two families separately (here $n \geq 2$):

$$\prod_{r=2}^{n} \frac{r-1}{r} = \frac{1}{2} \times \frac{2}{3} \times \cdots \times \frac{n-1}{n} = \frac{1}{n},$$
$$\prod_{r=2}^{n} \frac{r+1}{r} = \frac{3}{2} \times \frac{4}{3} \times \cdots \times \frac{n+1}{n} = \frac{n+1}{2}.$$

Both telescope exactly as in part (i). Multiplying,

$$\prod_{r=2}^{n} \frac{r^2-1}{r^2} = \frac{1}{n} \times \frac{n+1}{2} = \frac{n+1}{2n}.$$

Part (iii)

Write $\alpha = \dfrac{2\pi}{n}$ and $\beta_r = \dfrac{(2r-1)\pi}{n}$, so the rth factor is $\cos\alpha + \sin\alpha \cot\beta_r$.

Ask yourself How can $\cos\alpha + \sin\alpha \cot\beta$ be a ratio of two similar things?

Write $\cot\beta = \cos\beta / \sin\beta$ and use a common denominator. The numerator is then the expansion of a compound angle:

$$\cos\alpha + \sin\alpha \frac{\cos\beta}{\sin\beta} = \frac{\sin\beta\cos\alpha + \cos\beta\sin\alpha}{\sin\beta} = \frac{\sin(\alpha+\beta)}{\sin\beta}.$$

This needs $\sin\beta_r \neq 0$, and it is here that we use the condition that n is even. We would have $\sin\beta_r = 0$ only if $(2r-1)/n$ were an integer. But $2r-1$ is odd and n is even, so an odd number can't be a multiple of n. So every factor is defined, and every denominator below is non-zero.

Now $\alpha + \beta_r = \dfrac{2\pi + (2r-1)\pi}{n} = \dfrac{(2r+1)\pi}{n} = \beta_{r+1}$, so

$$\cos\alpha + \sin\alpha\cot\beta_r = \frac{\sin\beta_{r+1}}{\sin\beta_r}.$$

This is exactly the shape of part (i), with $g(r) = \sin\beta_r$. The product telescopes:

$$\prod_{r=1}^{n} \frac{\sin\beta_{r+1}}{\sin\beta_r} = \frac{\sin\beta_2}{\sin\beta_1} \times \frac{\sin\beta_3}{\sin\beta_2} \times \cdots \times \frac{\sin\beta_{n+1}}{\sin\beta_n}$$
$$= \frac{\sin\beta_{n+1}}{\sin\beta_1} = \frac{\sin\left(\frac{(2n+1)\pi}{n}\right)}{\sin\frac{\pi}{n}} = \frac{\sin\left(2\pi + \frac{\pi}{n}\right)}{\sin\frac{\pi}{n}}.$$

Since sin has period 2π, the numerator equals the denominator, and

$$\prod_{r=1}^{n}\left(\cos\tfrac{2\pi}{n} + \sin\tfrac{2\pi}{n}\cot\tfrac{(2r-1)\pi}{n}\right) = 1.$$

Checks

- Part (i) with $n = 3$: $2 \cdot \frac{3}{2} \cdot \frac{4}{3} = 4 = n + 1$.
- Part (ii) with $n = 2$: the single factor is $\frac{3}{4}$, and $\frac{n+1}{2n} = \frac{3}{4}$. As $n \to \infty$ the product tends to $\frac{1}{2}$, which is a well-known infinite product.
- Part (iii) with $n = 4$: $\cos \frac{\pi}{2} = 0$ and $\sin \frac{\pi}{2} = 1$, so the factors are $\cot \frac{\pi}{4}$, $\cot \frac{3\pi}{4}$, $\cot \frac{5\pi}{4}$ and $\cot \frac{7\pi}{4}$, that is, 1, -1, 1 and -1, with product 1. We also checked numerically that the product equals 1 (to within 10^{-8}) for every even n up to 60.
- If n is odd, the factor with $r = \frac{n+1}{2}$ involves $\cot \pi$, which is undefined. That explains the condition.

Common slips

- In part (ii), starting the cancellation from $r = 1$: the lower limit is 2, so the first factors are $\frac{1}{2}$ and $\frac{3}{2}$.
- In part (iii), stopping at $\sin \frac{(2n+1)\pi}{n} / \sin \frac{\pi}{n}$ without using periodicity to simplify it to 1.
- Not explaining where "n even" is used. It is exactly what makes every cotangent, and every denominator, well defined.

Beyond STEP

Telescoping is the multiplicative cousin of the telescoping sums you meet with partial fractions. Taking logarithms turns one into the other. Part (ii) is a finite version of the infinite product $\prod_{r \geq 2}(1 - 1/r^2) = \frac{1}{2}$. Infinite products like this, and Euler's product

$$\sin \pi x = \pi x \prod_{r \geq 1} \left(1 - \frac{x^2}{r^2}\right),$$

are studied properly in a first analysis course, where convergence of products is defined via the convergence of sums of logarithms. Abbott's *Understanding Analysis* is a good place to build the convergence ideas you'll need.

WORKED SOLUTION

STEP I 2005, Question 8

Pure • First Order Differential Equations

STEP I 2005, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q8

The key idea

The opening identity is a recipe for spotting equations of the form $2xy\frac{\mathrm{d}y}{\mathrm{d}x} = ky^2 +$ (something in x alone). Once an equation is in that shape, the substitution $y^2 = x^k f(x)$ cancels the y^2 terms and leaves $f'(x)$ equal to a function of x, which we simply integrate. The only work is choosing k and, in part (ii), dividing through so that the left-hand side becomes $2xy\frac{\mathrm{d}y}{\mathrm{d}x}$.

Solution

The identity

Differentiate $y^2 = x^k f(x)$ implicitly with respect to x, using the product rule on the right:

$$2y\frac{\mathrm{d}y}{\mathrm{d}x} = kx^{k-1}f(x) + x^k\frac{\mathrm{d}f}{\mathrm{d}x}.$$

Multiplying by x,

$$2xy\frac{\mathrm{d}y}{\mathrm{d}x} = k\,x^k f(x) + x^{k+1}\frac{\mathrm{d}f}{\mathrm{d}x} = ky^2 + x^{k+1}\frac{\mathrm{d}f}{\mathrm{d}x},$$

since $x^k f(x) = y^2$. This is the required result.

Part (i)

With $k = 1$ we write $y^2 = xf(x)$, and the identity says $2xy\frac{\mathrm{d}y}{\mathrm{d}x} = y^2 + x^2 f'(x)$. Comparing with the given equation, the y^2 terms match, so the equation holds exactly when

$$x^2 f'(x) = x^2 - 1, \qquad \text{that is,} \qquad f'(x) = 1 - \frac{1}{x^2} \quad (x \neq 0).$$

Integrating, $f(x) = x + \dfrac{1}{x} + c$, so

$$y^2 = xf(x) = x^2 + cx + 1.$$

The condition $y = 2$ when $x = 1$ gives $4 = 1 + c + 1$, so $c = 2$ and

$$y^2 = x^2 + 2x + 1 = (x+1)^2.$$

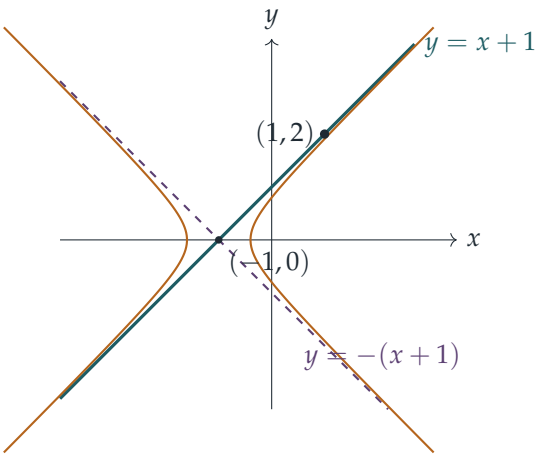
Hence $y = x + 1$ or $y = -(x+1)$; only $y = x + 1$ passes through $(1, 2)$, so the solution is

$$y = x + 1.$$

(Substituting back: $2x(x+1) \cdot 1 = 2x^2 + 2x$ and $(x+1)^2 + x^2 - 1 = 2x^2 + 2x$, so it satisfies the equation for every x, including $x = 0$ where we divided.)

Ask yourself What does "describe geometrically" want, when the answer is just $y = x + 1$?

The solution curve is a straight line: the line of gradient 1 through $(-1, 0)$ and $(0, 1)$, making an angle of $45°$ with the x-axis. It is worth saying why this is a little surprising. The general solution $y^2 - x^2 - cx = 1$ is a family of conics; completing the square gives $y^2 - (x + \frac{c}{2})^2 = 1 - \frac{c^2}{4}$, a rectangular hyperbola centred at $(-\frac{c}{2}, 0)$ with asymptotes $y = \pm(x + \frac{c}{2})$. The initial condition picks out $c = 2$, the case where the right-hand side is zero and the hyperbola degenerates into its own pair of asymptotes $y = \pm(x+1)$. Our solution is the branch $y = x + 1$ of that pair.



The solid line is the solution; the dashed line is the other half of $y^2 = (x+1)^2$. The amber curves sketch the shape of the non-degenerate members of the family: a rectangular hyperbola approaching a pair of lines like these as asymptotes.

Part (ii)

To use the identity we need $2xy\,\frac{\mathrm{d}y}{\mathrm{d}x}$ on the left, so divide the equation by x (the $\ln x$ term forces $x > 0$, so this is safe):

$$2xy\frac{\mathrm{d}y}{\mathrm{d}x} = -y^2 + \frac{2\ln x}{x}.$$

The coefficient of y^2 is -1, so take $k = -1$, that is $y^2 = f(x)/x$. Then $x^{k+1} = x^0 = 1$ and the identity reads $2xy\,\frac{\mathrm{d}y}{\mathrm{d}x} = -y^2 + f'(x)$. Comparing,

$$f'(x) = \frac{2\ln x}{x}.$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}(\ln x)^2 = \frac{2\ln x}{x}$, we get $f(x) = (\ln x)^2 + c$, and so

$$y^2 = \frac{(\ln x)^2 + c}{x}.$$

At $x = 1$, $y = 1$: $1 = (0 + c)/1$, so $c = 1$. Taking the positive square root, because $y = 1 > 0$ at $x = 1$ and y never vanishes (the numerator $1 + (\ln x)^2$ is at least 1), the solution is

$$y = \sqrt{\frac{1 + (\ln x)^2}{x}}, \qquad x > 0.$$

Checks

- In (i), the line $y = x + 1$ can be checked directly in the equation, as above; integrating the equation numerically from $(1, 2)$ (Runge–Kutta) gives $y = 3.000$ at $x = 2$ and $y = 6.000$ at $x = 5$.
- In (ii), $y^2 = (1 + (\ln x)^2)/x$ gives $y(3) \approx 0.8577$ and $y(0.2) \approx 4.2369$; a numerical integration of the differential equation from $(1, 1)$ agrees to many decimal places.
- The identity itself: try $k = 0$, where it collapses to $2xy\,y' = xf'$, which is just $(y^2)' = f'$ multiplied by x.

Common slips

- In (ii), using $k = 1$ out of habit. The sign of the y^2 term, after dividing by x, is what fixes $k = -1$.
- Forgetting to divide by x in (ii), and so trying to match $2x^2y\,y'$ with the identity.
- Leaving the answer to (i) as $y^2 = (x + 1)^2$ without choosing the branch through $(1, 2)$.

Beyond STEP

The substitution here is a disguised version of a standard trick: writing $u = y^2$ turns $2xy\,y' = ky^2 + g(x)$ into the *linear* equation $xu' - ku = g(x)$, and x^{-k} is its integrating factor. Equations that become linear after a power substitution are called Bernoulli equations, and they are among the first non-linear equations met in a university ODE course. Boyce and DiPrima, *Elementary Differential Equations*, treats them alongside integrating factors and exact equations.

WORKED SOLUTION

STEP I 2005, Question 9

Mechanics • Rigid Body Equilibrium

STEP I 2005, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q9

The key idea

Everything hinges on one unknown: where the centre of mass G of the non-uniform rod lies. The first two set-ups each give T in terms of that position, so equating them pins down G and T together. After that, the tilted set-up is a standard rigid-body problem: take moments about the point where the unknown contact forces act, then resolve.

Solution

We model the rod as rigid, with all its weight W acting at its centre of mass G, and the strings as light and inextensible. Let $AG = x$, so $GB = 3l - x$.

Finding G and showing $5T = 2W$

Strings at both ends. Taking moments about B removes the unknown tension at B:

$$T \cdot 3l = W(3l - x).$$

Pivot at distance l from A, string at B. Let P be the pivot, so $PB = 2l$. Taking moments about P removes the unknown pivot reaction. For the string at B to hold the rod up, G must lie on the B side of P, so $PG = x - l$ and

$$T \cdot 2l = W(x - l).$$

Eliminating T (multiply the first equation by 2, the second by 3, and compare):

$$2W(3l - x) = 3W(x - l) \quad \Longrightarrow \quad 9l = 5x \quad \Longrightarrow \quad x = \tfrac{9}{5}l.$$

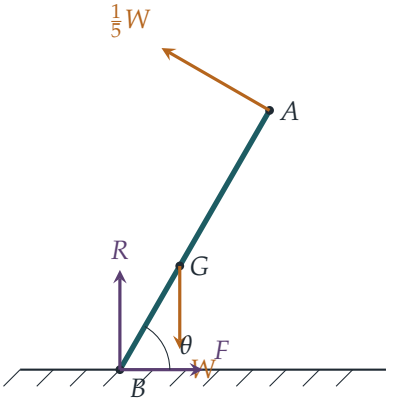
Then, from the second equation, $2lT = W \cdot \frac{4}{5}l$, so $T = \frac{2}{5}W$, that is,

$$5T = 2W,$$

as required. Note that $x = \frac{9}{5}l > l$, which agrees with our assumption that G is beyond the pivot. So G is at distance $GB = 3l - \frac{9}{5}l = \frac{6}{5}l$ from B.

The rod resting on the ground

Now B is on rough ground, the rod makes angle θ with the horizontal, and the string at A is perpendicular to the rod with tension $\frac{1}{2}T = \frac{1}{5}W$. At B there is a normal reaction R upwards and a friction force F along the ground.



Moments about B. This kills R and F. The string force is perpendicular to the rod at distance $3l$ from B, so its moment is $\frac{1}{5}W \cdot 3l$. The weight acts vertically at distance $\frac{6}{5}l$ along the rod, so its perpendicular distance from B is $\frac{6}{5}l\cos\theta$. Hence

$$\tfrac{1}{5}W \cdot 3l = W \cdot \tfrac{6}{5}l\cos\theta \quad\Longrightarrow\quad \cos\theta = \tfrac{1}{2},$$

so $\theta = 60^\circ$ (that is, $\frac{\pi}{3}$).

Ask yourself Which way do the components of the string's tension point?

The rod makes angle θ with the horizontal, and the string is at right angles to the rod, so the string makes angle θ with the *vertical*. Its tension $\frac{1}{5}W$ therefore has vertical component $\frac{1}{5}W\cos\theta$ (upwards) and horizontal component $\frac{1}{5}W\sin\theta$, pointing back over B, away from the side of A. Friction must balance this horizontal pull, so it acts at B towards the side of A, as drawn.

Resolving vertically:

$$R + \tfrac{1}{5}W\cos 60^\circ = W \quad\Longrightarrow\quad R = W - \tfrac{1}{10}W = \tfrac{9}{10}W.$$

Resolving horizontally:

$$F = \tfrac{1}{5}W\sin 60^\circ = \tfrac{\sqrt{3}}{10}W.$$

The rod does not slip provided $F \le \mu R$, that is,

$$\mu \ge \frac{F}{R} = \frac{\sqrt{3}/10}{9/10} = \frac{\sqrt{3}}{9}.$$

So the smallest coefficient of friction that prevents slipping is $\mu = \dfrac{\sqrt{3}}{9} = \dfrac{1}{3\sqrt{3}} \approx 0.192$ (with equality the rod is in limiting equilibrium).

Checks

- With $x = \frac{9}{5}l$, the two-string set-up gives $T = W \cdot \frac{6/5}{3} = \frac{2}{5}W$, and the pivot set-up gives $T = W \cdot \frac{4/5}{2} = \frac{2}{5}W$: the same, as it must be.
- In the tilted position, taking moments about A as well (with $W = l = 1$, $\theta = 60°$, $R = 0.9$, $F \approx 0.1732$) also gives zero resultant moment, a check on R and F together.
- Numerically $\mu_{\min} = \sqrt{3}/9 \approx 0.19245$, which a computer confirmed by building the force vectors and checking all three equilibrium equations.

Common slips

- Using $\frac{6}{5}l\sin\theta$ for the weight's moment arm. The horizontal distance from B to the line of the weight is $\frac{6}{5}l\cos\theta$.
- Forgetting that the perpendicular string has a vertical component, and taking $R = W$. That gives $\mu = \sqrt{3}/10$, which is too small.
- Assuming the rod is uniform, so that G is at the midpoint. The whole point of the first part is to locate G.

Beyond STEP

Locating a centre of mass from equilibrium measurements is the static version of a powerful idea: the moment of a force system about a point determines the system, up to a single resultant. At university this becomes the study of rigid-body statics and then dynamics, where the centre of mass and the moment of inertia are the two numbers that govern a body's motion. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats torque and rigid bodies with a wealth of harder problems in this style.

WORKED SOLUTION

STEP I 2005, Question 10

Mechanics • Collisions and Restitution

STEP I 2005, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

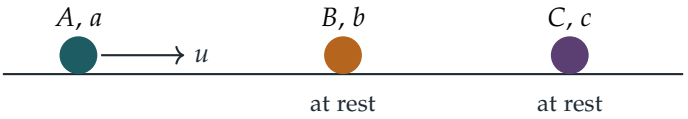
step.maths.org/questions/05-s1-q10

The key idea

Each collision is a direct impact between two smooth particles, so it gives us two equations: conservation of momentum along the line, and Newton's experimental law (separation speed $= e\times$ approach speed). Solving these once for a general pair gives formulas we can reuse for the second collision. For the special cases, the real question is whether the story ends after two collisions: we need the final velocities, so we must check that no particle catches up with the one in front.

Solution

We model A, B, C as particles moving along one line on a smooth surface, so no horizontal external forces act and momentum along the line is conserved in each impact. Take the direction of u as positive. Write p for the velocity of A after its collision with B, and q for the velocity of B after its collision with C.



The first collision: finding v

Before impact A has velocity u and B is at rest; afterwards they have velocities p and v. Momentum and Newton's law give

$$au = ap + bv, \qquad v - p = eu.$$

Substituting $p = v - eu$ into the first equation, $au = av - aeu + bv$, so

$$v = \frac{a(1+e)u}{a+b}, \qquad p = v - eu = \frac{(a-eb)u}{a+b}.$$

Note $v > 0$, so B really does move towards C.

The second collision: finding w

This is the same situation with B (mass b, speed v) striking C (mass c, at rest). So we can reuse the formulas above with $a \to b$, $b \to c$, $u \to v$:

$$w = \frac{b(1+e)v}{b+c}, \qquad q = \frac{(b-ec)v}{b+c}.$$

Substituting for v,

$$w = \frac{b(1+e)}{b+c} \cdot \frac{a(1+e)u}{a+b} = \frac{abu(1+e)^2}{(a+b)(b+c)},$$

as required.

After these two collisions the velocities are p, q, w. There is a further collision only if $p > q$ (A catches B) or $q > w$ (B catches C); since $q - w = -ev < 0$ by Newton's law, the second never happens, so we only need to compare p and q.

Ask yourself Why can't we just stop after two collisions and quote p, q, w?

Because the question asks for *final* velocities. If A were still moving faster than B afterwards, it would hit B again and everything would change. So in each case we must check that $p \le q$.

Part (i)

Here $a = eb$ and $b = ec$ (with $e > 0$, as the masses are positive). Then $a - eb = 0$ and $b - ec = 0$, so

$$p = 0, \qquad q = 0.$$

For w, use $a + b = b(1+e)$ and $b + c = b + b/e = b(1+e)/e$:

$$w = \frac{eb \cdot b\,u(1+e)^2}{b(1+e) \cdot b(1+e)/e} = e^2 u.$$

A and B are both at rest and C moves away, so there are no more collisions. The final velocities are

$$A:\ 0, \qquad B:\ 0, \qquad C:\ e^2 u.$$

Each collision passes all the momentum on to the next particle, like a Newton's cradle with lighter particles behind.

Part (ii)

Now $b = ea$ and $c = eb = e^2 a$. Then

$$v = \frac{a(1+e)u}{a(1+e)} = u, \qquad p = \frac{(a - e^2 a)u}{a(1+e)} = (1-e)u,$$

using $1 - e^2 = (1-e)(1+e)$. In the second collision the masses b, c are in the same ratio as a, b and B arrives with speed u, so it is an exact copy of the first collision:

$$w = u, \qquad q = (1-e)u.$$

(Directly: $w = \dfrac{a \cdot ea \cdot u(1+e)^2}{a(1+e) \cdot ea(1+e)} = u$.)

Now $p = q$, so A never catches B; and $q = (1-e)u < u = w$ for $e > 0$, so B never catches C. The final velocities are

$$A:\ (1-e)u, \qquad B:\ (1-e)u, \qquad C:\ u,$$

all in the original direction of motion of A.

Checks

- Momentum in (ii): $a(1-e)u + ea(1-e)u + e^2 a\, u = au$. In (i): $c \cdot e^2 u = e^2 c\, u = au$.
- With $e = 1$ in (ii) (perfectly elastic), A and B end at rest and C moves off with u; kinetic energy is conserved: $\frac{1}{2}au^2 = \frac{1}{2}cu^2$ needs $c = a$, which is exactly $c = e^2 a$ with $e = 1$.
- A numerical case: with $e = \frac{1}{2}$, masses $1, 2, 4$ and $u = 1$ (part (i)) the final velocities are $0, 0, \frac{1}{4}$; masses $4, 2, 1$ (part (ii)) give $\frac{1}{2}, \frac{1}{2}, 1$. We confirmed these, and the general formulas for v and w, with exact arithmetic over thousands of random masses and values of e, simulating any further impacts.

Common slips

- Writing Newton's law with the velocities the wrong way round: it is (speed of the front particle $-$ speed of the back particle) after $= e \times$ (back $-$ front) before.
- Quoting the velocities after two collisions without checking whether A catches B again.

Beyond STEP

Chains of collisions are the simplest model of how momentum and energy travel through a material, and the mass ratios in part (i) are chosen so that *all* the momentum ends up in the last particle. Ordering the masses the other way, heavy behind light, with e close to 1, makes the last particle move faster than the first: the idea behind the toy where a small ball balanced on a large one shoots high into the air when the pair is dropped. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions using the centre-of-mass frame, which makes formulas like v and w almost obvious.

WORKED SOLUTION

STEP I 2005, Question 11

Mechanics • Vector Kinematics

STEP I 2005, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q11

The key idea

Everything in this question is about how two vectors, $\mathbf{r}$ and $\mathbf{v} = \dot{\mathbf{r}}$, sit relative to each other. Perpendicular means the scalar product $\mathbf{r} \cdot \mathbf{v}$ is zero; parallel means the "cross" combination $xv_y - yv_x$ is zero. Both expressions collapse nicely once we write $\sin 2t = 2\sin t \cos t$ and $\cos 2t = 1 - 2\sin^2 t$. For the distance and the sketch, it pays to write everything in terms of $\cos t$, or to eliminate t altogether.

Solution

Write $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j}$ with

$$x = \sin 2t = 2\sin t \cos t, \qquad y = 2\cos t, \qquad 0 \leq t < 2\pi.$$

Differentiating, the velocity is

$$\mathbf{v} = \dot{\mathbf{r}} = (2\cos 2t)\,\mathbf{i} - (2\sin t)\,\mathbf{j}.$$

Part (i)

The particle is at the origin when $x = 0$ and $y = 0$. The condition $y = 0$ means $\cos t = 0$, so $t = \frac{\pi}{2}$ or $t = \frac{3\pi}{2}$ in the given range. At both of these times $x = 2\sin t \cos t = 0$ automatically. So the particle is at the origin at exactly these two times, $t = \frac{\pi}{2}$ and $t = \frac{3\pi}{2}$, and at no others, which is what we had to show.

Part (ii)

The velocity is perpendicular to the displacement when $\mathbf{r} \cdot \mathbf{v} = 0$ (with both vectors non-zero, so that "perpendicular" has a meaning). Now

$$\begin{aligned}\mathbf{r} \cdot \mathbf{v} &= 2\sin 2t \cos 2t - 4 \sin t \cos t \\ &= 2 \sin 2t \cos 2t - 2 \sin 2t \\ &= 2 \sin 2t \, (\cos 2t - 1) \\ &= -4 \sin 2t \, \sin^2 t,\end{aligned}$$

using $\cos 2t - 1 = -2\sin^2 t$. This vanishes when $\sin 2t = 0$ or $\sin t = 0$, that is, at

$$t = 0, \ \tfrac{\pi}{2}, \ \pi, \ \tfrac{3\pi}{2}.$$

Ask yourself Should $t = \frac{\pi}{2}$ and $t = \frac{3\pi}{2}$ really be on the list?

No. By part (i), at those times $\mathbf{r} = \mathbf{0}$, and the zero vector has no direction, so it can't be perpendicular to anything in a meaningful sense; the scalar product is zero for a trivial reason. At $t = 0$ we have $\mathbf{r} = 2\mathbf{j}$ and $\mathbf{v} = 2\mathbf{i}$, and at $t = \pi$ we have $\mathbf{r} = -2\mathbf{j}$ and $\mathbf{v} = 2\mathbf{i}$: genuinely perpendicular, non-zero vectors. So the velocity is perpendicular to the displacement at

$$t = 0 \quad \text{and} \quad t = \pi.$$

Part (iii)

First, $\mathbf{v}$ is never zero: $\mathbf{v} = \mathbf{0}$ would need $\sin t = 0$, but then $\cos 2t = 1 - 2\sin^2 t = 1 \neq 0$.

Two non-zero vectors (x, y) and (v_x, v_y) are parallel exactly when $xv_y - yv_x = 0$. Here

$$\begin{aligned}xv_y - yv_x &= \sin 2t \, (-2 \sin t) - 2 \cos t \, (2 \cos 2t) \\ &= -4 \sin^2 t \cos t - 4 \cos t \, (1 - 2\sin^2 t) \\ &= -4 \cos t \, (1 - \sin^2 t) \\ &= -4 \cos^3 t.\end{aligned}$$

This is zero only when $\cos t = 0$, which by part (i) happens only when the particle is at the origin. So whenever the particle is away from the origin, $\cos t \neq 0$, the combination is non-zero, and the velocity is not parallel to the displacement. That completes the proof.

Part (iv)

Let $c = \cos^2 t$, so $0 \leq c \leq 1$. Then

$$|\mathbf{r}|^2 = \sin^2 2t + 4\cos^2 t = 4 \sin^2 t \cos^2 t + 4 \cos^2 t = 4c(1 - c) + 4c = 4c(2 - c).$$

The quadratic $4c(2 - c)$ has its vertex at $c = 1$ and is increasing on $[0, 1]$, so its largest value on this interval is at $c = 1$, giving $|\mathbf{r}|^2 = 4$. The maximum distance from the

origin is therefore

$$|\mathbf{r}|_{\max} = 2,$$

reached when $\cos^2 t = 1$, that is, at $t = 0$ and $t = \pi$, at the points $(0, \pm 2)$. It's reassuring that these are exactly the times found in part (ii): at a maximum of distance, the radial component of the velocity must vanish.

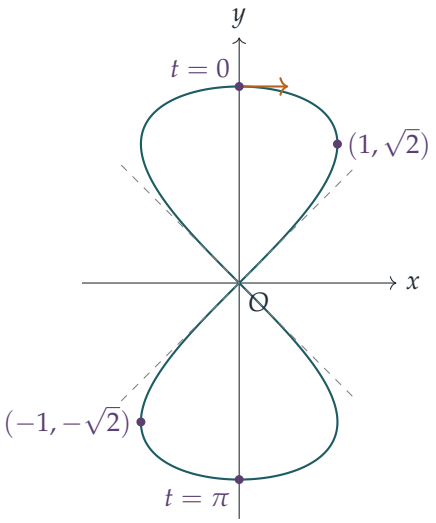
For the sketch, eliminate t. Since $x = y \sin t$ and $\sin^2 t = 1 - \cos^2 t = 1 - \frac{1}{4}y^2$,

$$x^2 = y^2\left(1 - \frac{y^2}{4}\right), \qquad \text{that is,} \qquad 4x^2 = y^2(4 - y^2).$$

Some features to put on the sketch:

- The curve is symmetric in both axes, and $-2 \le y \le 2$.
- $|x| = |\sin 2t| \le 1$, with $|x| = 1$ when $y = \pm\sqrt{2}$.
- Near the origin $4x^2 \approx 4y^2$, so the curve crosses itself there along the lines $y = \pm x$. This matches the velocities $-2\mathbf{i} - 2\mathbf{j}$ at $t = \frac{\pi}{2}$ and $-2\mathbf{i} + 2\mathbf{j}$ at $t = \frac{3\pi}{2}$.
- At $(0, \pm 2)$ the path is horizontal, as part (ii) showed.

Following t: the particle starts at $(0, 2)$ moving in the $+\mathbf{i}$ direction, goes down the right side of the upper loop to the origin, round the lower loop (left side first) back to the origin, and finally up the left side of the upper loop. The path is a figure of eight.



Checks

- At $t = \frac{\pi}{4}$: $\mathbf{r} = \mathbf{i} + \sqrt{2}\,\mathbf{j}$, which satisfies $4x^2 = 4 = y^2(4 - y^2) = 2 \cdot 2$, and $|\mathbf{r}| = \sqrt{3} < 2$, agreeing with $4c(2 - c) = 4 \cdot \frac{1}{2} \cdot \frac{3}{2} = 3$.
- Sampling t finely over $[0, 2\pi)$ gives a largest distance of 2.000, near-zero distances only at $t \approx 1.57$ and 4.71, and confirms the identities $\mathbf{r} \cdot \mathbf{v} = -4 \sin 2t \sin^2 t$ and

$xv_y - yv_x = -4\cos^3 t$ at random times. The smallest speed is about 1.32, so $\mathbf{v}$ is never zero.

Common slips

- Listing $t = \frac{\pi}{2}$ and $\frac{3\pi}{2}$ in part (ii): the scalar product vanishes there only because $\mathbf{r} = \mathbf{0}$.
- In part (iii), forgetting to check that $\mathbf{v}$ is never zero; a zero velocity would be "parallel" to everything.
- Drawing the loops side by side: the eight stands upright, with loops above and below the origin, since y ranges over $[-2, 2]$ but x only over $[-1, 1]$.

Beyond STEP

The quantities $\mathbf{r} \cdot \mathbf{v} = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}|\mathbf{r}|^2$ and $xv_y - yv_x$ are the radial part of the motion and (per unit mass) the angular momentum about the origin; in polar coordinates the second is $r^2\dot{\theta}$. Part (iii) says the angular momentum vanishes only at the origin, so the particle never moves straight towards or away from O. The curve itself is the lemniscate of Gerono, one of many classical figure-of-eight curves. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops polar coordinates and angular momentum in exactly this spirit.

WORKED SOLUTION

STEP I 2005, Question 12

Probability and Statistics • Venn Diagram Bounds

STEP I 2005, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q12

The key idea

Questions of the form "what range of values is consistent?" are about a Venn diagram whose regions are unknown but must all be *non-negative*. Write every region in terms of as few free unknowns as possible; each condition "this region ≥ 0" then becomes a bound, and the bounds together give the range. In part (ii), a counting argument (how many times each wizard is counted when we add the three single probabilities) gets the "exactly two" probability without needing the individual regions.

Solution

Part (i)

Let S be the event that a hobbit smokes a pipe and H the event that it wears a hat. Put $t = \mathrm{P}(S \cap H)$. The four regions of the Venn diagram are

$$\mathrm{P}(S \cap H) = t, \qquad \mathrm{P}(S \cap H') = 0.7 - t,$$
$$\mathrm{P}(S' \cap H) = 0.4 - t, \qquad \mathrm{P}(S' \cap H') = 1 - (0.7 + 0.4 - t) = t - 0.1.$$

Any choice of t that makes all four non-negative gives a genuine probability distribution, and conversely. The binding conditions are $t \geq 0.1$ (from the last region) and $t \leq 0.4$ (from the third); the others then follow. So $0.1 \leq t \leq 0.4$, and since $p = 0.7 - t$,

$$0.3 \leq p \leq 0.6.$$

Both ends are attained: $p = 0.6$ when as few pipe-smokers as possible wear hats (every hobbit does at least one of the two), and $p = 0.3$ when every hat-wearer smokes.

Part (ii): exactly two

Let H, C, R be the events that a wizard wears a hat, a cloak and a ring. Let a, b be the probabilities that a wizard wears exactly one and exactly two of the three items. We

know the probability of exactly none is 0.05 and of exactly three is 0.1, so

$$0.05 + a + b + 0.1 = 1 \quad \Longrightarrow \quad a + b = 0.85.$$

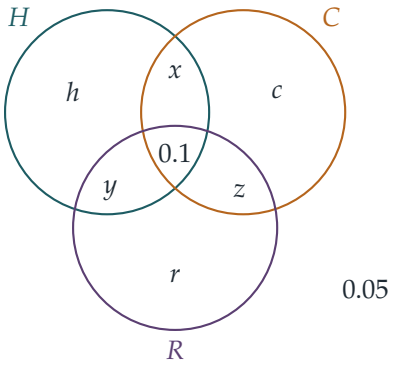
Now add the three given probabilities: $\mathrm{P}(H) + \mathrm{P}(C) + \mathrm{P}(R) = 0.7 + 0.8 + 0.4 = 1.9$. In this sum a wizard who wears exactly k items is counted k times, so

$$a + 2b + 3(0.1) = 1.9 \quad \Longrightarrow \quad a + 2b = 1.6.$$

Subtracting, $b = 0.75$ (and $a = 0.1$). The probability that a wizard wears exactly two of the items is $\frac{3}{4}$.

Part (ii): the range of q

Label the seven inside regions as in the diagram: h, c, r for exactly one item, x, y, z for exactly two, with 0.1 in the centre and 0.05 outside.



From the totals for H, C, R, and from part (ii) so far,

$$h + x + y = 0.6, \qquad c + x + z = 0.7,$$
$$r + y + z = 0.3, \qquad x + y + z = 0.75.$$

Using the last equation to eliminate a pair from each of the first three,

$$h = z - 0.15, \qquad c = y - 0.05, \qquad r = x - 0.45.$$

(As a check, $h + c + r = 0.75 - 0.65 = 0.1 = a$.) So the whole distribution is determined by x, y, z with $x + y + z = 0.75$, and it is a genuine distribution exactly when all six regions are non-negative:

$$x \geq 0.45, \qquad y \geq 0.05, \qquad z \geq 0.15.$$

Ask yourself How large can x be?

Since $x = 0.75 - y - z$ and $y + z \geq 0.05 + 0.15 = 0.2$, we get $x \leq 0.55$, with equality when $y = 0.05$, $z = 0.15$. The smallest value $x = 0.45$ is attained with, for example,

$y = 0.1$, $z = 0.2$. Every value in between is attained too, since we can move y and z continuously. Hence

$$0.45 \le x \le 0.55.$$

Finally, wearing a hat but not a ring while wearing a cloak is exactly the region x, so

$$q = \mathrm{P}(H \cap R' \mid C) = \frac{\mathrm{P}(H \cap C \cap R')}{\mathrm{P}(C)} = \frac{x}{0.8}, \qquad \text{giving} \qquad \frac{9}{16} \le q \le \frac{11}{16}.$$

Checks

- At the extreme $x = 0.55$, $y = 0.05$, $z = 0.15$ we get $h = 0$, $c = 0$, $r = 0.1$; the eight regions sum to 1 and give $\mathrm{P}(H) = 0.7$, $\mathrm{P}(C) = 0.8$, $\mathrm{P}(R) = 0.4$, as required.
- We enumerated every assignment of the eight regions in steps of 0.01 consistent with the data: every one has $x + y + z = 0.75$, and q runs from 0.5625 to 0.6875. Similarly for (i), p runs from 0.3 to 0.6.

Common slips

- In (i), forgetting the region "neither" and so getting the lower bound $t \ge 0$ instead of $t \ge 0.1$, which gives the wrong upper limit $p \le 0.7$.
- In (ii), dividing by 1 instead of by $\mathrm{P}(C) = 0.8$: the condition "given that he wears a cloak" changes the denominator.
- Treating H, C, R as independent. Nothing in the question says they are, and in fact $\mathrm{P}(H \cap C \cap R) = 0.1 \neq 0.7 \times 0.8 \times 0.4$.

Beyond STEP

Finding the range of a quantity that is linear in unknown probabilities, subject to linear equations and non-negativity, is a small *linear programme*: the extremes always occur at corners of the feasible region, which is why ours came from setting regions to zero. The same idea gives the Fréchet bounds $\max(0, \mathrm{P}(A) + \mathrm{P}(B) - 1) \le \mathrm{P}(A \cap B) \le \min(\mathrm{P}(A), \mathrm{P}(B))$ used in part (i). For more on inclusion–exclusion and conditional probability, read Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP I 2005, Question 13

Probability and Statistics • Symmetric Distributions

STEP I 2005, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q13

The key idea

We are told nothing about the shape of the distribution except two cumulative probabilities and symmetry, so everything has to be built out of those. Symmetry about μ gives two more facts for free: half the probability lies on each side of μ, and the tail below $\mu - k\sigma$ matches the tail above $\mu + k\sigma$. In part (ii) the trick is to notice that every volume mentioned (495, 500, 505 ml) sits at μ, $\mu \pm \frac{1}{2}\sigma$ or $\mu + \sigma$ for one type of milk or the other.

Solution

Throughout we treat X as continuous (it models a volume), so a single value has probability zero and it doesn't matter whether we write $<$ or $\leq$.

Symmetry about μ gives $\mathrm{P}(X \leq \mu) = \frac{1}{2}$ and $\mathrm{P}(X \leq \mu - k\sigma) = \mathrm{P}(X \geq \mu + k\sigma)$ for any k. In particular

$$\mathrm{P}\left(X \leq \mu - \tfrac{1}{2}\sigma\right) = 1 - b, \qquad \mathrm{P}\left(X \leq \mu - \sigma\right) = 1 - a.$$

It helps to record the probability of each strip, measured in units of σ from μ:



For example, $\mathrm{P}(\mu < X \leq \mu + \frac{1}{2}\sigma) = b - \frac{1}{2}$ and $\mathrm{P}(\mu + \frac{1}{2}\sigma < X \leq \mu + \sigma) = a - b$; the left-hand strips follow by symmetry.

Part (i)

Subtracting cumulative probabilities,

$$\mathrm{P}\left(\mu - \tfrac{1}{2}\sigma \leq X \leq \mu + \sigma\right) = \mathrm{P}(X \leq \mu + \sigma) - \mathrm{P}\left(X < \mu - \tfrac{1}{2}\sigma\right) = a - (1 - b) = a + b - 1.$$

For the conditional probability, the event in the numerator is $\{\mu - \frac{1}{2}\sigma \le X \le \mu + \frac{1}{2}\sigma\}$, which has probability $b - (1-b) = 2b - 1$, while $\mathrm{P}(X \ge \mu - \frac{1}{2}\sigma) = 1 - (1-b) = b$. Hence

$$\mathrm{P}\big(X \le \mu + \tfrac{1}{2}\sigma \mid X \ge \mu - \tfrac{1}{2}\sigma\big) = \frac{2b-1}{b}.$$

Part (ii)(a)

Let S be the event that the carton is skimmed and F that it is full-fat, so $\mathrm{P}(S) = 0.6$ and $\mathrm{P}(F) = 0.4$. Let V be the volume in ml.

Ask yourself Where do 500 and 505 ml sit for each type of milk, in units of σ?

For skimmed milk ($\mu = 500$, $\sigma = 10$), 500 is μ and 505 is $\mu + \frac{1}{2}\sigma$. For full-fat milk ($\mu = 495$, $\sigma = 10$), 500 is $\mu + \frac{1}{2}\sigma$ and 505 is $\mu + \sigma$. Reading off the strips:

$$\mathrm{P}(V < 505 \mid S) = b, \qquad \mathrm{P}(500 < V < 505 \mid S) = b - \tfrac{1}{2},$$
$$\mathrm{P}(V < 505 \mid F) = a, \qquad \mathrm{P}(500 < V < 505 \mid F) = a - b.$$

By the law of total probability,

$$\mathrm{P}(V < 505) = 0.6b + 0.4a,$$
$$\mathrm{P}(500 < V < 505) = 0.6\big(b - \tfrac{1}{2}\big) + 0.4(a-b) = 0.4a + 0.2b - 0.3.$$

The event "more than 500 ml and less than 505 ml" is the intersection we want, so

$$\mathrm{P}(V > 500 \mid V < 505) = \frac{0.4a + 0.2b - 0.3}{0.4a + 0.6b} = \frac{4a + 2b - 3}{4a + 6b}.$$

Part (ii)(b)

With a very large number of randomly chosen cartons, the observed proportions estimate the corresponding probabilities. The first piece of information says $\mathrm{P}(V \le 505) = 0.7$, which by part (a)'s calculation is

$$0.6b + 0.4a = 0.7, \quad \text{i.e.} \quad 4a + 6b = 7. \tag{1}$$

Ask yourself What is $\mathrm{P}(V \ge 495)$ for each type of milk?

For skimmed milk, 495 is $\mu - \frac{1}{2}\sigma$, so $\mathrm{P}(V \ge 495 \mid S) = b$ (this was the denominator in part (i)). For full-fat milk, 495 is μ, so $\mathrm{P}(V \ge 495 \mid F) = \frac{1}{2}$ by symmetry. By Bayes' theorem,

$$\mathrm{P}(F \mid V \ge 495) = \frac{0.4 \times \frac{1}{2}}{0.6b + 0.4 \times \frac{1}{2}} = \frac{0.2}{0.6b + 0.2}.$$

Setting this equal to $\frac{1}{3}$ gives $0.6b + 0.2 = 0.6$, so $b = \frac{2}{3}$. Then (1) gives $4a = 7 - 4 = 3$, so

$$a \approx \tfrac{3}{4}, \qquad b \approx \tfrac{2}{3}.$$

These are only estimates, because the observed proportions only approximate the true probabilities.

Checks

- The estimates are sensible: $\frac{1}{2} < b < a < 1$, as they must be, since $\mu < \mu + \frac{1}{2}\sigma < \mu + \sigma$.
- Substituting back: $0.6 \cdot \frac{2}{3} + 0.4 \cdot \frac{3}{4} = 0.4 + 0.3 = 0.7$, and $0.2/(0.4 + 0.2) = \frac{1}{3}$.
- For a Normal distribution $a \approx 0.841$ and $b \approx 0.691$; with these, part (i) gives 0.533 and 0.553. Simulating a (non-Normal) Laplace distribution with 10^5-plus samples matched all the formulae in parts (i) and (ii)(a); for instance, (ii)(a) gave 0.2518 against the formula's 0.2515.

Common slips

- Writing $\mathrm{P}(X \leq \mu - \frac{1}{2}\sigma) = b$ instead of $1 - b$: symmetry reflects the *tail*, not the cumulative probability.
- Forgetting that the two milk types have different means, so the same volume (say 500 ml) is a different number of standard deviations from the mean for each.
- Using $\mathrm{P}(V \geq 495) = b$ for both types; for full-fat milk 495 ml is the mean.

Beyond STEP

The point of the question is that you can get a long way with symmetry and the law of total probability without knowing the distribution at all; the mixture of two populations here is a simple *mixture model*, and part (ii)(b) is a baby version of estimating its parameters from data. A related distribution-free result is Chebyshev's inequality, which bounds $\mathrm{P}(|X - \mu| \geq k\sigma)$ using only the variance. See Blitzstein and Hwang, *Introduction to Probability*, or Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP I 2005, Question 14

Probability and Statistics • Mixed Random Variables

STEP I 2005, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q14

The key idea

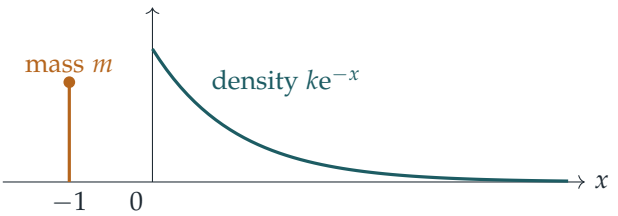
X is a *mixed* random variable: a lump of probability sitting at the single point -1, together with a continuous spread over $[0, \infty)$. The way to handle it is to split every calculation into the two pieces: a discrete term from $X = -1$ plus an integral against the density on $[0, \infty)$. Once we notice that the continuous piece is just a scaled exponential distribution, most of the integrals are ones we already know.

Solution

Differentiating the given cumulative probability, the continuous part of X has density

$$f(x) = \frac{\mathrm{d}}{\mathrm{d}x} k(1 - \mathrm{e}^{-x}) = k\mathrm{e}^{-x}, \qquad x \geq 0.$$

So X looks like this: a point mass m at -1, and a curve $k\mathrm{e}^{-x}$ on $[0, \infty)$.



Part (i)

The total probability is 1. Letting $x \to \infty$, $\mathrm{P}(X \geq 0) = k(1 - 0) = k$, so

$$m + k = 1, \qquad \text{that is,} \qquad k = 1 - m.$$

(Since $m \geq 0$ is a probability and $m < \frac{1}{2}$, we have $k > \frac{1}{2} > 0$, so the density is genuinely positive.)

Part (ii)

Splitting into the two pieces,

$$\begin{aligned}\mathrm{E}(X) &= (-1)\cdot m + \int_0^\infty x\,(1-m)\mathrm{e}^{-x}\,\mathrm{d}x \\ &= -m + (1-m)\cdot 1 = 1-2m,\end{aligned}$$

using $\int_0^\infty x\mathrm{e}^{-x}\,\mathrm{d}x = 1$ (integrate by parts, or recognise the mean of an exponential distribution with parameter 1). This is the required result.

Part (iii)

Variance. In the same way, using $\int_0^\infty x^2\mathrm{e}^{-x}\,\mathrm{d}x = 2$ (integrate by parts twice),

$$\mathrm{E}(X^2) = (-1)^2\cdot m + (1-m)\cdot 2 = 2-m.$$

Hence

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{E}(X^2) - \big(\mathrm{E}(X)\big)^2 = 2-m-(1-2m)^2 \\ &= 1+3m-4m^2 = (1-m)(1+4m).\end{aligned}$$

? **Ask yourself** Could the median be -1, or lie somewhere between -1 and 0?

Median. No: X never takes values in $(-1,0)$, and $\mathrm{P}(X\le -1) = m < \frac{1}{2}$, so the point mass alone is not enough to reach half the probability. This is exactly where the condition $m<\frac{1}{2}$ is used. So the median M satisfies $M\ge 0$ and $\mathrm{P}(X\le M) = \frac{1}{2}$:

$$\begin{aligned}m + (1-m)(1-\mathrm{e}^{-M}) &= \tfrac{1}{2} \\ 1-(1-m)\mathrm{e}^{-M} &= \tfrac{1}{2} \\ \mathrm{e}^{-M} &= \frac{1}{2(1-m)}.\end{aligned}$$

Therefore

$$M = \ln\big(2(1-m)\big) = \ln(2-2m).$$

As a consistency check, $0\le m<\frac{1}{2}$ gives $1<2-2m\le 2$, so $M>0$, as it must be. Because the distribution is continuous on $[0,\infty)$, this M also has $\mathrm{P}(X\ge M)=\frac{1}{2}$, so the median is unambiguous.

Part (iv)

At $X=-1$ we have $|X|^{1/2}=1$, and for $X\ge 0$ we have $|X|^{1/2}=\sqrt{X}$. So

$$\mathrm{E}\big(|X|^{1/2}\big) = 1\cdot m + (1-m)\int_0^\infty \sqrt{x}\,\mathrm{e}^{-x}\,\mathrm{d}x.$$

Ask yourself How do we turn $\int_0^\infty \sqrt{x}\,\mathrm{e}^{-x}\,\mathrm{d}x$ into the integral we were given?

The given integral has e^{-y^2}, which suggests putting $x = y^2$ (with $y \geq 0$), so $\sqrt{x} = y$ and $\mathrm{d}x = 2y\,\mathrm{d}y$; the limits 0 and ∞ are unchanged:

$$\int_0^\infty \sqrt{x}\,\mathrm{e}^{-x}\,\mathrm{d}x = \int_0^\infty y\,\mathrm{e}^{-y^2}\,2y\,\mathrm{d}y = 2\int_0^\infty y^2\mathrm{e}^{-y^2}\,\mathrm{d}y = 2\cdot\tfrac{1}{4}\sqrt{\pi} = \tfrac{1}{2}\sqrt{\pi}.$$

Hence

$$\mathrm{E}\big(|X|^{1/2}\big) = m + \tfrac{1}{2}\sqrt{\pi}\,(1-m).$$

Checks

- With $m = 0$, X is exactly exponential with parameter 1: our formulas give mean 1, variance 1, median $\ln 2$ and $\mathrm{E}(\sqrt{X}) = \frac{1}{2}\sqrt{\pi} = \Gamma(\frac{3}{2})$, all the standard values.
- As $m \to \frac{1}{2}$ the median tends to $\ln 1 = 0$, which makes sense: half the probability is at -1, so the median slides down to the bottom of the continuous part.
- A simulation with $m = 0.3$ and 4×10^5 samples gave mean 0.402 (formula 0.4), variance 1.551 (formula 1.54), median 0.336 (formula $\ln 1.4 \approx 0.336$) and $\mathrm{E}(|X|^{1/2}) \approx 0.922$ (formula 0.920).

Common slips

- Forgetting the point mass in $\mathrm{E}(X^2)$ or $\mathrm{E}(|X|^{1/2})$: it contributes $+m$, not $-m$, because $(-1)^2 = |-1|^{1/2} = 1$.
- Solving $(1-m)(1-\mathrm{e}^{-M}) = \frac{1}{2}$ for the median, which ignores the probability m already accumulated at -1.
- Changing variable without converting $\mathrm{d}x$ to $2y\,\mathrm{d}y$, which gives $\frac{1}{4}\sqrt{\pi}$ instead of $\frac{1}{2}\sqrt{\pi}$.

Beyond STEP

Distributions that mix point masses and densities are common in practice (an insurance claim is zero with some probability, otherwise continuous). At university they are handled uniformly by writing expectations as integrals against the distribution function, $\mathrm{E}(g(X)) = \int g(x)\,\mathrm{d}F(x)$, which covers discrete, continuous and mixed cases at once. The integral in part (iv) is the Gamma function value $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi}$. Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

WORKED SOLUTION

STEP II 2005, Question 1

Pure • Polynomials and Stationary Points

STEP II 2005, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q1

The key idea

Differentiating anything of the form $P(x)\mathrm{e}^{-x^2}$ by the product rule gives $\big(P'(x) - 2xP(x)\big)\mathrm{e}^{-x^2}$, and since e^{-x^2} is never zero, the stationary points are exactly the real roots of the polynomial $P'(x) - 2xP(x)$. So the second part becomes a reverse problem: decide which polynomial we want $P' - 2xP$ to be, then work backwards to P by comparing coefficients.

Solution

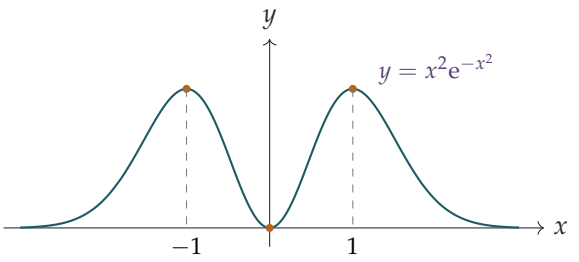
The first part

By the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2\mathrm{e}^{-x^2}\right) = 2x\,\mathrm{e}^{-x^2} + x^2\cdot(-2x)\,\mathrm{e}^{-x^2} = 2x\big(1-x^2\big)\mathrm{e}^{-x^2}.$$

Since $\mathrm{e}^{-x^2} > 0$ for every x, this is zero exactly when $x(1-x^2) = 0$, that is, at

$$x = 0, \qquad x = 1, \qquad x = -1.$$



The graph has a minimum at the origin and maxima of height e^{-1} at $x = \pm 1$.

The second part

For any polynomial P,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(P(x)\mathrm{e}^{-x^2}\right) = \left(P'(x) - 2xP(x)\right)\mathrm{e}^{-x^2},$$

so, again because $\mathrm{e}^{-x^2} > 0$, the derivative vanishes exactly where $P'(x) - 2xP(x) = 0$. The first part fits this pattern: $P(x) = x^2$ gave $P' - 2xP = -2x(x^2 - 1)$. So we aim for

$$P'(x) - 2xP(x) = -2x\left(x^2 - a^2\right)\left(x^2 - b^2\right), \qquad (*)$$

whose only real roots are 0, $\pm a$ and $\pm b$. These are five *distinct* values, because a and b are distinct and positive.

Ask yourself What degree should P have, and why should it contain only even powers?

If P has degree n with leading coefficient c, then P' has degree $n-1$, so $P' - 2xP$ has degree $n+1$ with leading coefficient $-2c$. The right-hand side of $(*)$ has degree 5 and leading coefficient -2, so we need $n = 4$ and $c = 1$. Moreover, the right-hand side is an odd function; if P is even, then P' and $2xP$ are both odd, so it is natural to try

$$P(x) = x^4 + \beta x^2 + \gamma.$$

Then $P'(x) = 4x^3 + 2\beta x$ and

$$\begin{aligned} P'(x) - 2xP(x) &= 4x^3 + 2\beta x - 2x^5 - 2\beta x^3 - 2\gamma x \\ &= -2x^5 + (4 - 2\beta)x^3 + (2\beta - 2\gamma)x, \end{aligned}$$

while the right-hand side of $(*)$ expands to

$$-2x^5 + 2\left(a^2 + b^2\right)x^3 - 2a^2b^2x.$$

Comparing the coefficients of x^3 and of x:

$$\begin{aligned} 4 - 2\beta = 2\left(a^2 + b^2\right) \qquad &\Longrightarrow \qquad \beta = 2 - a^2 - b^2, \\ 2\beta - 2\gamma = -2a^2b^2 \qquad &\Longrightarrow \qquad \gamma = \beta + a^2b^2 = a^2b^2 - a^2 - b^2 + 2. \end{aligned}$$

So one suitable polynomial is

$$\boxed{P(x) = x^4 + \left(2 - a^2 - b^2\right)x^2 + a^2b^2 - a^2 - b^2 + 2.}$$

With this choice $(*)$ holds identically, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(P(x)\mathrm{e}^{-x^2}\right) = -2x\left(x^2 - a^2\right)\left(x^2 - b^2\right)\mathrm{e}^{-x^2},$$

which is zero for $x = 0$, $x = \pm a$ and $x = \pm b$, and, since $\mathrm{e}^{-x^2} \neq 0$, for no other values of x. Any non-zero constant multiple of P works equally well.

Checks

- Take $a = 1$, $b = 2$: then $P(x) = x^4 - 3x^2 + 1$, and $P' - 2xP = 4x^3 - 6x - 2x^5 + 6x^3 - 2x = -2x^5 + 10x^3 - 8x = -2x(x^2 - 1)(x^2 - 4)$, as required.
- Numerically, for $a = 0.7$, $b = 1.9$, the derivative of $P(x)\mathrm{e}^{-x^2}$ changes sign only near $x = 0, \pm 0.7, \pm 1.9$ on $[-5, 5]$; the formula was also confirmed at thousands of random (a, b, x).
- Letting $b \to 0$ heuristically: $P \to x^4 + (2 - a^2)x^2 + 2 - a^2$, and the derivative becomes $-2x^3(x^2 - a^2)\mathrm{e}^{-x^2}$, with the roots at 0 and $\pm b$ merging.

Common slips

- Forgetting the chain-rule factor: the derivative of e^{-x^2} is $-2x\,\mathrm{e}^{-x^2}$, not e^{-x^2}.
- Guessing P of degree 5 to match the five roots: differentiating $P\mathrm{e}^{-x^2}$ *raises* the degree by one, so P has degree 4.
- Not saying why there are no other stationary points: this needs $\mathrm{e}^{-x^2} \neq 0$ and the fact that the quintic has only these roots.

Beyond STEP

The map $P \mapsto P' - 2xP$ is a linear operator on polynomials, and solving $(*)$ is solving a first-order linear differential equation with an integrating factor e^{-x^2}. Functions of the form $H(x)\mathrm{e}^{-x^2}$ with polynomial H lead to the Hermite polynomials, which appear in probability (the normal distribution) and in the quantum harmonic oscillator. Boyce and DiPrima, *Elementary Differential Equations*, covers integrating factors and Hermite's equation.

WORKED SOLUTION

STEP II 2005, Question 2

Pure • Euler's Totient Function

STEP II 2005, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q2

The key idea

Everything follows from writing N in terms of its prime factorisation. Each factor $\left(1-\frac{1}{p}\right)=\frac{p-1}{p}$ has a p in the denominator, and N certainly contains at least one p to cancel it. Once $\mathrm{f}(N)$ is written as a product over prime powers, you can see exactly when $\mathrm{f}(m)\mathrm{f}(n)$ and $\mathrm{f}(mn)$ agree: it depends on whether m and n share a prime factor.

Solution

Part (i)

(a) The primes dividing $12=2^2\cdot 3$ are 2 and 3, so

$$\mathrm{f}(12)=12\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)=12\cdot\frac{1}{2}\cdot\frac{2}{3}=4.$$

The primes dividing $180=2^2\cdot 3^2\cdot 5$ are 2, 3 and 5, so

$$\mathrm{f}(180)=180\cdot\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{4}{5}=48.$$

(b) For $N=1$ there are no prime factors, the product is empty, and $\mathrm{f}(1)=1$. For $N\geq 2$ write

$$N=p_1^{a_1}p_2^{a_2}\cdots p_k^{a_k},$$

where the p_i are the distinct primes dividing N and every $a_i\geq 1$. Then

$$\begin{aligned}\mathrm{f}(N)&=p_1^{a_1}\cdots p_k^{a_k}\cdot\frac{p_1-1}{p_1}\cdots\frac{p_k-1}{p_k}\\&=p_1^{a_1-1}(p_1-1)\,p_2^{a_2-1}(p_2-1)\cdots p_k^{a_k-1}(p_k-1).\end{aligned}$$

Since each $a_i\geq 1$, every exponent a_i-1 is a non-negative integer, so each factor $p_i^{a_i-1}(p_i-1)$ is a positive integer. Hence $\mathrm{f}(N)$ is an integer for all N. We will use this

formula,

$$\mathrm{f}(N) = \prod_{i=1}^{k} p_i^{a_i - 1}(p_i - 1), \qquad (*)$$

for the rest of the question.

Part (ii)

(a) This is false. Take $m = n = 2$. Then $\mathrm{f}(2) = 2 \cdot \frac{1}{2} = 1$, so $\mathrm{f}(2)\mathrm{f}(2) = 1$, but $\mathrm{f}(4) = 4 \cdot \frac{1}{2} = 2$.

The problem is that the factor $\left(1 - \frac{1}{2}\right)$ appears twice on the left but only once on the right, because 2 is counted only once as a prime factor of 4.

(b) This is true. If p and q are distinct primes, then $\mathrm{f}(p) = p\left(1 - \frac{1}{p}\right) = p - 1$, and similarly $\mathrm{f}(q) = q - 1$. The primes dividing pq are exactly p and q, and they are different, so

$$\mathrm{f}(pq) = pq\left(1 - \frac{1}{p}\right)\left(1 - \frac{1}{q}\right) = (p-1)(q-1) = \mathrm{f}(p)\mathrm{f}(q).$$

(c) The claim is that $\mathrm{f}(p)\mathrm{f}(q) = \mathrm{f}(pq)$ can hold *only* when p and q are distinct primes; that is, the equation forces p and q to be distinct primes. This is false. Take $p = 3$ and $q = 4$, neither of which needs to be prime for the equation to make sense. Then

$$\mathrm{f}(3)\mathrm{f}(4) = 2 \cdot 2 = 4 = \mathrm{f}(12),$$

by part (i)(a), but 4 is not prime. (An even simpler example is $p = 1$, since $\mathrm{f}(1)\mathrm{f}(q) = \mathrm{f}(q)$ for every q.)

Ask yourself What really makes the equation hold, if not primality?

What matters is that p and q share no prime factor. If $\gcd(m, n) = 1$, then the prime powers in m and in n are different, so the factorisation of mn is just the two factorisations side by side, and $(*)$ gives $\mathrm{f}(mn) = \mathrm{f}(m)\mathrm{f}(n)$. If some prime r divides both, then the factor $\left(1 - \frac{1}{r}\right)$ appears twice in $\mathrm{f}(m)\mathrm{f}(n)$ but once in $\mathrm{f}(mn)$, so $\mathrm{f}(m)\mathrm{f}(n) < \mathrm{f}(mn)$. So the equation holds exactly when $\gcd(m, n) = 1$, and 3, 4 is just one coprime pair that is not a pair of primes.

If instead you read p and q as standing for primes throughout part (ii), then (c) only asks whether $p = q$ can occur, and it cannot: $\mathrm{f}(p)^2 = (p-1)^2$, while $\mathrm{f}(p^2) = p(p-1)$, and these are never equal because $p - 1 \neq p$. Under that reading (c) is true. The general reading, with the counterexample above, is the more natural one.

Part (iii)

By $(*)$, a prime power has

$$\mathrm{f}(p^m) = p^{m-1}(p-1).$$

Factorising the target gives $146410 = 10 \times 14641 = 2 \cdot 5 \cdot 11^4$, because $14641 = 121^2 = 11^4$. This already has the form $p^{m-1}(p-1)$ with $p = 11$: $11^4 \cdot 10 = 11^4(11-1)$, so $m - 1 = 4$. Hence

$$p = 11, \quad m = 5: \qquad \mathrm{f}(11^5) = 11^4 \cdot 10 = 146410.$$

This is in fact the only answer. If $m \geq 2$, then p divides 146410, so $p \in \{2, 5, 11\}$. Taking $p = 2$ would give $2^{m-1} = 146410$, impossible because 146410 has odd prime factors. Taking $p = 5$ would give $4 \cdot 5^{m-1} = 146410$, impossible because 146410 is not divisible by 4. That leaves $p = 11$, which gives $m = 5$. If $m = 1$ we would need $p = 146411$ to be prime, but $146411 = 41 \times 3571$.

Checks

- The function is Euler's totient: $\mathrm{f}(N)$ counts the integers in $\{1, \ldots, N\}$ coprime to N. For example, $1, 5, 7, 11$ are the four such numbers for $N = 12$. A computer confirmed that the product formula agrees with this count, and is an integer, for every $N < 2000$.
- The worked example in the question gives $\mathrm{f}(80) = 80 \cdot \frac{1}{2} \cdot \frac{4}{5} = 32 = 2^3(2-1) \cdot 5^0(5-1)$, which matches $(*)$.
- The computer tested 20 000 random pairs $m, n \leq 400$ and found $\mathrm{f}(m)\mathrm{f}(n) = \mathrm{f}(mn)$ exactly when $\gcd(m, n) = 1$. A search over all primes $p \leq 146411$ found $(p, m) = (11, 5)$ as the only solution to (iii).

Common slips

- Including a prime more than once in the product: for $180 = 2^2 \cdot 3^2 \cdot 5$ there are three factors, not five.
- Treating "only if" as "if". Part (b) is the "if" direction and part (c) is the converse, which is a different statement.
- Missing that $14641 = 11^4$ in (iii). Spotting $11^2 = 121$ and $121^2 = 14641$, or recognising the row $1, 4, 6, 4, 1$ of Pascal's triangle, finds it quickly.

Beyond STEP

This function is Euler's totient function φ, and the facts in part (ii) say that it is *multiplicative*: $\varphi(mn) = \varphi(m)\varphi(n)$ whenever $\gcd(m, n) = 1$, though not in general. It appears in Euler's theorem, $a^{\varphi(n)} \equiv 1 \pmod{n}$ for a coprime to n, which generalises Fermat's little theorem and underpins RSA encryption. In group theory, $\varphi(n)$ is the order of the group of units modulo n. Davenport's *The Higher Arithmetic* develops all of this beautifully.

WORKED SOLUTION

STEP II 2005, Question 3

Pure • Integration by Parts

STEP II 2005, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q3

The key idea

Differentiate first: the derivative of $\sin x - x\cos x$ collapses to something with an obvious sign, and that gives the sketch and the bounds on y in one go. The two integrals are routine integration by parts, helped by double-angle formulae for the squared terms. The final inequality comes from comparing them: a number between 0 and 1 is at least its own square.

Solution

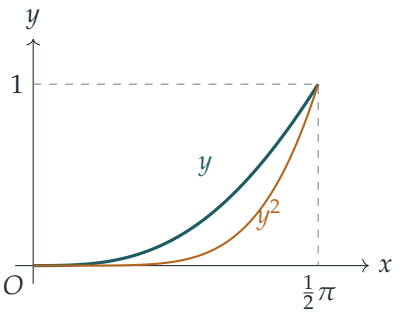
The sketch and the bounds

Differentiating with the product rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos x - (\cos x - x\sin x) = x\sin x.$$

For $0 < x \leq \frac{1}{2}\pi$ both x and $\sin x$ are positive, so y is strictly increasing on $[0, \frac{1}{2}\pi]$. Its least value is therefore $y(0) = 0 - 0 = 0$ and its greatest is $y(\frac{1}{2}\pi) = 1 - \frac{1}{2}\pi \cdot 0 = 1$. Hence $0 \leq y \leq 1$, and in fact $0 < y < 1$ for $0 < x < \frac{1}{2}\pi$.

For the shape, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \sin x + x\cos x > 0$ on $(0, \frac{1}{2}\pi]$, so the curve bends upwards throughout. At the origin the gradient is 0. Using the Maclaurin series, $y = \left(x - \frac{1}{6}x^3\right) - x\left(1 - \frac{1}{2}x^2\right) + \cdots = \frac{1}{3}x^3 + \cdots$, so the curve leaves the origin very flatly, like a cubic. At $x = \frac{1}{2}\pi$ the gradient is $\frac{1}{2}\pi$.



The amber curve y^2 is shown too, since it is the key to the last part.

Part (i)

We need $\int_0^{\pi/2} x\cos x\,\mathrm{d}x$. Integrating by parts with $u = x$, $\mathrm{d}v = \cos x\,\mathrm{d}x$,

$$\int_0^{\pi/2} x\cos x\,\mathrm{d}x = \Big[x\sin x\Big]_0^{\pi/2} - \int_0^{\pi/2} \sin x\,\mathrm{d}x = \tfrac{1}{2}\pi - 1.$$

Since $\int_0^{\pi/2} \sin x\,\mathrm{d}x = 1$,

$$\int_0^{\pi/2} y\,\mathrm{d}x = 1 - \left(\tfrac{1}{2}\pi - 1\right) = 2 - \frac{\pi}{2},$$

as required.

Part (ii)

Expanding,

$$y^2 = \sin^2 x - 2x\sin x\cos x + x^2\cos^2 x = \sin^2 x - x\sin 2x + \tfrac{1}{2}x^2 + \tfrac{1}{2}x^2\cos 2x,$$

using $2\sin x\cos x = \sin 2x$ and $\cos^2 x = \frac{1}{2}(1+\cos 2x)$. We take the four pieces in turn. First, $\sin^2 x = \frac{1}{2}(1-\cos 2x)$, and $\sin 2x$ vanishes at both limits, so

$$\int_0^{\pi/2} \sin^2 x\,\mathrm{d}x = \Big[\tfrac{1}{2}x - \tfrac{1}{4}\sin 2x\Big]_0^{\pi/2} = \frac{\pi}{4}.$$

Next, by parts with $u = x$, $\mathrm{d}v = \sin 2x\,\mathrm{d}x$, and using $\cos\pi = -1$,

$$\int_0^{\pi/2} x\sin 2x\,\mathrm{d}x = \Big[-\tfrac{1}{2}x\cos 2x\Big]_0^{\pi/2} + \tfrac{1}{2}\int_0^{\pi/2} \cos 2x\,\mathrm{d}x = \frac{\pi}{4} + 0 = \frac{\pi}{4}.$$

Third, $\int_0^{\pi/2} \tfrac{1}{2}x^2\,\mathrm{d}x = \tfrac{1}{6}\left(\tfrac{1}{2}\pi\right)^3 = \dfrac{\pi^3}{48}$.

Finally, by parts with $u = x^2$, $\mathrm{d}v = \cos 2x\,\mathrm{d}x$; the boundary term $\left[\tfrac{1}{2}x^2\sin 2x\right]_0^{\pi/2}$ is zero, and the integral left over is the one we have just found:

$$\int_0^{\pi/2} x^2\cos 2x\,\mathrm{d}x = 0 - \int_0^{\pi/2} x\sin 2x\,\mathrm{d}x = -\frac{\pi}{4},$$

so the last piece contributes $\frac{1}{2}\cdot\left(-\frac{1}{4}\pi\right) = -\frac{1}{8}\pi$.

Adding the four pieces,

$$\int_0^{\pi/2} y^2\,\mathrm{d}x = \frac{\pi}{4} - \frac{\pi}{4} + \frac{\pi^3}{48} - \frac{\pi}{8} = \frac{\pi^3}{48} - \frac{\pi}{8},$$

as required.

The deduction

? **Ask yourself** How can two integrals give an inequality about π?

Compare the integrands. Since $0 \le y \le 1$ we have $y - y^2 = y(1-y) \ge 0$ on the whole interval. Better, for $0 < x < \frac{1}{2}\pi$ we showed $0 < y < 1$, so $y(1-y) > 0$ there. A continuous function that is non-negative, and strictly positive on an open interval, has a strictly positive integral. Therefore

$$\int_0^{\pi/2} y^2\,\mathrm{d}x < \int_0^{\pi/2} y\,\mathrm{d}x, \qquad \text{that is,} \qquad \frac{\pi^3}{48} - \frac{\pi}{8} < 2 - \frac{\pi}{2}.$$

Multiplying by 48 (positive, so the inequality is preserved),

$$\pi^3 - 6\pi < 96 - 24\pi, \qquad \text{so} \qquad \pi^3 + 18\pi < 96.$$

This is the required result.

Checks

- Numerically, $2 - \frac{1}{2}\pi \approx 0.4292$ and $\frac{1}{48}\pi^3 - \frac{1}{8}\pi \approx 0.2533$; Simpson's rule on the original integrands gives the same values to twelve decimal places. Both lie between 0 and $\frac{1}{2}\pi \approx 1.571$, as they must since $0 \le y \le 1$.
- The second integral is smaller than the first, as the picture of y^2 under y suggests.
- With $\pi \approx 3.1416$, $\pi^3 + 18\pi \approx 31.006 + 56.549 = 87.55$, comfortably below 96.

Common slips

- Getting the sign of the boundary term wrong in $\int x\sin 2x\,\mathrm{d}x$: at $x = \frac{1}{2}\pi$, $\cos 2x = \cos\pi = -1$, so $-\frac{1}{2}x\cos 2x$ is $+\frac{1}{4}\pi$.
- Only claiming $\le$ in the final step. To get the strict inequality you need $y(1-y) > 0$ somewhere, not just ≥ 0.
- Forgetting to say why $y \le 1$: knowing $y(\frac{1}{2}\pi) = 1$ is not enough without the fact that y is increasing.

Beyond STEP

The final step is a small instance of a big idea: inequalities between functions integrate to inequalities between numbers, and a clever choice of function can trap a constant like π. The same strategy gives the famous proof that $\frac{22}{7} > \pi$ from $\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x > 0$, and, more seriously, Niven's proof that π is irrational. In a first analysis course you will prove carefully that a continuous, non-negative function which is positive somewhere has a positive integral. Abbott's *Understanding Analysis* is a friendly place to see this done properly.

WORKED SOLUTION

STEP II 2005, Question 4

Pure • Inverse Tangent Identities

STEP II 2005, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q4

The key idea

Sums of inverse tangents are handled by taking the tangent of both sides and using $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$. The condition $bc = a^2 + 1$ is exactly what makes the denominator factorise against the numerator. After that the question is about structure: the first result is a rule for *splitting* one inverse tangent into two, and the four-term identity is that rule applied twice, once at each level of a tree. For the numerical part, work backwards to find which splitting produced the four numbers.

Solution

The first identity

Let $A = \arctan\frac{1}{a+b}$ and $B = \arctan\frac{1}{a+c}$. Since $a, b, c > 0$, both $\frac{1}{a+b}$ and $\frac{1}{a+c}$ are positive, so $A, B \in (0, \frac{\pi}{2})$ and $A + B \in (0, \pi)$. Using $bc = a^2 + 1$,

$$\begin{aligned}(a+b)(a+c) - 1 &= a^2 + ab + ac + bc - 1 \\ &= 2a^2 + ab + ac = a(2a + b + c).\end{aligned}$$

Hence, by the addition formula,

$$\begin{aligned}\tan(A+B) &= \frac{\dfrac{1}{a+b} + \dfrac{1}{a+c}}{1 - \dfrac{1}{(a+b)(a+c)}} = \frac{(a+c)+(a+b)}{(a+b)(a+c)-1} \\ &= \frac{2a+b+c}{a(2a+b+c)} = \frac{1}{a},\end{aligned}$$

where we may cancel because $2a + b + c > 0$.

Ask yourself Does $\tan(A+B) = \frac{1}{a}$ really give $A + B = \arctan\frac{1}{a}$?

Not automatically: tan has period π, so we need $A+B$ to lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$, the range of arctan. We know $A+B \in (0,\pi)$. Also $(a+b)(a+c)-1 = a(2a+b+c) > 0$, so $\tan A \tan B < 1$ and the denominator in the addition formula is positive; together with a positive numerator this gives $\tan(A+B) > 0$, which rules out $A+B \in [\frac{\pi}{2}, \pi)$. (Indeed $\tan(A+B)$ is undefined at $\frac{\pi}{2}$ and negative on $(\frac{\pi}{2}, \pi)$.) So $A+B \in (0, \frac{\pi}{2})$, and since $\arctan \frac{1}{a}$ is the unique angle in that interval with tangent $\frac{1}{a}$,

$$\arctan \frac{1}{a+b} + \arctan \frac{1}{a+c} = \arctan \frac{1}{a}. \qquad \blacksquare$$

The four-term identity

Apply the first identity three times; each time all three numbers involved are positive.

- With $a = p+q$, $b = s$, $c = t$: the condition is $st = (p+q)^2 + 1$, which is given, so

$$\arctan \frac{1}{p+q+s} + \arctan \frac{1}{p+q+t} = \arctan \frac{1}{p+q}.$$

- With $a = p+r$, $b = u$, $c = v$: the condition $uv = (p+r)^2 + 1$ is given, so

$$\arctan \frac{1}{p+r+u} + \arctan \frac{1}{p+r+v} = \arctan \frac{1}{p+r}.$$

- With $a = p$, $b = q$, $c = r$: the condition $qr = p^2 + 1$ is given, so

$$\arctan \frac{1}{p+q} + \arctan \frac{1}{p+r} = \arctan \frac{1}{p}.$$

Adding the first two lines and using the third, the sum of the four inverse tangents is $\arctan \frac{1}{p}$, as required. $\blacksquare$

The numerical identity

We need $p = 7$, so $qr = 50$. Rather than guess q and r, work back from the four numbers. If 13 and 21 are the pair $a+b$, $a+c$ from one use of the first identity, then $b = 13-a$, $c = 21-a$ and $bc = a^2+1$ gives

$$(13-a)(21-a) = a^2+1 \iff 273 - 34a = 1 \iff a = 8.$$

Similarly, if 82 and 187 are a pair, then

$$(82-a)(187-a) = a^2+1 \iff 15334 - 269a = 1 \iff a = 57.$$

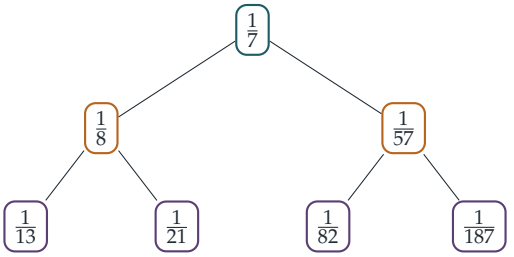
So try $p+q = 8$ and $p+r = 57$, that is $q = 1$, $r = 50$; and then $s = 13-8 = 5$, $t = 21-8 = 13$, $u = 82-57 = 25$, $v = 187-57 = 130$. All seven numbers are positive, and the three conditions hold:

$$qr = 50 = 7^2+1, \qquad st = 65 = 8^2+1, \qquad uv = 3250 = 57^2+1.$$

With these values $p+q+s=13$, $p+q+t=21$, $p+r+u=82$ and $p+r+v=187$, so the four-term identity gives

$$\arctan\frac{1}{13}+\arctan\frac{1}{21}+\arctan\frac{1}{82}+\arctan\frac{1}{187}=\arctan\frac{1}{7}. \qquad \blacksquare$$

The tree below shows how $\arctan\frac{1}{7}$ splits.



Each node's inverse tangent is the sum of its children's: $1\cdot 50=7^2+1$, $5\cdot 13=8^2+1$, $25\cdot 130=57^2+1$.

Checks

- Numerically, both sides of the final identity are $0.1418970546\ldots$.
- Exactly, with fractions: $\tan(A+B)$ for $\frac{1}{13},\frac{1}{21}$ is $\frac{34}{272}=\frac{1}{8}$; for $\frac{1}{82},\frac{1}{187}$ it is $\frac{269}{15333}=\frac{1}{57}$; and for $\frac{1}{8},\frac{1}{57}$ it is $\frac{65}{455}=\frac{1}{7}$.
- A special case of the first identity: $a=1$, $b=1$, $c=2$ gives the famous $\arctan\frac{1}{2}+\arctan\frac{1}{3}=\arctan 1=\frac{\pi}{4}$.

Common slips

- Concluding $A+B=\arctan\frac{1}{a}$ from $\tan(A+B)=\frac{1}{a}$ without checking that $A+B$ lies in $(-\frac{\pi}{2},\frac{\pi}{2})$.
- Searching blindly through factorisations of 50, 65 and 3250; working backwards from the pair 13, 21 finds $a=8$ in one line.

Beyond STEP

Identities like these are behind Machin-type formulae such as $\frac{\pi}{4}=4\arctan\frac{1}{5}-\arctan\frac{1}{239}$, which were used for centuries to compute π by hand, because $\arctan x$ has a rapidly converging series when x is small. The cleanest way to see them is through complex numbers: $\arctan\frac{1}{n}$ is the argument of $n+\mathrm{i}$, so adding inverse tangents corresponds to multiplying Gaussian integers, and the condition $bc=a^2+1$ is a factorisation of $a^2+1=(a+\mathrm{i})(a-\mathrm{i})$ in disguise. Needham's *Visual Complex Analysis*

develops this geometric view of multiplication, and Davenport's *The Higher Arithmetic* covers sums of two squares.

WORKED SOLUTION

STEP II 2005, Question 5

Pure • Incircle and Circumcircle

STEP II 2005, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q5

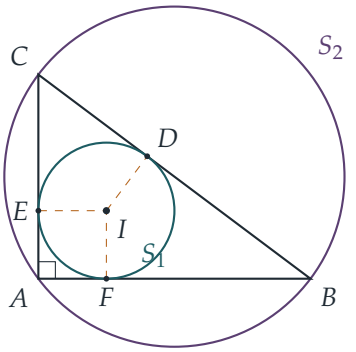
The key idea

The two tangents to a circle from an external point have equal length. Applied at each vertex, this splits every side of the triangle into two tangent lengths, and at the right angle the two tangent lengths are exactly r. That gives the formula for r almost at once. For the area ratio, the right angle tells us the hypotenuse is a diameter of S_2, and Pythagoras lets us write everything in terms of $b+c$ and a alone, which is why q appears.

Solution

The inradius

Let S_1 have centre I and touch BC, CA, AB at D, E, F respectively.



A radius meets a tangent at right angles, so $IE \perp AC$ and $IF \perp AB$. Together with the right angle at A, this makes $AEIF$ a rectangle with two adjacent sides $IE = IF = r$, so it is a square and

$$AE = AF = r.$$

Tangents from an external point are equal, so $BF = BD$ and $CE = CD$. Hence

$$c = AF + FB = r + BD, \qquad b = AE + EC = r + CD, \qquad a = BD + DC.$$

Adding the first two and subtracting the third,

$$b + c - a = 2r + BD + CD - (BD + DC) = 2r,$$

as required.

The area ratio

Since $\angle BAC = 90°$, the angle in a semicircle theorem (in reverse) says BC is a diameter of the circumcircle S_2. So S_2 has radius $\frac{1}{2}a$ and area $\frac{1}{4}\pi a^2$.

The region between S_1 and the triangle is the triangle with the disc of S_1 removed, so its area is $\frac{1}{2}bc - \pi r^2$. Therefore

$$R = \frac{\frac{1}{2}bc - \pi r^2}{\frac{1}{4}\pi a^2}, \qquad \text{so} \qquad \pi R = \frac{2bc - 4\pi r^2}{a^2} = \frac{2bc}{a^2} - \pi\,\frac{(b + c - a)^2}{a^2},$$

using $2r = b + c - a$.

? **Ask yourself** How do we turn $2bc/a^2$ into something involving only q?

Pythagoras gives $b^2 + c^2 = a^2$, so $2bc = (b + c)^2 - (b^2 + c^2) = (b + c)^2 - a^2$. Dividing by a^2,

$$\frac{2bc}{a^2} = q^2 - 1, \qquad \frac{(b + c - a)^2}{a^2} = \left(\frac{b + c}{a} - 1\right)^2 = (q - 1)^2.$$

Hence

$$\begin{aligned} \pi R &= q^2 - 1 - \pi(q^2 - 2q + 1) \\ &= -(\pi - 1)q^2 + 2\pi q - (\pi + 1), \end{aligned}$$

which is the required result.

The bound on R

The right-hand side is a quadratic in q with negative leading coefficient $-(\pi - 1)$, so we complete the square:

$$\begin{aligned} \pi R &= -(\pi - 1)\left(q^2 - \frac{2\pi}{\pi - 1}\,q\right) - (\pi + 1) \\ &= -(\pi - 1)\left(q - \frac{\pi}{\pi - 1}\right)^2 + \frac{\pi^2}{\pi - 1} - (\pi + 1) \\ &= -(\pi - 1)\left(q - \frac{\pi}{\pi - 1}\right)^2 + \frac{\pi^2 - (\pi^2 - 1)}{\pi - 1} \\ &= \frac{1}{\pi - 1} - (\pi - 1)\left(q - \frac{\pi}{\pi - 1}\right)^2. \end{aligned}$$

Since $\pi - 1 > 0$, the squared term is subtracted, so $\pi R \le \dfrac{1}{\pi - 1}$ for every real q, and dividing by $\pi > 0$,

$$R \le \frac{1}{\pi(\pi - 1)}.$$

This is what we were asked to deduce.

It is worth noticing that the bound is not actually attained. Equality would need $q = \pi/(\pi - 1) \approx 1.467$, but for a right-angled triangle $(b + c)^2 = a^2 + 2bc \le a^2 + (b^2 + c^2) = 2a^2$ (by AM–GM, $2bc \le b^2 + c^2$), so $q \le \sqrt{2} \approx 1.414$. The largest possible R comes from the isosceles triangle, $q = \sqrt{2}$, giving $R = (2\sqrt{2}\,\pi - 3\pi + 1)/\pi \approx 0.1467$, just below the bound $1/(\pi(\pi - 1)) \approx 0.1486$.

Checks

For the 3–4–5 triangle ($b = 4$, $c = 3$, $a = 5$), the formula gives $r = 1$, and indeed the area 6 divided by the semi-perimeter 6 is 1, the standard formula $r = \text{area}/s$. Then $R = (6 - \pi)/(\frac{25}{4}\pi) \approx 0.14558$, and $q = 1.4$ in the quadratic gives the same value. A random test of 10^5 right-angled triangles confirmed the identity for πR every time and found no R above 0.14674, consistent with the bound 0.14863.

Common slips

- Forgetting that the radius of S_2 is $\frac{1}{2}a$, not a: the hypotenuse is the *diameter*.
- Trying to expand $2bc/a^2$ directly instead of using $b^2 + c^2 = a^2$ to write $2bc = (b + c)^2 - a^2$.
- Claiming the bound is attained without checking that $q = \pi/(\pi - 1)$ is achievable; it is not, though the question only asks for the inequality.

Beyond STEP

The incircle and circumcircle of a general triangle are tied together by Euler's formula $d^2 = R_2(R_2 - 2r)$ for the distance d between their centres, where R_2 is the circumradius; it implies $R_2 \ge 2r$, with equality only for the equilateral triangle. Results like this, and the idea of optimising a ratio over a family of shapes, lead into the study of geometric inequalities. Coxeter and Greitzer's *Geometry Revisited* is a beautiful next step, and Siklos's *Advanced Problems in Mathematics* (free, CC BY) has more STEP-style practice.

WORKED SOLUTION

STEP II 2005, Question 6

Pure • Binomial Series Sums

STEP II 2005, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q6

The key idea

Every sum here is a power series evaluated at a particular x, so the job is to recognise which function it comes from. In part (i) the trick is to rewrite n and n^2 as combinations of 1, $n+1$ and $(n+1)(n+2)$, which are exactly the coefficients we have just written down. In part (ii) the same idea works once we notice how the coefficients of $(1-x)^{-1/2}$ and $(1-x)^{-3/2}$ are related.

Solution

Part (i)

By the binomial theorem for negative integer powers (valid for $|x|<1$), the coefficients of x^n, for $n \geq 0$, are

$$(1-x)^{-1}: \quad 1, \qquad (1-x)^{-2}: \quad n+1, \qquad (1-x)^{-3}: \quad \tfrac{1}{2}(n+1)(n+2).$$

For instance, the coefficient in $(1-x)^{-3}$ is $\dfrac{(-3)(-4)\cdots(-3-n+1)}{n!}(-1)^n = \dfrac{3\cdot 4\cdots(n+2)}{n!} = \dfrac{(n+1)(n+2)}{2}$.

The first sum. Since $n = (n+1)-1$, for $|x|<1$

$$\sum_{n=0}^{\infty} nx^n = \sum_{n=0}^{\infty}(n+1)x^n - \sum_{n=0}^{\infty} x^n = \frac{1}{(1-x)^2} - \frac{1}{1-x}.$$

(Splitting the sum is fine because both series on the right converge.) The $n=0$ term is zero, so putting $x = \frac{1}{2}$,

$$\sum_{n=1}^{\infty} n\,2^{-n} = 4 - 2 = 2.$$

The second sum. We want n^2 as a combination of $(n+1)(n+2)$, $n+1$ and 1. Since

$(n+1)(n+2) = n^2 + 3n + 2$,

$$n^2 = (n+1)(n+2) - 3(n+1) + 1.$$

Hence, for $|x| < 1$,

$$\sum_{n=0}^{\infty} n^2 x^n = \frac{2}{(1-x)^3} - \frac{3}{(1-x)^2} + \frac{1}{1-x},$$

where the factor 2 appears because the coefficient in $(1-x)^{-3}$ is *half* of $(n+1)(n+2)$. Again the $n = 0$ term vanishes, and at $x = \frac{1}{2}$

$$\sum_{n=1}^{\infty} n^2 2^{-n} = 2 \cdot 8 - 3 \cdot 4 + 2 = 6.$$

Part (ii)

By the binomial theorem with index $-\frac{1}{2}$, the coefficient of x^n in $(1-x)^{-1/2}$ is

$$\frac{(-\frac{1}{2})(-\frac{3}{2}) \cdots (-\frac{2n-1}{2})}{n!} (-1)^n = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^n \, n!}.$$

Ask yourself How do we turn a product of odd numbers into factorials?

Fill in the missing even numbers: $(2n)! = \big(1 \cdot 3 \cdots (2n-1)\big)\big(2 \cdot 4 \cdots 2n\big)$, and $2 \cdot 4 \cdots 2n = 2^n n!$. So $1 \cdot 3 \cdots (2n-1) = \dfrac{(2n)!}{2^n n!}$, and the coefficient is

$$a_n = \frac{(2n)!}{2^n n! \cdot 2^n n!} = \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}}.$$

This holds for $n = 0$ too (empty product, $a_0 = 1$), so for $|x| < 1$

$$(1-x)^{-1/2} = \sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} \frac{x^n}{2^{2n}},$$

as required.

The first sum. The sum asked for is $\sum a_n x^n$ at $x = \frac{1}{3}$, which is inside the interval of validity. So it equals

$$\left(1 - \tfrac{1}{3}\right)^{-1/2} = \sqrt{\tfrac{3}{2}} = \frac{\sqrt{6}}{2}.$$

The second sum. Now we need $\sum n a_n x^n$. In the spirit of part (i), relate na_n to a series we know. Exactly as above, the coefficient of x^m in $(1-x)^{-3/2}$ is

$$b_m = \frac{3 \cdot 5 \cdots (2m+1)}{2^m \, m!} = (2m+1)\, a_m.$$

From the formula for a_n, the ratio of consecutive coefficients is $a_n/a_{n-1} = \dfrac{(2n)(2n-1)}{n^2 \cdot 4} = \dfrac{2n-1}{2n}$, so for $n \geq 1$

$$n\,a_n = \tfrac{1}{2}(2n-1)\,a_{n-1} = \tfrac{1}{2}\,b_{n-1}.$$

Therefore, for $|x| < 1$,

$$\sum_{n=1}^{\infty} n a_n x^n = \frac{x}{2}\sum_{n=1}^{\infty} b_{n-1}x^{n-1} = \frac{x}{2}(1-x)^{-3/2}.$$

(Equivalently, differentiate the series for $(1-x)^{-1/2}$ term by term, which is allowed inside its interval of convergence, and multiply by x.) At $x = \frac{1}{3}$:

$$\sum_{n=1}^{\infty} \frac{n\,(2n)!}{(n!)^2\,2^{2n}\,3^n} = \frac{1}{6}\left(\frac{3}{2}\right)^{3/2} = \frac{1}{6}\cdot\frac{3}{2}\sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{8}.$$

Checks

- The first few terms of $\sum n2^{-n}$ are $\frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{1}{4} + \cdots$, already 1.625 after four terms and creeping towards 2; partial sums of $\sum n^2 2^{-n}$ to 200 terms agree with 6 to many decimal places.
- Numerically, $\sqrt{6}/2 \approx 1.224745$ and $\sqrt{6}/8 \approx 0.306186$; the partial sums of the two series in part (ii) match both to 12 decimal places.
- A quick sanity check on the general term: $n = 1$ gives $a_1 = \frac{2}{4} = \frac{1}{2}$, the familiar coefficient in $(1-x)^{-1/2} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \cdots$.

Common slips

- Forgetting the $\frac{1}{2}$ in the coefficient of $(1-x)^{-3}$, which gives $8 - 12 + 2 = -2$ for $\sum n^2 2^{-n}$, impossible for a sum of positive terms.
- Starting the sum for $(1-x)^{-2}$ with coefficient n rather than $n+1$: the series is $1 + 2x + 3x^2 + \cdots$.
- In the last sum, losing a factor of x: $\sum n a_n x^{n-1}$ is the derivative, but the sum asked for is $\sum n a_n x^n$.

Beyond STEP

The manoeuvres here (splitting a series into pieces, differentiating term by term) are legitimate because a power series converges absolutely inside its radius of convergence, and can be differentiated there term by term. Proving that carefully is a standard part of a first course in analysis. The coefficients $\binom{2n}{n}/4^n$ also turn up in probability, as the chance that a simple random walk is back at its start after $2n$ steps. For the analysis, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 2005, Question 7

Pure • Vectors and Circular Motion

STEP II 2005, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q7

The key idea

Each position vector has the form $\cos(\text{angle})\,\mathbf{a} + \sin(\text{angle})\,\mathbf{b}$, and when $\mathbf{a}$ and $\mathbf{b}$ are perpendicular and of equal length this traces out a circle in the plane of $\mathbf{a}$ and $\mathbf{b}$. Once we know $|OP|$ and $|OQ|$ are constant, the angle POQ comes straight from the scalar product, and product-to-sum formulae tidy it into a single cosine. The last part is then an inequality for one cosine over two full periods.

Solution

Part (i)

Write $\mathbf{p}$ and $\mathbf{q}$ for the position vectors of P and Q.

For P, $\mathbf{p} = \cos t\,\mathbf{i} + \sin t\,\mathbf{j}$, so $|\mathbf{p}| = 1$ and $\mathbf{p}$ lies in the xy-plane. So P moves round the circle of radius 1, centre O, in the plane $z = 0$, anticlockwise when viewed from above (from the positive z-axis), with constant angular speed 1 (so speed 1). It starts at $(1, 0, 0)$ and makes exactly one revolution as t runs from 0 to 2π.

For Q, put $s = t + \frac{1}{4}\pi$ and

$$\mathbf{a} = \tfrac{3}{2}\mathbf{i} + \tfrac{3\sqrt{3}}{2}\mathbf{k}, \qquad \mathbf{b} = 3\mathbf{j}, \qquad \text{so that} \qquad \mathbf{q} = \cos s\,\mathbf{a} + \sin s\,\mathbf{b}.$$

Now $|\mathbf{a}|^2 = \frac{9}{4} + \frac{27}{4} = 9$, $|\mathbf{b}|^2 = 9$ and $\mathbf{a} \cdot \mathbf{b} = 0$. Hence

$$|\mathbf{q}|^2 = 9\cos^2 s + 9\sin^2 s + 2\cos s \sin s\,(\mathbf{a} \cdot \mathbf{b}) = 9,$$

and $\mathbf{q}$ always lies in the plane through O spanned by $\mathbf{a}$ and $\mathbf{j}$. That plane contains the y-axis and the direction $\mathbf{i} + \sqrt{3}\,\mathbf{k}$, so its equation is $z = \sqrt{3}\,x$; its normal $-\sqrt{3}\,\mathbf{i} + \mathbf{k}$ makes an angle $\arccos\frac{1}{2} = \frac{1}{3}\pi$ with $\mathbf{k}$.

So Q moves round a circle of radius 3, centre O, in the plane $z = \sqrt{3}\,x$, which contains the y-axis and is inclined at 60° to the plane of P's circle. It too has angular speed 1 (speed 3) and makes one revolution, rotating from the direction of $\mathbf{a}$ towards $\mathbf{j}$, and its phase is always $\frac{1}{4}\pi$ ahead of P's: at $t = 0$ it is at angle $\frac{1}{4}\pi$ from $\mathbf{a}$, at $\frac{3}{2\sqrt{2}}(1,\, 2,\, \sqrt{3})$.

Part (ii)

Since $|\mathbf{p}| = 1$ and $|\mathbf{q}| = 3$,

$$\cos\theta = \frac{\mathbf{p}\cdot\mathbf{q}}{3} = \frac{\frac{3}{2}\cos t\cos s + 3\sin t\sin s}{3} = \tfrac{1}{2}\cos t\cos s + \sin t\sin s,$$

where $s = t + \frac{1}{4}\pi$. The product-to-sum formulae, with $t + s = 2t + \frac{1}{4}\pi$ and $s - t = \frac{1}{4}\pi$, give

$$\cos t\cos s = \tfrac{1}{2}\big[\cos(2t + \tfrac{1}{4}\pi) + \cos\tfrac{1}{4}\pi\big],$$
$$\sin t\sin s = \tfrac{1}{2}\big[\cos\tfrac{1}{4}\pi - \cos(2t + \tfrac{1}{4}\pi)\big].$$

Writing $C = \cos(2t + \frac{1}{4}\pi)$ and using $\cos\frac{1}{4}\pi = \frac{\sqrt{2}}{2}$,

$$\begin{aligned}\cos\theta &= \tfrac{1}{4}\big(C + \tfrac{\sqrt{2}}{2}\big) + \tfrac{1}{2}\big(\tfrac{\sqrt{2}}{2} - C\big)\\ &= \tfrac{\sqrt{2}}{8} + \tfrac{\sqrt{2}}{4} - \tfrac{1}{4}C = \tfrac{3\sqrt{2}}{8} - \tfrac{1}{4}\cos(2t + \tfrac{1}{4}\pi),\end{aligned}$$

as required.

Part (iii)

? **Ask yourself** How do we turn $\theta \geq \frac{1}{4}\pi$ into a condition on $\cos\theta$?

Cosine is strictly decreasing on $[0, \pi]$, and both θ and $\frac{1}{4}\pi$ lie in that interval, so

$$\theta \geq \tfrac{1}{4}\pi \iff \cos\theta \leq \cos\tfrac{1}{4}\pi = \tfrac{\sqrt{2}}{2} = \tfrac{4\sqrt{2}}{8}.$$

By part (ii) this is

$$\tfrac{3\sqrt{2}}{8} - \tfrac{1}{4}\cos(2t + \tfrac{1}{4}\pi) \leq \tfrac{4\sqrt{2}}{8} \iff \cos(2t + \tfrac{1}{4}\pi) \geq -\tfrac{\sqrt{2}}{2}.$$

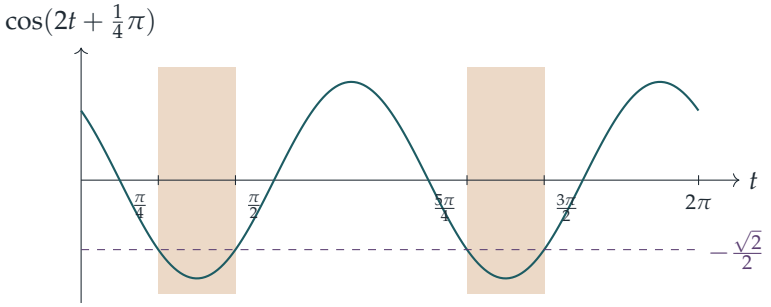
Let $u = 2t + \frac{1}{4}\pi$. As t runs over $[0, 2\pi]$, u runs over $[\frac{1}{4}\pi, \frac{17}{4}\pi]$, an interval of length 4π: exactly two periods of $\cos u$. In any period, $\cos u < -\frac{\sqrt{2}}{2}$ exactly on an open interval of length $\frac{1}{2}\pi$ (between $\frac{3}{4}\pi$ and $\frac{5}{4}\pi$, modulo 2π). Here the failing values are

$$u \in (\tfrac{3}{4}\pi, \tfrac{5}{4}\pi) \cup (\tfrac{11}{4}\pi, \tfrac{13}{4}\pi) \iff t \in (\tfrac{1}{4}\pi, \tfrac{1}{2}\pi) \cup (\tfrac{5}{4}\pi, \tfrac{3}{2}\pi),$$

each of length $\frac{1}{4}\pi$ in t. So $\theta < \frac{1}{4}\pi$ for a total time of $\frac{1}{2}\pi$, and the total time for which $\theta \geq \frac{1}{4}\pi$ is

$$2\pi - \tfrac{1}{2}\pi = \tfrac{3}{2}\pi,$$

as required.



The shaded strips are where the curve dips below $-\frac{\sqrt{2}}{2}$, that is, where $\theta < \frac{1}{4}\pi$.

Checks

- At $t = 0$: $\mathbf{p} = \mathbf{i}$ and $\mathbf{q} = \frac{3}{2\sqrt{2}}(\mathbf{i} + 2\mathbf{j} + \sqrt{3}\,\mathbf{k})$, so directly $\cos\theta = \frac{3}{2\sqrt{2}} \cdot \frac{1}{3} = \frac{\sqrt{2}}{4}$. The formula gives $\frac{3\sqrt{2}}{8} - \frac{1}{4} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}}{4}$. They agree.
- $\cos\theta$ lies between $\frac{3\sqrt{2}}{8} - \frac{1}{4} \approx 0.280$ and $\frac{3\sqrt{2}}{8} + \frac{1}{4} \approx 0.780$, so θ is never 0. That makes sense: $\mathbf{p}$ and $\mathbf{q}$ could only be parallel along the y-axis, where the two planes meet, but P is on that axis at $t = \frac{1}{2}\pi, \frac{3}{2}\pi$ and Q at $t = \frac{1}{4}\pi, \frac{5}{4}\pi$.
- Sampling t at two million points in $[0, 2\pi]$ and computing θ directly from the vectors gives a total time of 4.712389 with $\theta \geq \frac{1}{4}\pi$, matching $\frac{3}{2}\pi \approx 4.712389$.

Common slips

- Forgetting to divide by $|\mathbf{p}||\mathbf{q}| = 3$, or assuming $|\mathbf{q}|$ varies because the coefficients of $\mathbf{i}$ and $\mathbf{j}$ differ.
- Reversing the inequality: $\theta \geq \frac{1}{4}\pi$ means $\cos\theta \leq \frac{\sqrt{2}}{2}$, because cosine is decreasing on $[0, \pi]$.
- Counting only one period of $\cos(2t + \frac{1}{4}\pi)$: the $2t$ means it goes round twice while t goes round once.

Beyond STEP

Any circle centred at the origin can be written as $\cos s\,\mathbf{a} + \sin s\,\mathbf{b}$ with $\mathbf{a} \perp \mathbf{b}$ and $|\mathbf{a}| = |\mathbf{b}|$; if we drop those conditions we get an ellipse, and in general the image of the unit circle under a linear map. Understanding curves this way, and measuring angles between moving vectors with the scalar product, is the start of the linear algebra and vector calculus met in the first year. Strang's *Introduction to Linear Algebra* develops these ideas clearly.

WORKED SOLUTION

STEP II 2005, Question 8

Pure • Separable Differential Equations

STEP II 2005, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q8

The key idea

The equation is separable: all the y-dependence is a single factor y^2, so dividing by it gives $\frac{1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x}$, the derivative of $-1/y$. The x-integral yields to the substitution $u = 1 + x^2$, because the odd power x^3 supplies the $x\,\mathrm{d}x$ that u needs. For the second curve, notice that the right-hand side is z^3 rather than y^2: the quantity to look at is then $1/z^2$, and that turns out to obey exactly the same equation as $1/y$.

Solution

Solving for y

On C we have $y(0) = 1 > 0$, and the right-hand side of the equation is never negative for $x \geq 0$, so y never decreases. Hence $y \geq 1$ throughout, and we may divide by y^2:

$$\frac{1}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x^3}{(1+x^2)^{5/2}} \quad\Longrightarrow\quad -\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{y}\right) = \frac{x^3}{(1+x^2)^{5/2}}.$$

For the integral put $u = 1 + x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$ and $x^3\,\mathrm{d}x = x^2 \cdot x\,\mathrm{d}x = \frac{1}{2}(u-1)\,\mathrm{d}u$:

$$\begin{aligned}\int \frac{x^3}{(1+x^2)^{5/2}}\,\mathrm{d}x &= \frac{1}{2}\int \left(u^{-3/2} - u^{-5/2}\right)\mathrm{d}u = -u^{-1/2} + \tfrac{1}{3}u^{-3/2} + \text{const}\\ &= -\frac{3u-1}{3u^{3/2}} + \text{const} = -\frac{2+3x^2}{3(1+x^2)^{3/2}} + \text{const}.\end{aligned}$$

Therefore

$$\frac{1}{y} = \frac{2+3x^2}{3(1+x^2)^{3/2}} + k$$

for some constant k. At $x = 0$ this reads $1 = \frac{2}{3} + k$, so $k = \frac{1}{3}$, which is the required result:

$$\frac{1}{y} = \frac{2+3x^2}{3(1+x^2)^{3/2}} + \frac{1}{3}.$$

Large x

Take out the dominant powers, $3x^2$ from the numerator and x^3 from $(1+x^2)^{3/2}$:

$$\frac{2+3x^2}{3(1+x^2)^{3/2}} = \frac{3x^2\left(1+\frac{2}{3x^2}\right)}{3x^3\left(1+\frac{1}{x^2}\right)^{3/2}} = \frac{1}{x}\left(1+\frac{2}{3x^2}\right)\left(1-\frac{3}{2x^2}+\cdots\right) = \frac{1}{x}+O\left(\frac{1}{x^3}\right),$$

using the binomial expansion, which is valid since $1/x^2 < 1$ for large x. So

$$\frac{1}{y} = \frac{1}{3}+\frac{1}{x}+O\left(\frac{1}{x^3}\right) = \frac{1}{3}\left(1+\frac{3}{x}+O\left(\frac{1}{x^3}\right)\right),$$

and inverting, with $(1+t)^{-1} = 1-t+O(t^2)$,

$$y = 3\left(1-\frac{3}{x}+O\left(\frac{1}{x^2}\right)\right) \approx 3-\frac{9}{x},$$

as required.

? **Ask yourself** Why can we throw away the $\frac{2}{3x^2}$ and $-\frac{3}{2x^2}$ terms but not the $\frac{1}{x}$?

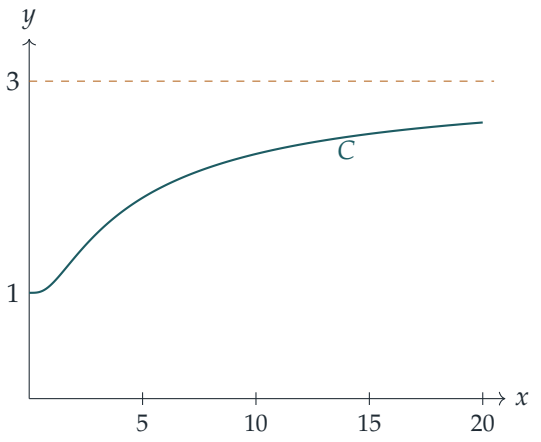
The target $3-\frac{9}{x}$ is accurate to order $\frac{1}{x}$. In $1/y$ the correction terms are a further factor $\frac{1}{x^2}$ smaller than $\frac{1}{x}$, so they only affect y at order $\frac{1}{x^2}$ or beyond. The $\frac{1}{x}$ term itself is what produces the $-\frac{9}{x}$.

Sketch of C

We now collect the features of the curve.

- $y(0) = 1$, and $\mathrm{d}y/\mathrm{d}x = 0$ at $x = 0$. Near the origin $\mathrm{d}y/\mathrm{d}x \approx x^3$, so $y \approx 1+\frac{1}{4}x^4$: the curve starts out very flat.
- For $x > 0$ the gradient is strictly positive, so y is increasing.
- The first term of $1/y$ is positive, so $1/y > \frac{1}{3}$, giving $1 \le y < 3$ for all x. The curve never blows up.
- As $x \to \infty$, $y \to 3$ from below, like $3-\frac{9}{x}$. The approach is slow: $y(10) \approx 2.31$.

Since the curve starts out concave up and then levels off towards $y = 3$, it must have a point of inflection.



The curves for z

Wherever $z \neq 0$ we can divide by z^3. Since $\dfrac{\mathrm{d}}{\mathrm{d}x}(z^{-2}) = -2z^{-3}\dfrac{\mathrm{d}z}{\mathrm{d}x}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{z^2}\right) = -\frac{2}{z^3}\cdot\frac{x^3z^3}{2(1+x^2)^{5/2}} = -\frac{x^3}{(1+x^2)^{5/2}},$$

which is exactly the equation satisfied by $1/y$. On both curves $1/z^2 = 1$ at $x = 0$, the same value as $1/y$. So

$$\frac{1}{z^2} = \frac{1}{y}, \qquad \text{that is} \qquad z^2 = y.$$

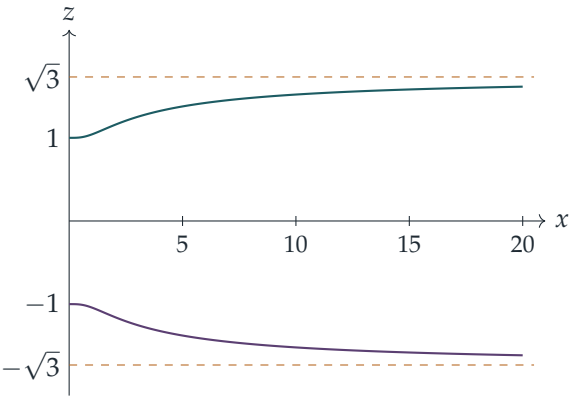
This also confirms that z never reaches 0, which justifies the division. While $z \neq 0$ we have $z^2 = y \geq 1$, so z can't get close to 0. By continuity, z therefore keeps the sign it starts with, and the two curves are

$$z = \sqrt{y} \quad (z(0) = 1), \qquad z = -\sqrt{y} \quad (z(0) = -1).$$

The upper curve rises from 1 towards $\sqrt{3}$, again starting flat (now $z \approx 1 + \frac{1}{8}x^4$), and for large x

$$z = \sqrt{3}\left(1 - \frac{3}{x} + \cdots\right)^{1/2} \approx \sqrt{3} - \frac{3\sqrt{3}}{2x}.$$

The lower curve is its reflection in the x-axis: it falls from -1 towards the asymptote $z = -\sqrt{3}$.



Ask yourself Why isn't the curve with $z(0) = -1$ just a copy of the other one, translated or otherwise transformed?

The equation is odd in z: replacing z by $-z$ changes both sides by a factor -1. So if $z(x)$ is a solution then so is $-z(x)$, and the solution starting at -1 is the reflection of the one starting at $+1$. This matches $z = \pm\sqrt{y}$.

Checks

- Differentiate the answer: with $1/y = u^{-1/2} - \frac{1}{3}u^{-3/2} + \frac{1}{3}$, we get $\mathrm{d}(1/y)/\mathrm{d}x = 2x\left(-\frac{1}{2}u^{-3/2} + \frac{1}{2}u^{-5/2}\right) = -x^3u^{-5/2}$, as needed.
- A numerical (Runge–Kutta) solution of the original equation gives $y(1) = 1.08391$ and $y(5) = 1.89777$, agreeing with the formula to 10 or more significant figures. Solving the z-equation numerically gives $z(5) = \pm 1.37759 = \pm\sqrt{y(5)}$.
- At $x = 100$ the formula gives $y = 2.9126$, against $3 - \frac{9}{x} = 2.91$. The difference is about $27/x^2$, consistent with an $O(1/x^2)$ error.

Common slips

- Losing the minus sign: $\int y^{-2}\,\mathrm{d}y = -y^{-1}$.
- Concluding that $y \to 3$ quickly because $3 - \frac{9}{x}$ "looks close" to 3. At $x = 10$ the curve is still only at about 2.3.
- Writing $z = \sqrt{y}$ for both curves, or failing to explain why z cannot change sign.

Beyond STEP

Both equations have the form $\mathrm{d}y/\mathrm{d}x = f(x)\,y^n$, a special case of a *Bernoulli equation*, which the substitution $w = y^{1-n}$ always turns into a linear equation. We used exactly this substitution: $w = 1/y$ and $w = 1/z^2$. The step "same equation, same initial value,

so same function" is an instance of the existence and uniqueness theorem (Picard–Lindelöf) for first-order equations. Boyce and DiPrima, *Elementary Differential Equations*, cover both.

WORKED SOLUTION

STEP II 2005, Question 9

Mechanics • Friction on Inclined Planes

STEP II 2005, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q9

The key idea

Treat the two particles separately. A has no applied force, so its equation fixes the tension the string must provide. All the freedom is in the *direction* of P on B. Tilting P changes the normal reaction on B, and so the friction on B. Tilting P off the slope wastes some of its component along the slope but changes the friction, and the best tilt balances these two effects. It comes out as an expression $a\cos\theta + b\sin\theta$ to maximise.

Solution

We model A and B as particles and the string as light and inextensible, lying along ℓ. We take P to act in the vertical plane containing ℓ. A component of P across the slope would do nothing to help. Note that

$$\sin 30^\circ = \tfrac{1}{2}, \qquad \cos 30^\circ = \tfrac{\sqrt{3}}{2}, \qquad \tan 30^\circ = \tfrac{1}{\sqrt{3}} = \tfrac{1}{3}\sqrt{3}.$$

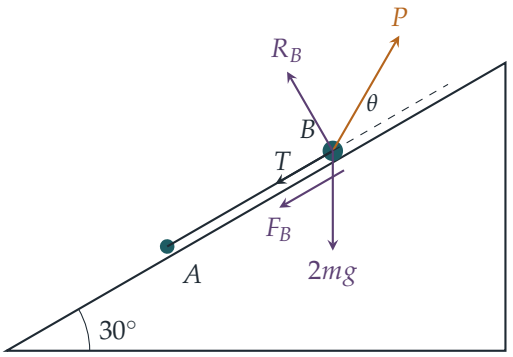
So $\mu_B = \tan 30^\circ$: on its own, B would be exactly on the point of slipping. Also $\mu_A = \frac{1}{2}\tan 30^\circ$, so A on its own would slip.

Particle A. Resolving perpendicular to the plane, $R_A = mg\cos 30^\circ = \frac{\sqrt{3}}{2}mg$, so the greatest friction on A is

$$\mu_A R_A = \tfrac{\sqrt{3}}{6}\cdot\tfrac{\sqrt{3}}{2}mg = \tfrac{1}{4}mg.$$

The weight component down the slope is $mg\sin 30^\circ = \frac{1}{2}mg$.

Particle B. Let P make an angle θ with the upward direction of ℓ, measured away from the plane. (A negative θ means P pushes into the plane.) Let T be the tension.



The diagram shows friction F_B acting down the slope, as in part (i). Perpendicular to the plane,

$$R_B + P\sin\theta = 2mg\cos 30^\circ \quad\Longrightarrow\quad R_B = \sqrt{3}\,mg - P\sin\theta.$$

So the greatest friction available at B is

$$\mu_B R_B = \tfrac{1}{\sqrt{3}}\left(\sqrt{3}\,mg - P\sin\theta\right) = mg - \tfrac{1}{\sqrt{3}}P\sin\theta.$$

The weight component of B down the slope is $2mg\sin 30^\circ = mg$.

Part (i)

For the particles to move up ℓ, friction on each acts down the slope. At the least P both are on the point of moving, so friction is limiting on both.

For A to move up, the string must overcome the weight component and the friction:

$$T = \tfrac{1}{2}mg + \tfrac{1}{4}mg = \tfrac{3}{4}mg.$$

For B, along the slope, the component of P must overcome T, the weight component and limiting friction:

$$P\cos\theta = \tfrac{3}{4}mg + mg + \left(mg - \tfrac{1}{\sqrt{3}}P\sin\theta\right),$$

$$P\left(\cos\theta + \tfrac{1}{\sqrt{3}}\sin\theta\right) = \tfrac{11}{4}mg.$$

Ask yourself Why does tilting P away from the slope help, when it reduces the component of P along the slope?

Lifting B slightly reduces R_B, and so the friction that P has to beat. For a small tilt θ, $\cos\theta$ falls only like $1 - \frac{1}{2}\theta^2$ while $\sin\theta$ grows like θ. So at first the gain from reduced friction wins. To find the best angle, write

$$\cos\theta + \tfrac{1}{\sqrt{3}}\sin\theta = \tfrac{2}{\sqrt{3}}\left(\tfrac{\sqrt{3}}{2}\cos\theta + \tfrac{1}{2}\sin\theta\right) = \tfrac{2}{\sqrt{3}}\cos(\theta - 30^\circ).$$

This is at most $\frac{2}{\sqrt{3}}$, with equality exactly when $\theta = 30°$. Since P times this factor must equal $\frac{11}{4}mg$, P is least when the factor is greatest:

$$P_{\min} = \tfrac{11}{4}mg \cdot \tfrac{\sqrt{3}}{2} = \tfrac{11}{8}\sqrt{3}\,mg,$$

as required. We must check that B stays on the plane:

$$R_B = \sqrt{3}\,mg - \tfrac{11}{8}\sqrt{3}\,mg \cdot \tfrac{1}{2} = \tfrac{5}{16}\sqrt{3}\,mg > 0.$$

So P acts in the vertical plane through ℓ, at 30° to ℓ on the side away from the plane. That is, it acts **at** 60° **above the horizontal**, pulling up the slope. (Notice that the best angle with the slope is $\arctan \mu_B$, the angle of friction for B.)

Part (ii)

Now P only has to stop the system slipping down, so friction on each particle acts up the slope. Without P, A cannot stay at rest alone, because its weight component $\frac{1}{2}mg$ exceeds the greatest friction $\frac{1}{4}mg$. So the string must hold A back:

$$T \geq \tfrac{1}{2}mg - \tfrac{1}{4}mg = \tfrac{1}{4}mg.$$

In particular the string must be taut. For B not to slip down, the forces up the slope must be able to balance T and the weight component:

$$P\cos\theta + \left(mg - \tfrac{1}{\sqrt{3}}P\sin\theta\right) \geq T + mg.$$

The left-hand side does not depend on T, so the least P comes with the least tension, $T = \frac{1}{4}mg$. Then

$$P\left(\cos\theta - \tfrac{1}{\sqrt{3}}\sin\theta\right) \geq \tfrac{1}{4}mg.$$

Now $\cos\theta - \frac{1}{\sqrt{3}}\sin\theta = \frac{2}{\sqrt{3}}\cos(\theta + 30°) \leq \frac{2}{\sqrt{3}}$, with equality when $\theta = -30°$. This time the best direction pushes B *into* the plane, which increases R_B and so the friction helping to hold B. Hence the least magnitude is

$$P_{\min} = \tfrac{1}{4}mg \cdot \tfrac{\sqrt{3}}{2} = \tfrac{1}{8}\sqrt{3}\,mg.$$

It acts at 30° below the upward direction of ℓ. Since ℓ rises at 30°, that direction is horizontal: a horizontal push on B towards the slope. At this value, A and B are both in limiting equilibrium, with $T = \frac{1}{4}mg$.

Checks

- We checked numerically with $m = g = 1$. A scan over every direction of P in the vertical plane found the least P for each one by bisection. In (i) the minimum is $2.3816 \approx \frac{11}{8}\sqrt{3}$, at 30.0° to the slope. In (ii) it is $0.2165 \approx \frac{1}{8}\sqrt{3}$, at −30.0°, which is horizontal. We also confirmed that a full equilibrium exists at the minimum in (ii).

- Total bookkeeping for (i) with P along the slope ($\theta = 0$): we need $P = \frac{1}{2}mg + \frac{1}{4}mg + mg + mg = \frac{11}{4}mg \approx 2.75mg$. This is larger than $2.38mg$, as it should be.
- For (ii) with P along the slope, the system needs $\frac{3}{2}mg - \frac{1}{4}mg - mg = \frac{1}{4}mg$. The horizontal push does better, at $\frac{\sqrt{3}}{8}mg \approx 0.217mg$.

Common slips

- Assuming P acts along the slope. The question asks for the least *magnitude*, so the direction is yours to choose.
- Forgetting that $P\sin\theta$ changes R_B, and hence the limiting friction on B.
- In (ii), keeping the friction directions from (i). When the danger is slipping down, friction acts up the slope.

Beyond STEP

Both parts rest on the same principle: the most efficient force against friction makes the angle of friction $\lambda = \arctan\mu$ with the line of motion. In (i) it is tilted away from the plane, and in (ii) it is tilted into it. This is because the normal reaction and limiting friction combine into a single "total reaction" at angle λ to the normal. Thinking in terms of this total reaction turns many friction problems into triangle-of-forces geometry. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and constrained motion carefully at first-year level.

WORKED SOLUTION

STEP II 2005, Question 10

Mechanics • Colliding Projectiles

STEP II 2005, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q10

The key idea

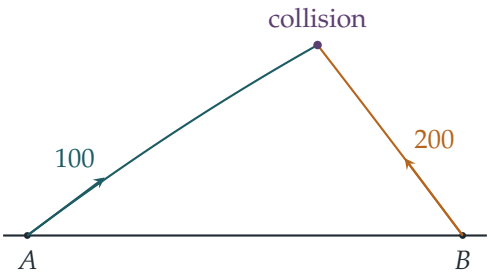
Both missiles are projectiles, so we can write down where each one is at any time. A collision means the two positions agree at the *same* instant, which gives two equations (horizontal and vertical) in two unknowns: the collision time and T. The horizontal equation is linear, so it lets us eliminate the collision time, and the vertical one then becomes a quadratic in T. We then have to decide which root makes physical sense.

Solution

Setting up

We model both missiles as particles moving freely under gravity, with no air resistance. Take A as the origin, the x-axis along AB (so B is at $x = 180$) and the y-axis vertically upwards. Measure time t in seconds from the launch of the first missile.

For the first missile, $\sin\alpha = \frac{3}{5}$ gives $\cos\alpha = \frac{4}{5}$, so its initial velocity components are $100 \cdot \frac{4}{5} = 80$ and $100 \cdot \frac{3}{5} = 60$. For the second, $\sin\beta = \frac{4}{5}$ gives $\cos\beta = \frac{3}{5}$. It travels towards A, so its components are $-200 \cdot \frac{3}{5} = -120$ and $200 \cdot \frac{4}{5} = 160$.



Let $s = t - T$ be the time since the second launch. Then, while both are in flight,

$$\begin{aligned} &\text{first missile:} && x = 80t, && y = 60t - 5t^2, \\ &\text{second missile:} && x = 180 - 120s, && y = 160s - 5s^2. \end{aligned}$$

The collision

At the collision the x-coordinates agree:

$$80t = 180 - 120(t - T) \quad \Longrightarrow \quad 200t = 180 + 120T \quad \Longrightarrow \quad t = 0.9 + 0.6T,$$

and so $s = t - T = 0.9 - 0.4T$.

The y-coordinates also agree: $60t - 5t^2 = 160s - 5s^2$, that is,

$$60t - 160s = 5(t^2 - s^2) = 5(t - s)(t + s) = 5T(t + s),$$

using $t - s = T$. Substituting the expressions for t and s:

$$\begin{aligned} 60t - 160s &= 60(0.9 + 0.6T) - 160(0.9 - 0.4T) = 100T - 90, \\ 5T(t + s) &= 5T(1.8 + 0.2T) = 9T + T^2. \end{aligned}$$

Equating these,

$$T^2 - 91T + 90 = 0 \quad \Longrightarrow \quad (T - 1)(T - 90) = 0,$$

so $T = 1$ or $T = 90$.

Ask yourself Both roots solve the equations. Why can't $T = 90$ be the answer?

For $T = 90$ we get $s = 0.9 - 36 = -35.1$, which means the "collision" would happen 35.1 seconds *before* the second missile is launched. (It is also far below ground level: the equations describe the parabolas extended beyond the actual flights.) So this root is not physical. The first missile is in the air for $2 \cdot 60/10 = 12$ seconds anyway, so it would have landed long before.

For $T = 1$ we get $t = 1.5$ and $s = 0.5 > 0$, so both missiles are in flight. The heights are $60(1.5) - 5(1.5)^2 = 78.75$ and $160(0.5) - 5(0.5)^2 = 78.75$, both positive, and both missiles are at $x = 120$. Hence

$$T = 1 \text{ second}.$$

Checks

Substitute $T = 1$ back in: at $t = 1.5\,\text{s}$ both missiles are at $(120, 78.75)$ metres, which we confirmed with exact fractions. Stepping both motions numerically with launch delay 1 s, the distance between the missiles falls to essentially zero at $t = 1.5\,\text{s}$. As a sanity check, the collision is at $x = 120$, between A and B, and the first missile is still rising, since its vertical velocity there is $60 - 15 = 45\,\text{m}\,\text{s}^{-1}$.

Common slips

- Using the same time variable for both missiles. The second missile has been flying for $t - T$, not t.
- Forgetting that the second missile moves in the negative x-direction.
- Keeping $T = 90$ without checking that $t - T > 0$.

Beyond STEP

There is a slicker way to see this. Both missiles have the same acceleration g, so once both are in flight their *relative* velocity is constant and each sees the other moving in a straight line. A collision happens exactly when that relative velocity points along the line joining them at time T, and writing this down gives the same quadratic. Changing to a freely falling frame like this is the first step towards Einstein's equivalence principle. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats relative motion and non-inertial frames carefully.

WORKED SOLUTION

STEP II 2005, Question 11

Mechanics • Friction on Inclined Plane

STEP II 2005, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

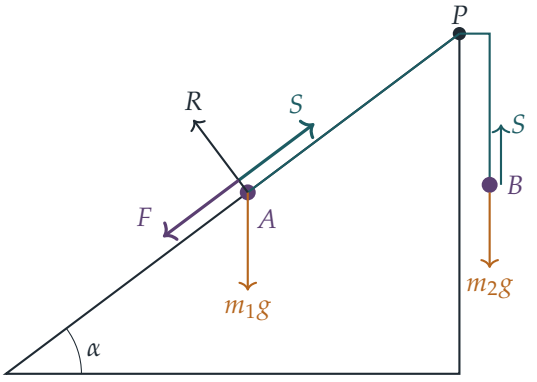
step.maths.org/questions/05-s2-q11

The key idea

With $\tan\alpha = \frac{3}{4}$ we have a 3–4–5 triangle, so $\sin\alpha = \frac{3}{5}$ and $\cos\alpha = \frac{4}{5}$, and every force on A becomes a tidy multiple of $m_1 g$. The motion of A then splits into three stages with constant acceleration: pulled up by the string, sliding up freely, and sliding back down. The direction of friction changes between the last two, and that is where the question hides its subtlety.

Solution

We model A and B as particles, the string as light and inextensible, and the pulley as smooth, so the tension S is the same on both sides. Let $\alpha = \arctan\frac{3}{4}$, so that $\sin\alpha = \frac{3}{5}$ and $\cos\alpha = \frac{4}{5}$.



The acceleration while the string is taut

Perpendicular to the plane, A has no acceleration, so

$$R = m_1 g\cos\alpha = \tfrac{4}{5}m_1 g, \qquad \text{and limiting friction is} \qquad \mu R = \tfrac{1}{2}\cdot\tfrac{4}{5}m_1 g = \tfrac{2}{5}m_1 g.$$

First we check that the system moves. If B were to descend, A would move up the slope, opposed by its weight component $\frac{3}{5}m_1g$ and by friction of at most $\frac{2}{5}m_1g$: a total of at most m_1g. Pulling the other way is B's weight m_2g, and $m_2 > m_1$, so friction cannot hold the system: B descends, A moves up the slope, and friction on A is limiting, acting down the slope.

Let a be the common acceleration. Newton's second law for A (up the slope) and for B (downwards) gives

$$m_1a = S - \tfrac{3}{5}m_1g - \tfrac{2}{5}m_1g = S - m_1g,$$
$$m_2a = m_2g - S.$$

Adding, $(m_1 + m_2)a = (m_2 - m_1)g$, so

$$a = \frac{m_2 - m_1}{m_2 + m_1}\,g,$$

as required. It will be convenient to write $a = kg$, where $k = \dfrac{m_2 - m_1}{m_2 + m_1}$ and $0 < k < 1$.

The time of maximum height

At time T, A is moving up the slope with speed $u = aT = kgT$, having travelled $\frac{1}{2}kgT^2$. When the string breaks, S disappears. A is still moving up the slope, so friction still acts down the slope, and the forces down the slope total $\frac{3}{5}m_1g + \frac{2}{5}m_1g = m_1g$. So A decelerates at exactly g, and comes to rest after a further time

$$\frac{u}{g} = kT.$$

Hence A reaches its maximum height at time

$$T + kT = (1 + k)T = \frac{2m_2}{m_1 + m_2}\,T$$

after release. (We are told A never reaches the pulley, so this stage is not cut short.) During this stage A travels a further $\dfrac{u^2}{2g} = \frac{1}{2}k^2gT^2$, so its total distance up the slope from the point of release is

$$d = \tfrac{1}{2}kgT^2 + \tfrac{1}{2}k^2gT^2 = \tfrac{1}{2}k(1 + k)gT^2.$$

? **Ask yourself** Once A stops, does it stay there?

It stays only if friction can hold it. The weight component down the slope is $\frac{3}{5}m_1g$, while friction can provide at most $\frac{2}{5}m_1g$. Equivalently, $\tan\alpha = \frac{3}{4} > \frac{1}{2} = \mu$. So A slides back down, and now friction acts *up* the slope. The net force down the slope is $\frac{3}{5}m_1g - \frac{2}{5}m_1g = \frac{1}{5}m_1g$, so A accelerates down the slope at $\frac{1}{5}g$, a much gentler rate than its deceleration on the way up.

The ratio of the masses

Starting from rest at its highest point, A takes a time t_d to slide back the distance d to its point of release, where

$$\tfrac{1}{2}\cdot\tfrac{1}{5}g\,t_d^2 = d = \tfrac{1}{2}k(1+k)gT^2, \qquad \text{so} \qquad t_d^2 = 5k(1+k)T^2.$$

We are told that t_d equals the time of maximum height, $(1+k)T$. Squaring, this means

$$(1+k)^2T^2 = 5k(1+k)T^2.$$

Dividing by $(1+k)T^2$, which is positive, gives $1+k=5k$, so $k=\tfrac{1}{4}$. Notice that T has cancelled: this is why the equality holds regardless of when the string breaks, and it pins down k uniquely. Then

$$\frac{m_2-m_1}{m_2+m_1} = \frac{1}{4} \iff 4m_2-4m_1 = m_2+m_1 \iff 3m_2 = 5m_1,$$

so

$$m_1 : m_2 = 3 : 5.$$

Checks

- Limiting cases: if $m_2 \to m_1$ then $k \to 0$ and the time of maximum height tends to T (there is barely any speed to lose); if $m_2 \gg m_1$ then $k \to 1$ and A coasts for as long again as it was pulled, as we'd expect when it reaches speed gT and then decelerates at g.
- With $m_1 = 3$, $m_2 = 5$ and $T = 1\,\text{s}$, we have $k=\tfrac{1}{4}$; maximum height comes at 1.25 s, $d = \tfrac{5}{32}g$, and the descent takes $\sqrt{2d/(g/5)} = \sqrt{25/16} = 1.25\,\text{s}$. A step-by-step numerical simulation of all three stages gave both times as 1.250 s, for several values of T.

Common slips

- Forgetting that friction reverses direction when A starts sliding back, and so using g (or $\tfrac{3}{5}g$) for the downward acceleration instead of $\tfrac{1}{5}g$.
- Measuring the descent only from the point where the string broke, rather than from the point of release.
- Not checking that A actually slides back: this needs $\tan\alpha > \mu$.

Beyond STEP

Friction is the classic example of a force that depends on the state of motion: it switches direction with the velocity and its magnitude is capped while the body is at rest. This makes the motion piecewise, with a different equation of motion in each

stage, and the same idea governs stick–slip phenomena such as squeaking hinges and earthquake faults. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and constrained motion with the care that first-year physics demands.

WORKED SOLUTION

STEP II 2005, Question 12

Probability and Statistics • Bayes' Theorem

STEP II 2005, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q12

The key idea

This is Bayes' theorem with a fair coin as the prior. Because the coin is fair, the probability of heads given the answers is $\frac{1}{2}$ exactly when the answers are *equally likely* whether the coin showed heads or tails. So in each part we just compare two likelihoods, $\mathrm{P}(\text{answers} \mid H)$ and $\mathrm{P}(\text{answers} \mid T)$, conditioning on which twin is online to work them out.

Solution

Let H and T be the events that the coin shows heads and tails, so $\mathrm{P}(H) = \mathrm{P}(T) = \frac{1}{2}$. Let A be the event that Anna is online and B the event that Bella is online, with $\mathrm{P}(A) = p$, $\mathrm{P}(B) = q = 1 - p$. Who is online has nothing to do with the coin, so A and B are independent of H. In parts (ii) and (iii) we take the coin to be tossed once and both e-mails to be answered by the same twin, the one online when we write. (See the remark at the end for the other reading.)

Write $t = ap + bq$, the overall probability that a single answer is truthful.

For any record of answers E, Bayes' theorem with $\mathrm{P}(H) = \mathrm{P}(T)$ gives

$$\mathrm{P}(H \mid E) = \frac{\frac{1}{2}\mathrm{P}(E \mid H)}{\frac{1}{2}\mathrm{P}(E \mid H) + \frac{1}{2}\mathrm{P}(E \mid T)} = \frac{\mathrm{P}(E \mid H)}{\mathrm{P}(E \mid H) + \mathrm{P}(E \mid T)}.$$

This equals $\frac{1}{2}$ if and only if $\mathrm{P}(E \mid H) = \mathrm{P}(E \mid T)$. That one observation does most of the work.

Part (i)

Let Y be the event that the answer is 'yes'. If the coin shows heads, 'yes' is the truth. Conditioning on who is online,

$$\mathrm{P}(Y \mid H) = pa + qb = t.$$

If the coin shows tails, 'yes' is a lie, so

$$\mathrm{P}(Y \mid T) = p(1-a) + q(1-b) = (p+q) - (ap+bq) = 1 - t.$$

Hence

$$\mathrm{P}(H \mid Y) = \frac{t}{t + (1-t)} = t = ap + bq.$$

So $\mathrm{P}(H \mid Y) = \frac{1}{2}$ if and only if $ap + bq = \frac{1}{2}$, that is, if and only if $2(ap+bq) = 1$, as required.

Part (ii)

Let E be the event 'yes' then 'no'. Suppose Anna is online. Her two answers are independent, and exactly one of them is true whichever way the coin landed:

$$\mathrm{P}(E \mid H \cap A) = a(1-a) \qquad \text{(truthful 'yes', then a lie)},$$
$$\mathrm{P}(E \mid T \cap A) = (1-a)a \qquad \text{(a lie, then truthful 'no')}.$$

These are equal. In the same way $\mathrm{P}(E \mid H \cap B) = b(1-b) = \mathrm{P}(E \mid T \cap B)$. Therefore

$$\mathrm{P}(E \mid H) = pa(1-a) + qb(1-b) = \mathrm{P}(E \mid T),$$

and by the observation above $\mathrm{P}(H \mid E) = \frac{1}{2}$, whatever the values of p, a and b.

Ask yourself Why do we get exactly $\frac{1}{2}$ here with no condition at all?

Contradictory answers mean that, whatever the coin showed, the twin told the truth once and lied once. That combination is equally likely under heads and under tails, so it carries no information about the coin, and we are left with the prior, $\frac{1}{2}$.

Part (iii)

Now let E be the event 'yes' then 'yes'. Given heads, both answers are true; given tails, both are lies. Conditioning on who is online,

$$\mathrm{P}(E \mid H) = pa^2 + qb^2,$$
$$\mathrm{P}(E \mid T) = p(1-a)^2 + q(1-b)^2.$$

Subtracting, and using $(1-a)^2 = 1 - 2a + a^2$,

$$\begin{aligned}\mathrm{P}(E \mid H) - \mathrm{P}(E \mid T) &= p\left(a^2 - (1-a)^2\right) + q\left(b^2 - (1-b)^2\right) \\ &= p(2a-1) + q(2b-1) \\ &= 2(ap+bq) - (p+q) \\ &= 2(ap+bq) - 1.\end{aligned}$$

So if $2(ap+bq) = 1$, the two likelihoods are equal and $\mathrm{P}(H \mid E) = \frac{1}{2}$, as required.

(In fact the calculation shows more: the difference of the likelihoods is exactly $2(ap+bq) - 1$, so $\mathrm{P}(H \mid E) = \frac{1}{2}$ holds *if and only if* $2(ap+bq) = 1$, the same condition as in part (i).)

Remark: the other reading

If instead whichever twin happens to be online is chosen afresh for each e-mail, independently, then each answer is truthful with probability t, independently. Then in (ii) $\mathrm{P}(E \mid H) = t(1-t) = \mathrm{P}(E \mid T)$, and in (iii) $\mathrm{P}(E \mid H) = t^2$ and $\mathrm{P}(E \mid T) = (1-t)^2$, which are equal when $t = \frac{1}{2}$. The conclusions are the same.

Checks

- If both twins are always truthful ($a = b = 1$), then $t = 1 \neq \frac{1}{2}$, and in (i) a 'yes' makes heads certain: $\mathrm{P}(H \mid Y) = t = 1$. Contradictory answers as in (ii) are then impossible, which is consistent with $a(1-a) = b(1-b) = 0$.
- With $p = 0.3$, $a = 0.8$ and $b = \frac{0.5-0.24}{0.7} \approx 0.371$ (so $2(ap + bq) = 1$), a simulation of 2×10^5 rounds in the same-twin model gave $\mathrm{P}(H \mid \text{yes, yes}) \approx 0.496$. Exact enumeration with fractions over thousands of random (p, a, b) confirmed all three results in both readings.
- If $p = \frac{1}{2}$ and $a = b = 0.9$, the condition fails and two 'yes' answers give $\mathrm{P}(H \mid E) = \frac{81}{82}$, far from $\frac{1}{2}$, so the condition in (iii) really matters.

Common slips

- Writing $\mathrm{P}(Y \mid T) = 1 - \mathrm{P}(Y \mid H)$ without justification. It happens to be true here, but only because a 'yes' under tails is a lie; say so.
- In (iii), treating the two answers as independent truthful answers with probability t each while also assuming the same twin answers both. Given the twin they are independent, but unconditionally they are not, since both depend on who is online.

Beyond STEP

The single principle here is that the posterior odds equal the prior odds times the likelihood ratio: evidence that is equally likely under both hypotheses tells you nothing. The same-twin model is a simple *mixture*: answers are independent given a hidden variable (who is online) but dependent overall, an idea at the heart of hidden-variable and Bayesian models. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of conditioning and Bayes' rule in exactly this spirit.

WORKED SOLUTION

STEP II 2005, Question 13

Probability and Statistics • Poisson and Binomial Distributions

STEP II 2005, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q13

The key idea

Each sampled page either has fewer than two errors or it doesn't, and the pages behave independently, so Y counts successes in n independent trials: it is binomial. Everything after that is about what happens when λ is small, and the tool for that is the Maclaurin series of $\mathrm{e}^{-\lambda}$ (or of $\ln(1+\lambda)$), keeping only the first term that doesn't cancel.

Solution

The distribution of Y

Let X be the number of errors on a particular page, so $X \sim \mathrm{Po}(\lambda)$. Call a page *good* if $X \leq 1$. Then

$$\mathrm{P}(\text{good}) = \mathrm{P}(X=0) + \mathrm{P}(X=1) = \mathrm{e}^{-\lambda} + \lambda\mathrm{e}^{-\lambda} = (1+\lambda)\mathrm{e}^{-\lambda} = p.$$

The sample consists of n different pages. The error counts on different pages are independent and all have the same $\mathrm{Po}(\lambda)$ distribution, so whichever n pages are chosen, the events "page j is good" are independent, each with probability p. (The condition $n \ll N$ just reassures us that there are plenty of pages to sample from; it isn't needed for the independence, which comes from the model.)

So Y is the number of successes in n independent trials with success probability p. A particular set of k good pages and $n-k$ bad ones has probability $p^k q^{n-k}$, and there are $\binom{n}{k}$ ways to choose which k pages are good, so

$$\mathrm{P}(Y=k) = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, \ldots, n,$$

with $p = (1+\lambda)\mathrm{e}^{-\lambda}$ and $q = 1-p$, as required. That is, $Y \sim \mathrm{B}(n, p)$.

Part (i)

Expand $e^{-\lambda} = 1 - \lambda + \frac{1}{2}\lambda^2 - \frac{1}{6}\lambda^3 + \cdots$ and multiply by $1 + \lambda$:

$$\begin{aligned} p = (1+\lambda)e^{-\lambda} &= \left(1 - \lambda + \tfrac{1}{2}\lambda^2 - \tfrac{1}{6}\lambda^3 + \cdots\right) + \left(\lambda - \lambda^2 + \tfrac{1}{2}\lambda^3 - \cdots\right) \\ &= 1 - \tfrac{1}{2}\lambda^2 + \tfrac{1}{3}\lambda^3 - \cdots . \end{aligned}$$

The terms in λ cancel, so

$$q = 1 - p = \tfrac{1}{2}\lambda^2 - \tfrac{1}{3}\lambda^3 + \cdots \approx \tfrac{1}{2}\lambda^2$$

when λ is small, since the neglected terms are smaller than $\frac{1}{2}\lambda^2$ by a factor of order λ.

Ask yourself Why isn't $q \approx \lambda$, as the first term of $e^{-\lambda}$ might suggest?

Because the $-\lambda$ from $e^{-\lambda}$ is exactly cancelled by the $+\lambda$ from multiplying by $1 + \lambda$. A page is bad only if it has at least two errors, and $P(X \geq 2) \approx P(X = 2) = \frac{1}{2}\lambda^2 e^{-\lambda} \approx \frac{1}{2}\lambda^2$. That is a nice independent confirmation of (i).

Part (ii)

$P(Y = n) = p^n$, and we need the largest n with $p^n \geq 1 - \lambda$. Both sides are positive (for $0 < \lambda < 1$), and ln is increasing, so this is equivalent to

$$n \ln p \geq \ln(1 - \lambda).$$

Now $\ln p = \ln(1 + \lambda) - \lambda = \left(\lambda - \frac{1}{2}\lambda^2 + \cdots\right) - \lambda \approx -\frac{1}{2}\lambda^2$, and $\ln(1 - \lambda) \approx -\lambda$. Since $\ln p < 0$, dividing by it reverses the inequality:

$$n \leq \frac{\ln(1-\lambda)}{\ln p} \approx \frac{-\lambda}{-\frac{1}{2}\lambda^2} = \frac{2}{\lambda}.$$

So the largest such n is approximately $2/\lambda$.

Equivalently, using (i): $p^n = (1 - q)^n \approx 1 - nq \approx 1 - \frac{1}{2}n\lambda^2$, and $1 - \frac{1}{2}n\lambda^2 \geq 1 - \lambda$ exactly when $n \leq 2/\lambda$. The step $(1 - q)^n \approx 1 - nq$ needs nq small; here $nq \approx \frac{2}{\lambda} \cdot \frac{1}{2}\lambda^2 = \lambda$, which is small, so the approximation is self-consistent.

Part (iii)

Since $Y > 1$ implies $Y > 0$,

$$P(Y > 1 \mid Y > 0) = \frac{P(Y > 1)}{P(Y > 0)} = \frac{P(Y > 0) - P(Y = 1)}{P(Y > 0)} = 1 - \frac{npq^{n-1}}{1 - q^n}.$$

(Here $n \geq 2$ matters: it makes $Y > 1$ a possible event and the power q^{n-1} at least q.) Now let λ be small with n fixed. Then $q \approx \frac{1}{2}\lambda^2$ is small, so $p \approx 1$ and $1 - q^n \approx 1$, giving

$$P(Y > 1 \mid Y > 0) \approx 1 - nq^{n-1} \approx 1 - n\left(\frac{\lambda^2}{2}\right)^{n-1},$$

using (i) in the last step. This is the required result.

Checks

- A simulation of 200,000 samples of $n = 5$ Poisson pages with $\lambda = 0.8$ gave relative frequencies for $Y = 0, \ldots, 5$ agreeing with $\mathrm{B}(5, p)$ to within 0.002.
- For $\lambda = 0.01$, $q = 4.967 \times 10^{-5}$, against $\frac{1}{2}\lambda^2 = 5 \times 10^{-5}$.
- For $\lambda = 0.01$, the exact largest n with $p^n \geq 0.99$ is 202, against $2/\lambda = 200$. (Keeping one more term in each series gives $2/\lambda + \frac{7}{3}$, so the error stays bounded while $2/\lambda$ grows.)
- For $\lambda = 0.01$, $n = 3$: $1 - \mathrm{P}(Y > 1 \mid Y > 0) = 7.40 \times 10^{-9}$ exactly, against $3(\lambda^2/2)^2 = 7.5 \times 10^{-9}$.

Common slips

- Taking "fewer than two errors" as $X = 0$ only, or as $X \leq 2$.
- Stopping the expansion of $(1 + \lambda)\mathrm{e}^{-\lambda}$ at the linear term and concluding $q \approx 0$: the first surviving term is the quadratic one.
- Forgetting to reverse the inequality when dividing by $\ln p < 0$ in (ii).

Beyond STEP

Part (ii) is an example of asymptotic analysis: the condition is controlled by the product $n\lambda^2$, so the natural scale for n is $1/\lambda$, and quantities like p^n behave like $\mathrm{e}^{-n\lambda^2/2}$. The same reasoning underlies the Poisson approximation to the binomial and many "rare events" limits in probability. Grimmett and Welsh, *Probability: An Introduction*, develops these limit theorems carefully.

WORKED SOLUTION

STEP II 2005, Question 14

Probability and Statistics • Mixture Distributions

STEP II 2005, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q14

The key idea

The density is a weighted mixture of two densities we already know well: a standard normal and a uniform on $[0, \lambda]$. Every integral of f therefore splits into a normal piece and a uniform piece, and we can read off each piece from standard facts (total probability 1, mean, $\mathrm{E}(X^2)$) without doing any hard integration. The variance then comes from $\mathrm{E}(X^2) - \mu^2$.

Solution

The constant k, the mean and the variance

Since ϕ and g are both probability density functions, each integrates to 1 over the real line. So

$$1 = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = k\,[1 + \lambda \cdot 1] \quad \Longrightarrow \quad k = \frac{1}{1+\lambda}.$$

For the mean, we use the means of the two components: the standard normal has mean 0, and the uniform density g on $[0, \lambda]$ has mean $\frac{1}{2}\lambda$. Hence

$$\mu = \int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x = k\left[\int x\phi(x)\,\mathrm{d}x + \lambda \int x g(x)\,\mathrm{d}x\right] = \frac{1}{1+\lambda}\left[0 + \lambda \cdot \frac{\lambda}{2}\right] = \frac{\lambda^2}{2(1+\lambda)}.$$

In the same way, the standard normal has $\int x^2\phi(x)\,\mathrm{d}x = 1$ (its variance, since its mean is 0), and

$$\int x^2 g(x)\,\mathrm{d}x = \frac{1}{\lambda}\int_0^{\lambda} x^2\,\mathrm{d}x = \frac{\lambda^2}{3}.$$

So

$$\mathrm{E}(X^2) = \frac{1}{1+\lambda}\left[1 + \lambda \cdot \frac{\lambda^2}{3}\right] = \frac{3+\lambda^3}{3(1+\lambda)}.$$

Ask yourself Why can we split $\mathrm{E}(X^2)$ like this, but not the variance?

Expectations are integrals, and integrals are linear, so $\mathrm{E}(X^2)$ is the same weighted combination of the components' second moments as f is of their densities. The variance is not linear (it involves μ^2), so we must combine the second moments first and only then subtract μ^2. Doing so,

$$\begin{aligned}\sigma^2 = \mathrm{E}(X^2) - \mu^2 &= \frac{3+\lambda^3}{3(1+\lambda)} - \frac{\lambda^4}{4(1+\lambda)^2} \\ &= \frac{4(3+\lambda^3)(1+\lambda) - 3\lambda^4}{12(1+\lambda)^2} \\ &= \frac{12+12\lambda+4\lambda^3+4\lambda^4-3\lambda^4}{12(1+\lambda)^2} = \frac{\lambda^4+4\lambda^3+12\lambda+12}{12(1+\lambda)^2},\end{aligned}$$

which is the required result.

The case $\lambda = 2$

Now $k = \frac{1}{3}$ and $\lambda g(x) = 1$ on $[0,2]$, so

$$f(x) = \begin{cases} \frac{1}{3}\phi(x) & x < 0 \text{ or } x > 2, \\ \frac{1}{3}(\phi(x)+1) & 0 \le x \le 2. \end{cases}$$

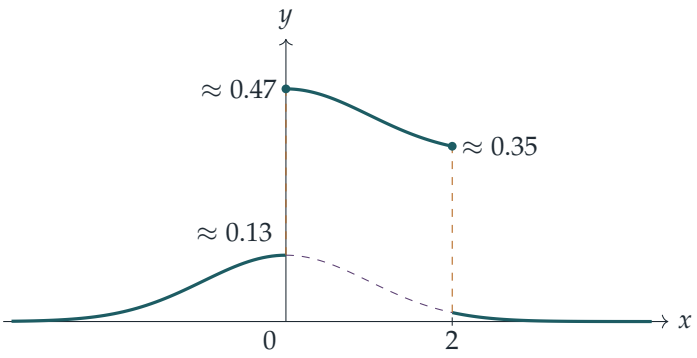
Also

$$\mu = \frac{4}{6} = \frac{2}{3}, \qquad \sigma^2 = \frac{16+32+24+12}{12\cdot 9} = \frac{84}{108} = \frac{7}{9}, \qquad \sigma = \frac{\sqrt{7}}{3}.$$

Part (i)

The curve is a third of the standard normal bell curve, lifted by $\frac{1}{3}$ on the interval $[0,2]$. So it has jumps up at $x = 0$ and down at $x = 2$, each of height $\frac{1}{3}$. Using $\phi(0) = 1/\sqrt{2\pi} \approx 0.399$ and $\phi(2) \approx 0.054$:

- just left of 0, $f \approx 0.133$; just right of 0, $f \approx 0.466$ (the maximum value);
- just left of 2, $f \approx 0.351$; just right of 2, $f \approx 0.018$;
- on $[0,2]$ the curve decreases, following the shape of the normal curve;
- $f \to 0$ as $x \to \pm\infty$, and $f > 0$ everywhere.



The dashed plum curve on $[0,2]$ is $\frac{1}{3}\phi(x)$, shown for comparison; the solid curve is $y = f(x)$. (The vertical scale is stretched.)

Part (ii)

Let $F(x) = \mathrm{P}(X \le x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t$. Integrating $\frac{1}{3}\phi$ gives $\frac{1}{3}\Phi$, and the extra $\frac{1}{3}$ on $[0,2]$ contributes $\frac{1}{3}$ times the length of $[0,2]$ that lies to the left of x. So

$$F(x) = \begin{cases} \frac{1}{3}\Phi(x) & x < 0, \\ \frac{1}{3}\big(\Phi(x) + x\big) & 0 \le x \le 2, \\ \frac{1}{3}\big(\Phi(x) + 2\big) & x > 2. \end{cases}$$

The pieces agree at $x = 0$ and $x = 2$, as they must (a CDF is continuous when there are no point masses), and $F(x) \to \frac{1}{3}(1+2) = 1$ as $x \to \infty$.

Part (iii)

Here $\mu + 2\sigma = \frac{2}{3} + \frac{2}{3}\sqrt{7} \approx 2.43$, which is greater than 2, so we use the last branch of F there. Since X is continuous,

$$\begin{aligned} \mathrm{P}(0 < X < \mu + 2\sigma) &= F\big(\tfrac{2}{3} + \tfrac{2}{3}\sqrt{7}\big) - F(0) \\ &= \tfrac{1}{3}\big(\Phi(\tfrac{2}{3} + \tfrac{2}{3}\sqrt{7}) + 2\big) - \tfrac{1}{3}\Phi(0) \\ &= \tfrac{1}{3}\big(0.9921 + 2 - 0.5\big) = \frac{2.4921}{3} = 0.8307. \end{aligned}$$

Checks

- With $\lambda \to 0$ the uniform part disappears: the formulae give $\mu \to 0$ and $\sigma^2 \to \frac{12}{12} = 1$, the standard normal values, as they should.
- For $\lambda = 2$, the law of total variance (see below) gives the same answer another way. The components have weights $\frac{1}{3}, \frac{2}{3}$, variances $1, \frac{1}{3}$ and means $0, 1$. The average variance is $\frac{1}{3} + \frac{2}{9} = \frac{5}{9}$, the variance of the means is $\frac{2}{3} - \frac{4}{9} = \frac{2}{9}$, and the total is $\frac{7}{9}$.

- Integrating f numerically (midpoint rule, split at 0 and λ) for $\lambda = 0.5, 1, 2, 3.7$ reproduced total probability 1, the mean and the variance formula to better than 10^{-6}; for $\lambda = 2$ it gave $\mu = 0.6667$ and $\sigma^2 = 0.7778$.
- Computing $\Phi(2.4305)$ directly gives 0.99246 rather than the quoted 0.9921, and the probability 0.8308 rather than 0.8307. The tiny difference comes from the value given in the question, which we should use.

Common slips

- Forgetting the factor $k = \frac{1}{3}$, so that the total probability is 3.
- Computing the variance as $k[\,1 + \lambda \cdot \frac{\lambda^2}{12}\,]$, that is, combining the components' variances. Variances of a mixture do not combine linearly; second moments do.
- Drawing f as a continuous curve: it jumps at $x = 0$ and $x = \lambda$.

Beyond STEP

A density of the form $\sum_i w_i f_i(x)$ with weights $w_i \geq 0$ summing to 1 is a *mixture distribution*: pick component i with probability w_i, then sample from f_i. Here the weights are $\frac{1}{1+\lambda}$ and $\frac{\lambda}{1+\lambda}$. The variance of a mixture is the law of total variance in disguise: the average of the component variances plus the variance of the component means. Mixtures are the basis of clustering methods in statistics and machine learning. Blitzstein and Hwang, *Introduction to Probability*, covers conditional expectation and the law of total variance clearly.

WORKED SOLUTION

STEP III 2005, Question 1

Pure • Trigonometric Equations and Graphs

STEP III 2005, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q1

The key idea

Every equation of the form sin(something) = cos(something else) can be turned into a statement about the two *arguments*, using $\cos B = \sin\left(\frac{\pi}{2} - B\right)$. Applied to $\sin(\sin x) = \cos(\cos x)$, this says that $\sin x \mp \cos x$ would have to equal an odd multiple of $\frac{\pi}{2}$, and the bound on $|\sin x \pm \cos x|$ shows that it is simply too small to reach one. The sketches then follow from thinking about what the outer function does to an inner argument that only ranges over a short interval.

Solution

The equation $\sin A = \cos B$

Since $\cos B = \sin\left(\frac{\pi}{2} - B\right)$, we need the general solution of $\sin A = \sin C$ with $C = \frac{\pi}{2} - B$. The sine function takes each value in $[-1, 1]$ exactly twice in each period 2π, at C and at $\pi - C$ (these coincide when the value is ± 1), so

$$\sin A = \sin C \iff A = C + 2k\pi \quad \text{or} \quad A = \pi - C + 2k\pi \quad \text{for some integer } k.$$

Substituting $C = \frac{\pi}{2} - B$:

$$\begin{aligned} & A = \tfrac{\pi}{2} - B + 2k\pi = (4k+1)\tfrac{\pi}{2} - B, \\ \text{or} \quad & A = \pi - \tfrac{\pi}{2} + B + 2k\pi = (4k+1)\tfrac{\pi}{2} + B. \end{aligned}$$

So $\sin A = \cos B$ if and only if $A = (4n+1)\frac{\pi}{2} \pm B$ for some integer n. Both directions are covered, because each step above is an "if and only if". (For the "if" direction directly: $\sin\left((4n+1)\frac{\pi}{2} \pm B\right) = \sin\left(\frac{\pi}{2} \pm B\right) = \cos B$, using the period 2π.)

The bound and the equation $\sin(\sin x) = \cos(\cos x)$

Using the compound angle formula,

$$\sin x \pm \cos x = \sqrt{2}\left(\sin x \cos\tfrac{\pi}{4} \pm \cos x \sin\tfrac{\pi}{4}\right) = \sqrt{2}\sin\left(x \pm \tfrac{\pi}{4}\right),$$

and since $|\sin\theta| \le 1$ for every θ, we get $|\sin x \pm \cos x| \le \sqrt{2}$ for all x, as required.

Now suppose $\sin(\sin x) = \cos(\cos x)$ for some x. By the first result with $A = \sin x$ and $B = \cos x$, there is an integer n with

$$\sin x = (4n+1)\tfrac{\pi}{2} \pm \cos x, \qquad \text{that is,} \qquad \sin x \mp \cos x = (4n+1)\tfrac{\pi}{2}.$$

Ask yourself Why can't the right-hand side be small?

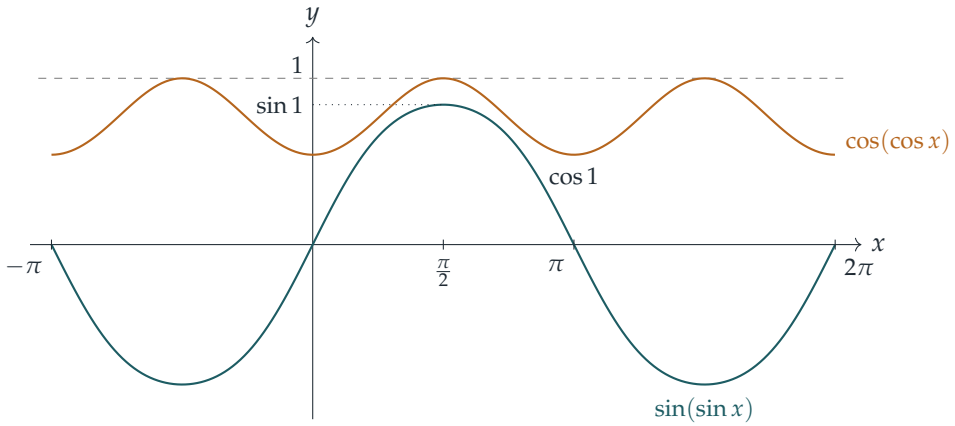
Because $4n+1$ is an odd integer, it is never 0, so $|4n+1| \ge 1$ and the right-hand side has modulus at least $\frac{\pi}{2} \approx 1.571$. The left-hand side has modulus at most $\sqrt{2} \approx 1.414$. Since $\sqrt{2} < \frac{\pi}{2}$ (square both: $2 < \frac{\pi^2}{4}$ because $\pi^2 > 9 > 8$), this is impossible. Hence $\sin(\sin x) = \cos(\cos x)$ has no solutions.

Sketching $y = \sin(\sin x)$ and $y = \cos(\cos x)$

$y = \sin(\sin x)$**.** The inner function $\sin x$ runs over $[-1, 1]$, and on that interval sin is increasing (as $[-1,1] \subset [-\frac{\pi}{2}, \frac{\pi}{2}]$). So $\sin(\sin x)$ rises and falls exactly when $\sin x$ does: it has period 2π, is odd, is zero at $x = n\pi$, has maximum $\sin 1 \approx 0.841$ at $x = \frac{\pi}{2} + 2n\pi$ and minimum $-\sin 1$ at $x = -\frac{\pi}{2} + 2n\pi$. Its gradient $\cos x \cos(\sin x)$ equals 1 at the origin. It looks like a slightly flattened sine curve.

$y = \cos(\cos x)$**.** Since $\cos(-u) = \cos u$ and $\cos(x + \pi) = -\cos x$, we have $\cos(\cos(x + \pi)) = \cos(\cos x)$: the graph has period π and is even. On $[-1, 1]$, $\cos u$ is largest ($= 1$) at $u = 0$ and smallest ($= \cos 1 \approx 0.540$) at $u = \pm 1$. So $\cos(\cos x)$ has maxima 1 where $\cos x = 0$, i.e. at $x = \frac{\pi}{2} + n\pi$, and minima $\cos 1$ where $\cos x = \pm 1$, i.e. at $x = n\pi$. It always lies between 0.54 and 1.

By the previous part the curves never meet, and since $\cos(\cos 0) = \cos 1 > 0 = \sin(\sin 0)$, continuity tells us that the cosine curve lies strictly above the sine curve everywhere. They come closest (a gap of about 0.107) near $x \approx 0.88$, where $\sin(\sin x)$ is nearly at its peak and $\cos(\cos x)$ is still climbing.



Sketching $y = \sin(2\sin x)$

This again has period 2π and is odd, with zeros at $x = n\pi$, so it is enough to understand $0 \le x \le \pi$, where it is symmetric about $x = \frac{\pi}{2}$ (because $\sin(\pi - x) = \sin x$).

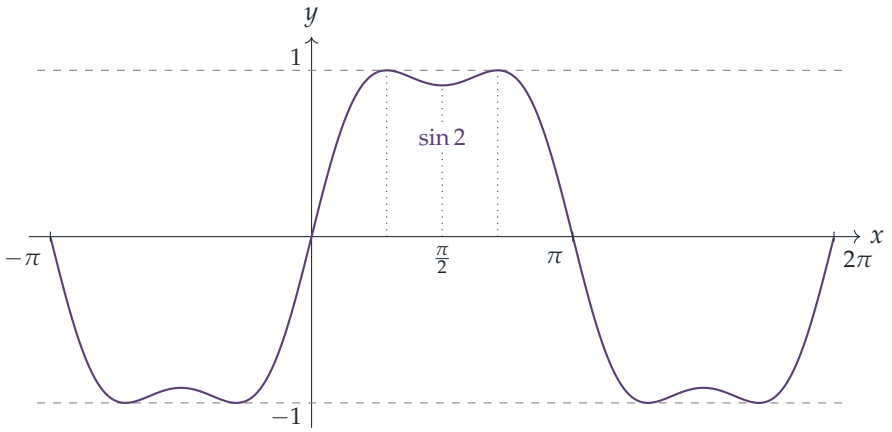
The difference now is that the inner argument $u = 2\sin x$ runs over $[0, 2]$ on this interval, and $2 > \frac{\pi}{2}$. So as x increases from 0 to $\frac{\pi}{2}$:

- u climbs from 0 to $\frac{\pi}{2}$ while $\sin x$ climbs to $\frac{\pi}{4}$, at $x = \arcsin\frac{\pi}{4} \approx 0.903$; meanwhile $y = \sin u$ rises from 0 to its maximum 1;
- u carries on from $\frac{\pi}{2}$ to 2, where $\sin u$ is decreasing, so y falls back to $\sin 2 \approx 0.909$ at $x = \frac{\pi}{2}$.

By symmetry, y rises again to 1 at $x = \pi - \arcsin\frac{\pi}{4} \approx 2.238$ and falls to 0 at $x = \pi$. Checking with the derivative,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 2\cos x \cos(2\sin x),$$

which is zero exactly when $\cos x = 0$ (the local minimum at $\frac{\pi}{2}$) or $2\sin x = \pm\frac{\pi}{2}$ (the maxima of 1); the gradient at the origin is 2. On $[\pi, 2\pi]$ the picture is reflected: two minima of -1 with a local maximum of $-\sin 2$ at $x = \frac{3\pi}{2}$ between them. Each hump of the sine curve has become a camel's back with a shallow dip.



Checks

- Take $B = 0.3$ and $n = 1$: $A = \frac{5\pi}{2} - 0.3$ gives $\sin A = \cos 0.3 \approx 0.9553$, as it should.
- Evaluating $\cos(\cos x) - \sin(\sin x)$ on a fine grid over a period gives a minimum of about 0.107 (near $x \approx 0.878$), never zero, which confirms that the curves don't meet.
- For $\sin(2\sin x)$, a numerical scan of $[0, \pi]$ finds exactly three turning points: maxima of 1 at $x \approx 0.903$ and 2.238, and a minimum of $\sin 2 \approx 0.909$ at $\frac{\pi}{2}$.

Common slips

- Writing only $A = \frac{\pi}{2} - B + 2n\pi$ and forgetting the second family of solutions $A = \frac{\pi}{2} + B + 2n\pi$, which comes from $\sin(\pi - C) = \sin C$.
- Forgetting that $n = 0$ is allowed and checking only $|4n + 1| \geq 5$: the crucial comparison is with $\frac{\pi}{2}$ itself.
- Drawing $\sin(2\sin x)$ as a plain sine curve of height 1: because $2 > \frac{\pi}{2}$, the inner argument overshoots the peak of sine, which causes the dip at $x = \frac{\pi}{2}$.

Beyond STEP

The sketches here are all about composition: the shape of $f(g(x))$ depends on which part of the graph of f the range of g covers. Iterating this idea ($\sin(\sin(\sin x))$, and so on) leads to fixed points and the dynamics of iterated maps, where one asks when $f(f(\cdots f(x)))$ settles down. The inequality argument, comparing the possible ranges of two sides, is a simple example of a technique used throughout analysis; Abbott's *Understanding Analysis* is a good place to see such arguments made rigorous.

WORKED SOLUTION

STEP III 2005, Question 2

Pure • Separable Differential Equations

STEP III 2005, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q2

The key idea

The differential equation is separable, and the integration gives logarithms that combine into the stated form. For the distance question, minimise the *square* of the distance, $D = x^2 + y^2$, which avoids square roots. On the curve y^2 is an explicit function of x, so D becomes a function of x alone, and its derivative factorises neatly. Which stationary points appear depends on how c compares with a^2.

Solution

The general solution

Separating the variables (for $y \neq 0$),

$$\int \frac{1}{y}\,\mathrm{d}y = -\int \frac{x}{x^2+a^2}\,\mathrm{d}x \quad \Longrightarrow \quad \ln|y| = -\tfrac{1}{2}\ln(x^2+a^2) + k,$$

where k is a constant. Note that $x^2 + a^2 > 0$ for every x because $a \neq 0$, so the logarithm on the right is defined everywhere. Doubling and taking exponentials,

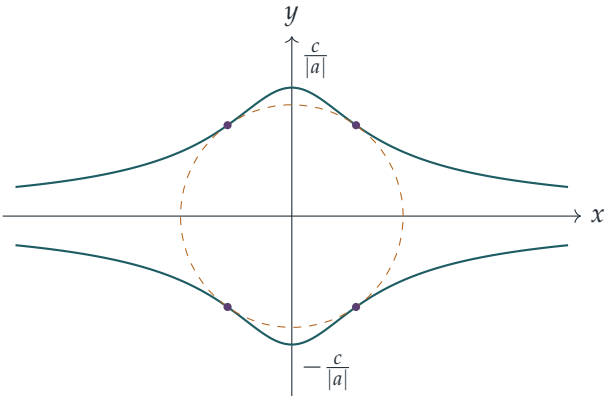
$$y^2 = \frac{\mathrm{e}^{2k}}{x^2+a^2}, \qquad \text{so} \qquad y^2(x^2+a^2) = c^2,$$

with $c^2 = \mathrm{e}^{2k}$. The solution $y = 0$, which we lost when dividing by y, is the case $c = 0$, so c is indeed an arbitrary constant. Explicitly, the general solution is $y = c/\sqrt{x^2+a^2}$, where c may have either sign. Since only c^2 appears in the curve's equation, we take $c > 0$ from now on.

Sketching the curve

The curve $y^2(x^2+a^2) = c^2$ consists of the two branches $y = \pm c/\sqrt{x^2+a^2}$. It is symmetric in both axes (only x^2 and y^2 appear) and never meets the x-axis. On the upper branch, y is greatest when $x^2 + a^2$ is least, giving the maximum $c/|a|$ at $x = 0$.

As $|x| \to \infty$ we have $y \approx \pm c/|x| \to 0$, so the x-axis is an asymptote. The figure shows the case $a = 1$, $c = 2$, together with the circle that we'll find in part (ii).



The derivatives of $x^2 + y^2$

Let $D = x^2 + y^2$. Using the differential equation and then $y^2 = c^2/(x^2 + a^2)$,

$$\frac{\mathrm{d}D}{\mathrm{d}x} = 2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - \frac{2xy^2}{x^2 + a^2} = 2x\left(1 - \frac{c^2}{(x^2 + a^2)^2}\right).$$

This holds on both branches, since it involves only y^2. Differentiating again with the product rule, and using $\frac{\mathrm{d}}{\mathrm{d}x}(x^2 + a^2)^{-2} = -4x(x^2 + a^2)^{-3}$,

$$\frac{\mathrm{d}^2D}{\mathrm{d}x^2} = 2\left(1 - \frac{c^2}{(x^2 + a^2)^2}\right) + 2x \cdot \frac{4c^2x}{(x^2 + a^2)^3} = 2\left(1 - \frac{c^2}{(x^2 + a^2)^2}\right) + \frac{8c^2x^2}{(x^2 + a^2)^3},$$

as required.

The stationary points of D are where $x = 0$ or $(x^2 + a^2)^2 = c^2$. Since $x^2 + a^2 > 0$ and $c > 0$, the second condition means $x^2 + a^2 = c$, that is, $x^2 = c - a^2$.

Part (i)

Suppose $0 < c < a^2$. For every x we have $x^2 + a^2 \geq a^2 > c$, so $(x^2 + a^2)^2 > c^2$ and the bracket in $\mathrm{d}D/\mathrm{d}x$ is strictly positive. Hence $\mathrm{d}D/\mathrm{d}x$ has the same sign as x: D decreases for $x < 0$ and increases for $x > 0$. So D has a *global* minimum at $x = 0$ and nowhere else. (Consistently, the second derivative there is $2(1 - c^2/a^4) > 0$.)

At $x = 0$, $y^2 = c^2/a^2$, so the nearest points are

$$\left(0,\ \pm\frac{c}{a}\right),$$

at distance $c/|a|$ from the origin. This is the required result.

Part (ii)

Now suppose $c > a^2$. There are three stationary points: $x = 0$ and $x = \pm\sqrt{c - a^2}$.

Ask yourself Is $x = 0$ still the nearest point?

No. At $x = 0$ the second derivative is $2(1 - c^2/a^4)$, which is negative because $c > a^2 > 0$ gives $c^2 > a^4$. So $x = 0$ is now a local *maximum* of D. At $x^2 = c - a^2$ we have $x^2 + a^2 = c$, so the bracket vanishes and

$$\frac{\mathrm{d}^2 D}{\mathrm{d}x^2} = \frac{8c^2(c - a^2)}{c^3} = \frac{8(c - a^2)}{c} > 0,$$

giving local minima. To see that they are global, look at the sign of $\mathrm{d}D/\mathrm{d}x$. For $x > 0$ the bracket is negative when $x^2 < c - a^2$ and positive when $x^2 > c - a^2$, so on $x \geq 0$ the function D decreases to $x = \sqrt{c - a^2}$ and increases after it. Hence D is least on $x \geq 0$ at $x = \sqrt{c - a^2}$, and by symmetry it is least on $x \leq 0$ at $x = -\sqrt{c - a^2}$.

At these points $y^2 = c^2/(x^2 + a^2) = c^2/c = c$. So the points on the curve nearest the origin are the four points

$$\left(\pm\sqrt{c - a^2},\ \pm\sqrt{c}\right)$$

(all four sign combinations), at distance $\sqrt{2c - a^2}$ from the origin.

As a cross-check, the distance squared at $x = 0$ is c^2/a^2, and

$$\frac{c^2}{a^2} - (2c - a^2) = \frac{(c - a^2)^2}{a^2} > 0,$$

so the points on the y-axis are indeed farther away.

Checks

- Substituting $y = c/\sqrt{x^2 + a^2}$ back into the differential equation works: $y' = -cx(x^2 + a^2)^{-3/2} = -xy/(x^2 + a^2)$.
- With $a = 1$, $c = 2$, a brute-force search of $x^2 + 4/(x^2 + 1)$ finds the least value $3 = 2c - a^2$ at $x = \pm 1 = \pm\sqrt{c - a^2}$, with $y^2 = 2 = c$. These are the four dots on the figure, where the dashed circle of radius $\sqrt{3}$ touches the curve.
- At $c = a^2$ the two answers agree: $\sqrt{c - a^2} = 0$ and $\sqrt{c} = c/|a| = |a|$, so the four points merge into $(0, \pm|a|)$.

Common slips

- Concluding from "$\mathrm{d}D/\mathrm{d}x = 0$ and $\mathrm{d}^2D/\mathrm{d}x^2 > 0$" that a point is the *least* distance. That only gives a local minimum; the sign of $\mathrm{d}D/\mathrm{d}x$ settles the global question.
- Forgetting the lower branch, and so giving only two points in part (ii) instead of four.
- Writing $x^2 + a^2 = \pm c$: the negative option is impossible, since $x^2 + a^2 > 0$.

Beyond STEP

Minimising $x^2 + y^2$ subject to $y^2(x^2 + a^2) = c^2$ is a constrained optimisation problem. At university you'd meet Lagrange multipliers, which handle such problems without solving for one variable: at the nearest point, the curve's normal points along the radius, which is why the dashed circle touches the curve. The way the single minimum at $x = 0$ splits into two as c passes a^2 is a simple example of a *pitchfork bifurcation*. For more on separable equations and their solution curves, see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 2005, Question 3

Pure • Composition of Quadratics

STEP III 2005, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q3

The key idea

Composing two monic quadratics gives a monic quartic whose coefficients are tied together: four parameters p, q, r, s have to produce four coefficients a, b, c, d, but p and s only ever appear in the combination $2s + p$ in two of them, so one relation must hold. The second form is really the same thing in disguise: completing the square on the *outer* quadratic turns $\mathrm{f}(\mathrm{g}(x))$ into (a square) $-$ (a constant). That form is exactly what lets us solve the final quartic with nothing more than the quadratic formula.

Solution

Composing the quadratics

Substituting $\mathrm{g}(x)$ for x in f,

$$\begin{aligned}\mathrm{f}(\mathrm{g}(x)) &= (x^2 + rx + s)^2 + p(x^2 + rx + s) + q \\ &= x^4 + 2rx^3 + (r^2 + 2s + p)x^2 + r(2s + p)x + (s^2 + ps + q).\end{aligned}$$

So $x^4 + ax^3 + bx^2 + cx + d = \mathrm{f}(\mathrm{g}(x))$ for all x exactly when

$$a = 2r, \qquad b = r^2 + (2s + p), \qquad c = r(2s + p), \qquad d = s^2 + ps + q.$$

Necessity. If such p, q, r, s exist, then $r = \frac{1}{2}a$ and $2s + p = b - r^2 = b - \frac{1}{4}a^2$, so

$$c = \tfrac{1}{2}a\left(b - \tfrac{1}{4}a^2\right), \qquad \text{that is} \qquad a^3 - 4ab + 8c = 0.$$

Sufficiency. Conversely, suppose $a^3 - 4ab + 8c = 0$. Take $r = \frac{1}{2}a$, take s to be anything (say $s = 0$), and then put $p = b - \frac{1}{4}a^2 - 2s$ and $q = d - s^2 - ps$. The first, second and fourth equations hold by construction, and the third reads $c = \frac{1}{2}a(b - \frac{1}{4}a^2)$, which is our condition rearranged. Notice that d is never constrained, since q appears only in the constant term.

So the required necessary and sufficient condition is

$$\boxed{a^3 - 4ab + 8c = 0}$$

(with d arbitrary).

The form $(x^2 + vx + w)^2 - k$

Ask yourself How can a sum like $\mathrm{g}^2 + p\,\mathrm{g} + q$ be rewritten as a single square minus a constant?

Complete the square in g, treating $\mathrm{g}(x)$ as a single variable:

$$\mathrm{f}(\mathrm{g}(x)) = \big(\mathrm{g}(x) + \tfrac{1}{2}p\big)^2 + q - \tfrac{1}{4}p^2 = \big(x^2 + rx + s + \tfrac{1}{2}p\big)^2 - \big(\tfrac{1}{4}p^2 - q\big).$$

Now we can prove both directions.

If the condition holds, the first part gives p, q, r, s with the quartic equal to $\mathrm{f}(\mathrm{g}(x))$, and the display above shows it equals $(x^2 + vx + w)^2 - k$ with

$$v = r, \qquad w = s + \tfrac{1}{2}p, \qquad k = \tfrac{1}{4}p^2 - q.$$

Conversely, $(x^2 + vx + w)^2 - k$ is itself $\mathrm{f}(\mathrm{g}(x))$ with $\mathrm{g}(x) = x^2 + vx + w$ and $\mathrm{f}(x) = x^2 - k$ (that is, $p = 0$, $q = -k$, $r = v$, $s = w$). By the necessity argument of the first part, the condition $a^3 - 4ab + 8c = 0$ must then hold.

This completes the "if and only if". For actual use it helps to have the values explicitly: expanding,

$$(x^2 + vx + w)^2 - k = x^4 + 2vx^3 + (v^2 + 2w)x^2 + 2vw\,x + (w^2 - k),$$

so we need

$$v = \tfrac{1}{2}a, \qquad w = \tfrac{1}{2}\big(b - \tfrac{1}{4}a^2\big) = \tfrac{1}{8}(4b - a^2), \qquad k = w^2 - d,$$

and then the x coefficient is $2vw = \tfrac{1}{2}a(b - \tfrac{1}{4}a^2)$, which equals c precisely when the condition holds.

Solving the quartic

Here $a = -4$, $b = 10$, $c = -12$, $d = 4$, and

$$a^3 - 4ab + 8c = -64 + 160 - 96 = 0,$$

so the method applies. The formulas give

$$v = -2, \qquad w = \tfrac{1}{8}(40 - 16) = 3, \qquad k = 3^2 - 4 = 5,$$

and indeed $(x^2 - 2x + 3)^2 - 5 = x^4 - 4x^3 + 10x^2 - 12x + 4$ (worth expanding once to be sure). The equation becomes

$$(x^2 - 2x + 3)^2 = 5 \quad \Longleftrightarrow \quad x^2 - 2x + 3 = \sqrt{5} \ \text{ or } \ x^2 - 2x + 3 = -\sqrt{5}.$$

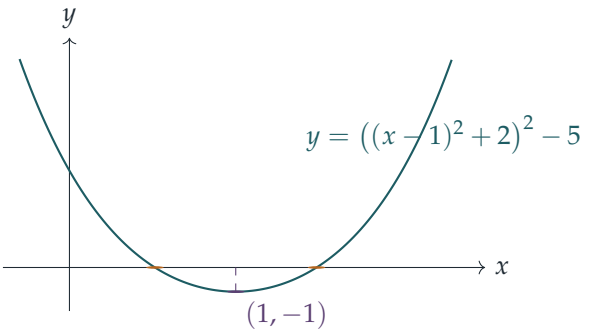
Writing $x^2 - 2x + 3 = (x - 1)^2 + 2$:

- $(x-1)^2 = \sqrt{5}-2$. Since $\sqrt{5} > 2$ the right-hand side is positive, giving two real roots $x = 1 \pm \sqrt{\sqrt{5}-2}$.
- $(x-1)^2 = -2-\sqrt{5} < 0$, giving the complex pair $x = 1 \pm \mathrm{i}\sqrt{2+\sqrt{5}}$.

So the roots are

$$\boxed{x = 1 \pm \sqrt{\sqrt{5}-2}, \qquad x = 1 \pm \mathrm{i}\sqrt{2+\sqrt{5}}}$$

with only the first pair real (if the question is read as asking for real roots, these are the two).



The graph is symmetric about $x = 1$ (it depends only on $(x-1)^2$), with minimum value $2^2 - 5 = -1$ at $x = 1$, so it crosses the axis exactly twice.

Checks

- Numerically, $1 \pm \sqrt{\sqrt{5}-2} \approx 1.4859$ and 0.5141; substituting either into $x^4 - 4x^3 + 10x^2 - 12x + 4$ gives zero to within 10^{-14}, as does $1 \pm 2.0582\,\mathrm{i}$.
- Sum of roots: the four roots add to $4 = -a$, as they must.
- Try a composition you build yourself, e.g. $\mathrm{f}(x) = x^2 + x$, $\mathrm{g}(x) = x^2 + 2x$: you get $x^4 + 4x^3 + 5x^2 + 2x$, and $a^3 - 4ab + 8c = 64 - 80 + 16 = 0$.

Common slips

- Including a condition on d: the constant term can always be matched using q (or k), so d is free.
- Proving only one direction of the "if and only if"; the observation that $(x^2 + vx + w)^2 - k$ is itself of the form $\mathrm{f}(\mathrm{g}(x))$ gives the converse in one line.
- Stopping after the real roots without noticing that the other factor gives a complex pair.

Beyond STEP

The condition $a^3 - 4ab + 8c = 0$ has a neat meaning: substituting $x = y - \frac{1}{4}a$ removes the cubic term, and the condition says exactly that the resulting *depressed* quartic has no y term, so it is a quadratic in y^2. That is the starting point of Ferrari's general solution of the quartic, and more broadly of the question of when a polynomial decomposes as a composition, which leads into Galois theory. Allenby's *Rings, Fields and Groups* is a good place to see the algebra behind which polynomial equations can be solved by radicals.

WORKED SOLUTION

STEP III 2005, Question 4

Pure • Non-linear Recurrence Relations

STEP III 2005, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q4

The key idea

The recurrence is not linear, but it is really about *ratios* of consecutive terms. If we write $u_{n+1} = \rho\, u_n$, the recurrence says the next ratio is $k - \rho$. So the ratios flip back and forth between two values whose sum is k, and everything else follows from that. The three special cases then become conditions on those two ratios.

Solution

The induction

Write $r = \frac{b}{a}$. Dividing the recurrence by u_{n+1} (when it is non-zero) gives

$$\frac{u_{n+2}}{u_{n+1}} = k - \frac{u_{n+1}}{u_n},$$

so each ratio of consecutive terms is k minus the previous ratio. The first ratio is $u_2/u_1 = r$, so the next should be

$$c = k - \frac{b}{a} = \frac{ka - b}{a}.$$

Note that $c \neq 0$, because $b \neq ka$. Also $r \neq 0$, because $b \neq 0$. Together these keep every term non-zero, so we never divide by zero.

Let $P(n)$ be the statement that $u_{2n-1} \neq 0$ and $u_{2n} = r\, u_{2n-1}$. $P(1)$ holds, since $u_1 = a \neq 0$ and $u_2 = b = ra$.

Suppose $P(n)$ holds for some $n \geq 1$. Then $u_{2n} = ru_{2n-1} \neq 0$, and the recurrence gives

$$u_{2n+1} = \frac{u_{2n}}{u_{2n-1}}\left(ku_{2n-1} - u_{2n}\right) = r\,(k-r)\,u_{2n-1} = (k-r)\,u_{2n} = c\,u_{2n},$$

which is the second required relation. Hence $u_{2n+1} \neq 0$, and

$$u_{2n+2} = \frac{u_{2n+1}}{u_{2n}}\left(ku_{2n} - u_{2n+1}\right) = c\,(k-c)\,u_{2n} = (k-c)\,u_{2n+1} = r\,u_{2n+1},$$

because $k - c = r$. That is $P(n+1)$. By induction, for every $n \geq 1$,

$$u_{2n} = \frac{b}{a}\, u_{2n-1}, \qquad u_{2n+1} = c\, u_{2n}, \qquad c = k - \frac{b}{a}.$$

Ask yourself Why put "$u_{2n-1} \neq 0$" into the induction statement?

The recurrence divides by u_n, so the sequence only makes sense if no term is zero. Carrying non-zeroness through the induction proves the terms are defined at the same time as it proves the formulae. This is where both conditions $b \neq 0$ and $b \neq ka$ are used.

Closed forms

Going two steps multiplies by c and then by $\frac{b}{a}$, so $u_{2n+1} = \frac{bc}{a}\, u_{2n-1}$. The odd terms form a geometric progression with first term a and ratio $\frac{bc}{a}$, and each even term is $\frac{b}{a}$ times the odd term before it:

$$u_{2n-1} = a\left(\frac{bc}{a}\right)^{n-1} = \frac{b^{n-1}c^{n-1}}{a^{n-2}}, \qquad u_{2n} = b\left(\frac{bc}{a}\right)^{n-1} = \frac{b^n c^{n-1}}{a^{n-1}}.$$

Part (i)

The ratios of consecutive terms alternate $\frac{b}{a}, c, \frac{b}{a}, c, \ldots$, so the sequence is geometric exactly when $c = \frac{b}{a}$, that is $k - \frac{b}{a} = \frac{b}{a}$:

$$ka = 2b.$$

The common ratio is then $\frac{b}{a}$. This is consistent with $b \neq ka$, because $ka = 2b \neq b$ when $b \neq 0$.

Part (ii)

From the closed forms, $u_{n+2} = \frac{bc}{a}\, u_n$ for every n. So $u_{n+2} = u_n$ for all n exactly when $\frac{bc}{a} = 1$, that is $b(ka - b) = a^2$:

$$kab = a^2 + b^2.$$

The sequence is then $a, b, a, b, \ldots$. If we want the period to be exactly 2 (not a constant sequence) we also need $a \neq b$. If $a = b$ the condition becomes $k = 2$, and every term equals a.

Part (iii)

Similarly $u_{n+4} = \left(\frac{bc}{a}\right)^2 u_n$, so $u_{n+4} = u_n$ for all n exactly when $\left(\frac{bc}{a}\right)^2 = 1$, that is $\frac{bc}{a} = \pm 1$. The value $+1$ is part (ii), where the period is 2 (or 1). For a period of exactly 4 we need $\frac{bc}{a} = -1$, that is $b(ka - b) = -a^2$:

$$kab = b^2 - a^2.$$

The sequence is then a, b, $-a$, $-b$, a, b, $\ldots$. Since $a \neq 0$ we have $-a \neq a$, so it really does have period 4 and not less.

In both (ii) and (iii) the condition $b \neq ka$ holds automatically. Putting $b = ka$ into either condition gives $a^2 = 0$, which is impossible.

Checks

Try $a = 1$, $b = 2$. With $k = 4$ (part (i)) the recurrence gives $1, 2, 4, 8, 16, \ldots$. With $k = \frac{5}{2}$ (part (ii)) it gives $1, 2, 1, 2, \ldots$. With $k = \frac{3}{2}$ (part (iii)) it gives $1, 2, -1, -2, 1, 2, \ldots$. We also ran the recurrence in exact rational arithmetic for about 2000 random choices of a, b and k. Every one matched the closed forms, and the sequence was geometric, 2-periodic or 4-periodic exactly when the conditions above held.

Common slips

- Forgetting to say why no term is ever zero, when the recurrence divides by u_n.
- Giving $\frac{bc}{a} = \pm 1$ as the answer to (iii). The $+1$ case has period 2.
- In (ii), not noticing that $a = b$, $k = 2$ gives a constant sequence.

Beyond STEP

The substitution $\rho_n = u_{n+1}/u_n$ turns this non-linear recurrence into the first-order map $\rho \mapsto k - \rho$. That map is an involution: applying it twice gets you back where you started, which is why the period is 2 in the ratios. Changing variables to make a non-linear recurrence or differential equation simpler is a standard first-year technique, and the study of periodic orbits of maps begins discrete dynamical systems. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2005, Question 5

Pure • Common Tangents to Parabolas

STEP III 2005, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q5

The key idea

Rather than labelling a tangent by its point of contact, label it by its gradient. A parabola has exactly one tangent of each gradient m, and the first part tells us its y-intercept. Two parabolas share a tangent of gradient m precisely when those two intercepts agree, so counting common tangents becomes counting the roots m of a single equation, which is a quadratic when $a_1 \neq a_2$ and linear when $a_1 = a_2$.

Solution

The tangent of gradient m

The gradient of $y = ax^2 + bx + c$ is $2ax + b$. Since $a \neq 0$, this equals m at exactly one point, $x_0 = \dfrac{m-b}{2a}$, with $y_0 = ax_0^2 + bx_0 + c$. The tangent there is $y - y_0 = m(x - x_0)$, that is $y - mx = y_0 - mx_0$. Using $m = 2ax_0 + b$,

$$\begin{aligned} y_0 - mx_0 &= ax_0^2 + bx_0 + c - (2ax_0 + b)x_0 \\ &= c - ax_0^2 = c - a \cdot \frac{(m-b)^2}{4a^2} = c - \frac{(m-b)^2}{4a}, \end{aligned}$$

so the tangent at P is $y - mx = c - \dfrac{(m-b)^2}{4a}$, as required.

The condition for a common tangent

By the first part, each curve has exactly one tangent of gradient m, namely

$$y = mx + k_1, \quad k_1 = c_1 - \frac{(m-b_1)^2}{4a_1}, \qquad \text{and} \qquad y = mx + k_2, \quad k_2 = c_2 - \frac{(m-b_2)^2}{4a_2}.$$

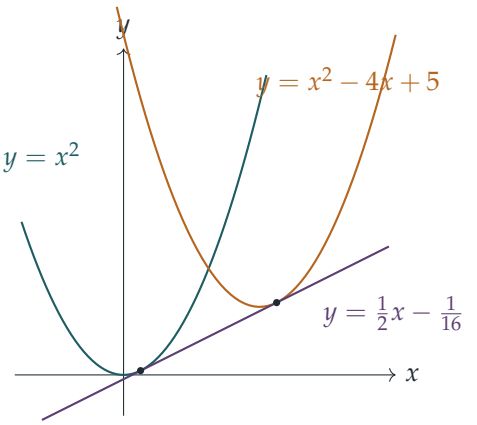
Two lines of the same gradient coincide exactly when their intercepts are equal, so the curves have a common tangent of gradient m if and only if $k_1 = k_2$. Multiplying by

$4a_1a_2 \neq 0$ (which doesn't change whether the equation holds),

$$\begin{aligned} 4a_1a_2(k_2 - k_1) &= 4a_1a_2(c_2 - c_1) - a_1(m - b_2)^2 + a_2(m - b_1)^2 \\ &= (a_2 - a_1)m^2 + 2(a_1b_2 - a_2b_1)m + 4a_1a_2(c_2 - c_1) + a_2b_1^2 - a_1b_2^2. \end{aligned}$$

Hence there is a common tangent of gradient m if and only if this expression is zero, which is the required condition. Call the left-hand side of that condition $Q(m)$.

Since each root m of $Q(m) = 0$ gives exactly one common tangent, and different roots give different lines (they have different gradients), *the number of common tangents equals the number of distinct real roots of* $Q(m) = 0$. Note also that no common tangent can be vertical, since neither curve has a vertical tangent.



Two parabolas with $a_1 = a_2$ and $b_1 \neq b_2$ have exactly one common tangent; here it has gradient $\frac{1}{2}$.

The case $a_1 \neq a_2$

Now Q is a genuine quadratic in m, so it has exactly one distinct real root if and only if its discriminant is zero. The discriminant is

$$\Delta = 4(a_1b_2 - a_2b_1)^2 - 4(a_2 - a_1)\big(4a_1a_2(c_2 - c_1) + a_2b_1^2 - a_1b_2^2\big).$$

The terms without c's simplify nicely:

$$\begin{aligned} &(a_1b_2 - a_2b_1)^2 - (a_2 - a_1)(a_2b_1^2 - a_1b_2^2) \\ &= \big(a_1^2b_2^2 - 2a_1a_2b_1b_2 + a_2^2b_1^2\big) - \big(a_2^2b_1^2 - a_1a_2b_2^2 - a_1a_2b_1^2 + a_1^2b_2^2\big) \\ &= a_1a_2\big(b_1^2 - 2b_1b_2 + b_2^2\big) = a_1a_2(b_1 - b_2)^2. \end{aligned}$$

Therefore

$$\Delta = 4a_1a_2\Big[(b_1 - b_2)^2 - 4(a_1 - a_2)(c_1 - c_2)\Big],$$

and since $a_1a_2 \neq 0$, there is exactly one common tangent if and only if

$$(b_1 - b_2)^2 = 4(a_1 - a_2)(c_1 - c_2). \qquad (*)$$

Ask yourself What exactly does it mean for the two curves to *touch*, and how can we turn that into an equation?

The curves touch when they meet at a point and have the same gradient there. Let

$$d(x) = (a_1 - a_2)x^2 + (b_1 - b_2)x + (c_1 - c_2),$$

the difference of the two quadratics. The curves meet at x_0 when $d(x_0) = 0$, and have equal gradients there when $d'(x_0) = 0$. So they touch if and only if d has a root x_0 with $d'(x_0) = 0$, that is, a repeated root. As $a_1 \neq a_2$, d is a quadratic, and it has a repeated root exactly when its discriminant $(b_1 - b_2)^2 - 4(a_1 - a_2)(c_1 - c_2)$ is zero, which is condition $(*)$.

So, for $a_1 \neq a_2$, the curves have exactly one common tangent if and only if $(*)$ holds, if and only if they touch. (In that case the single common tangent is, naturally, the tangent at the point of contact.)

The case $a_1 = a_2$

Write a for the common value. The m^2 term of Q vanishes and, dividing by $a \neq 0$, the condition becomes

$$2(b_2 - b_1)m + 4a(c_2 - c_1) + (b_1 - b_2)(b_1 + b_2) = 0.$$

- If $b_1 \neq b_2$, this linear equation has exactly one root,

  $$m = \frac{b_1 + b_2}{2} - \frac{2a(c_2 - c_1)}{b_2 - b_1},$$

  so there is exactly one common tangent.
- If $b_1 = b_2$, it reduces to $4a(c_2 - c_1) = 0$. If $c_1 \neq c_2$ there is no common tangent at all (one curve is the other shifted vertically, so their tangents of each gradient are parallel and distinct); if $c_1 = c_2$ the curves are identical and every tangent is common, giving infinitely many.

Hence, when $a_1 = a_2$, the curves have exactly one common tangent if and only if $b_1 \neq b_2$.

Checks

For $y = x^2$ and $y = -x^2 + 2x - \frac{1}{2}$, the difference is $2(x - \frac{1}{2})^2$, so they touch at $x = \frac{1}{2}$; here $Q(m) = -2m^2 + 4m - 2 = -2(m - 1)^2$, with the single root $m = 1$, the gradient of $y = x^2$ at $x = \frac{1}{2}$. For the pictured pair $y = x^2$ and $y = x^2 - 4x + 5$, the formula gives $m = -2 + \frac{5}{2} = \frac{1}{2}$, and both tangent intercepts come out as $-\frac{1}{16}$. We also checked the identity $Q(m) = 4a_1a_2(k_2 - k_1)$ and the discriminant simplification with exact arithmetic on thousands of random coefficient sets.

Common slips

- Claiming that $Q = 0$ having a repeated root "means" the curves touch, without explaining why: the link goes through the *different* quadratic $d(x)$, and the discriminant identity is what connects them.
- In the case $a_1 = a_2$, forgetting the sub-case $b_1 = b_2$, $c_1 = c_2$ (identical curves, infinitely many common tangents), or answering "$b_1 \neq b_2$ or $c_1 \neq c_2$".

Beyond STEP

Describing a curve by its family of tangent lines, rather than its points, is the idea of *duality* in projective geometry: the tangents $y = mx + k$ of a conic satisfy their own quadratic relation between m and k, and two conics in general position share four common tangents, just as they meet in four points. For two parabolas with vertical axes, the line at infinity touches both and uses up two of those four, which is why at most two ordinary common tangents remain. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 2005, Question 6

Pure • Solving Cubic Equations

STEP III 2005, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q6

The key idea

The identity $4\cosh^3 y - 3\cosh y = \cosh 3y$ has exactly the shape of a "depressed" cubic $x^3 - 3bx$, once we scale x by the right constant. So a cubic with no x^2 term can be solved by matching it to the triple-angle formula, and the arcosh logarithm then turns the hyperbolic answer back into cube roots. Once one root is known, factorising gives a quadratic for the other two, and the cube roots of unity tidy them up.

Solution

The hyperbolic root

Put $x = 2a\cosh\frac{1}{3}T$. Then

$$\begin{aligned} x^3 - 3a^2x &= 8a^3\cosh^3\tfrac{1}{3}T - 6a^3\cosh\tfrac{1}{3}T \\ &= 2a^3\left(4\cosh^3\tfrac{1}{3}T - 3\cosh\tfrac{1}{3}T\right) \\ &= 2a^3\cosh T, \end{aligned}$$

using the given identity with $y = \frac{1}{3}T$. So $2a\cosh\frac{1}{3}T$ satisfies the equation, as required.

One root of $x^3 - 3bx = 2c$

Suppose first that $c > 0$. Since $b > 0$ we may take $a = \sqrt{b} > 0$, so that $3a^2 = 3b$, and we want $2a^3\cosh T = 2c$, that is,

$$\cosh T = \frac{c}{b^{3/2}}.$$

This needs $c/b^{3/2} \geq 1$, which is exactly where the condition is used: $c^2 \geq b^3$ with $c > 0$ gives $c \geq b^{3/2}$. So we may take

$$T = \operatorname{arcosh}\frac{c}{b^{3/2}} = \ln\left(\frac{c}{b^{3/2}} + \sqrt{\frac{c^2}{b^3} - 1}\right) = \ln\frac{c + \sqrt{c^2 - b^3}}{b^{3/2}}.$$

Hence
$$e^{T/3} = \frac{(c+\sqrt{c^2-b^3})^{1/3}}{b^{1/2}} = \frac{u}{\sqrt{b}}, \qquad e^{-T/3} = \frac{\sqrt{b}}{u},$$
and by the first part one root is
$$2\sqrt{b}\cosh\tfrac{1}{3}T = \sqrt{b}\left(\frac{u}{\sqrt{b}} + \frac{\sqrt{b}}{u}\right) = u + \frac{b}{u}.$$

Ask yourself What if $c < 0$? Then $c/b^{3/2} \le -1$ and no real T has that cosh.

Replace x by $-x$: the equation becomes $x^3 - 3bx = -2c$ with $-c > 0$, so by the case just done it has the root $u' + b/u'$ where $u' = (-c+\sqrt{c^2-b^3})^{1/3}$, and $-(u'+b/u')$ is a root of the original equation. Now
$$u^3u'^3 = \left(\sqrt{c^2-b^3}+c\right)\left(\sqrt{c^2-b^3}-c\right) = (c^2-b^3) - c^2 = -b^3,$$
so, taking real cube roots, $u' = -b/u$ and $-(u'+b/u') = u + b/u$. So the formula holds for either sign of c. (Note $u \neq 0$ in both cases, since $\sqrt{c^2-b^3} < |c|$ when $b > 0$.)

The other two roots

Write $r = u + \frac{b}{u}$. Since r is a root, $x - r$ is a factor, and comparing coefficients in
$$x^3 - 3bx - 2c = (x-r)(x^2+rx+q) = x^3 + (q-r^2)x - rq$$
(the x^2 terms cancel automatically) forces $q = r^2 - 3b$. Expanding,
$$q = u^2 + 2b + \frac{b^2}{u^2} - 3b = u^2 + \frac{b^2}{u^2} - b,$$
so the other two roots satisfy
$$x^2 + \left(u + \frac{b}{u}\right)x + u^2 + \frac{b^2}{u^2} - b = 0,$$
as required. (As a consistency check, the constant term also matches: $rq = u^3 + b^3/u^3$, and $b^3/u^3 = b^3/(c+\sqrt{c^2-b^3}) = c - \sqrt{c^2-b^3}$ after rationalising, so $rq = 2c$.)

To solve the quadratic, write $v = b/u$, so $uv = b$ and the quadratic is
$$x^2 + (u+v)x + (u^2 - uv + v^2) = 0.$$
Now $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. Try $x_1 = \omega u + \omega^2 v$ and $x_2 = \omega^2 u + \omega v$:
$$x_1 + x_2 = (\omega+\omega^2)(u+v) = -(u+v),$$
$$x_1x_2 = \omega^3u^2 + \omega^2uv + \omega^4uv + \omega^3v^2 = u^2 + (\omega^2+\omega)uv + v^2 = u^2 - uv + v^2.$$
These are the sum and product of the roots of the quadratic, so its roots are
$$x = \omega u + \omega^2\frac{b}{u} \qquad \text{and} \qquad x = \omega^2 u + \omega\frac{b}{u}.$$
Since $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $\omega^2 = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$, these can also be written $-\frac{1}{2}\left(u + \frac{b}{u}\right) \pm \frac{\sqrt{3}}{2}\left(u - \frac{b}{u}\right)\mathrm{i}$.

Solving $x^3 - 6x = 6$

This is $x^3 - 3bx = 2c$ with $b = 2$, $c = 3$, and $c^2 = 9 \geq 8 = b^3 > 0$, so the results apply. Here

$$u = \left(3 + \sqrt{9-8}\right)^{1/3} = 4^{1/3} = 2^{2/3}, \qquad \frac{b}{u} = \frac{2}{2^{2/3}} = 2^{1/3}.$$

So the roots are

$$x = 2^{2/3} + 2^{1/3}, \qquad x = -\tfrac{1}{2}\left(2^{2/3} + 2^{1/3}\right) \pm \tfrac{\sqrt{3}}{2}\left(2^{2/3} - 2^{1/3}\right)\mathrm{i},$$

the complex pair being $\omega\, 2^{2/3} + \omega^2\, 2^{1/3}$ and its conjugate.

Checks

- The real root is $2^{2/3} + 2^{1/3} \approx 2.847322$, and $2.847322^3 - 6(2.847322) \approx 23.083930 - 17.083932 \approx 6.0000$.
- The three roots should sum to 0 (no x^2 term): $(u+v) - (u+v) = 0$. Their product should be 6: $r(u^2 - uv + v^2) = u^3 + v^3 = 4 + 2 = 6$.
- If $c^2 = b^3$ then $u = \sqrt{b} = b/u$, and the two complex roots both become $-\sqrt{b}$, a repeated root, as $x^3 - 3bx - 2b^{3/2} = (x - 2\sqrt{b})(x + \sqrt{b})^2$ confirms.

Common slips

- Forgetting that $\cosh T \geq 1$, so the hyperbolic argument only works directly for $c > 0$; the case $c < 0$ needs the substitution $x \to -x$.
- Writing the complex roots as $\omega u + \omega \frac{b}{u}$: the product of the two cube roots must stay equal to b, so the powers of ω must multiply to 1.

Beyond STEP

This is Cardano's formula for the cubic, and the form $u + b/u$ with u^3 a root of a quadratic is how it was first found in the sixteenth century. When $c^2 < b^3$ the square root is imaginary even though all three roots are real (the "irreducible case"), and the trigonometric version $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ takes over. Why no formula of this kind exists for the general quintic is the story of Galois theory; Allenby, *Rings, Fields and Groups*, is a good way in.

WORKED SOLUTION

STEP III 2005, Question 7

Pure • Integration by Substitution

STEP III 2005, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q7

The key idea

The opening result says that a factor of x^m hidden inside an integrand can be treated as a single new variable $u = x^m$, as long as the integrand also has a lone factor of $\frac{1}{x}$ outside. So for each integral we pull out a factor of x from the denominator, read off which power x^m is left and what the function f is, and do the much easier u-integral instead.

Solution

The general result

We are told that $\mathrm{F}'(u) = \dfrac{1}{u\,\mathrm{f}(u)}$. Differentiating $\mathrm{F}(x^m)$ by the chain rule (for $x > 0$, so that x^m makes sense for any real m),

$$\frac{\mathrm{d}}{\mathrm{d}x}\mathrm{F}(x^m) = \mathrm{F}'(x^m)\cdot mx^{m-1} = \frac{mx^{m-1}}{x^m\,\mathrm{f}(x^m)} = \frac{m}{x\,\mathrm{f}(x^m)}.$$

So $\mathrm{F}(x^m)$ is an antiderivative of $\dfrac{m}{x\,\mathrm{f}(x^m)}$, which is exactly the result required:

$$\int \frac{m}{x\,\mathrm{f}(x^m)}\,\mathrm{d}x = \mathrm{F}(x^m) + c.$$

Equivalently, this is the substitution $u = x^m$, $\mathrm{d}u = mx^{m-1}\,\mathrm{d}x$, read backwards. The condition $m \neq 0$ matters in what follows: we shall divide by m.

In practice we use it as

$$\int \frac{1}{x\,\mathrm{f}(x^m)}\,\mathrm{d}x = \frac{1}{m}\,\mathrm{F}(x^m) + c. \qquad (*)$$

Part (i)

Taking out a factor of x,

$$\frac{1}{x^n - x} = \frac{1}{x\left(x^{n-1} - 1\right)}.$$

This has the form in $(*)$ with $m = n - 1$ and $\mathrm{f}(u) = u - 1$. We need $n \neq 1$; when $n = 1$ the denominator $x^n - x$ is identically zero and there is no integral to find.

Now we need F, an antiderivative of $\dfrac{1}{u(u-1)}$. Partial fractions give

$$\frac{1}{u(u-1)} = \frac{1}{u-1} - \frac{1}{u}, \qquad \text{so} \qquad \mathrm{F}(u) = \ln|u-1| - \ln|u| = \ln\left|\frac{u-1}{u}\right|.$$

Putting $u = x^{n-1}$ into $(*)$:

$$\boxed{\int \frac{1}{x^n - x}\,\mathrm{d}x = \frac{1}{n-1}\ln\left|\frac{x^{n-1}-1}{x^{n-1}}\right| + c, \qquad n \neq 1.}$$

The modulus signs let this hold on any interval where the integrand is defined (for instance on $0 < x < 1$ as well as $x > 1$; and for $x < 0$ when n is an integer).

Part (ii)

Here we must be a little careful taking x out of the square root. We work with $x > 0$, where x^n is defined for every n and $\sqrt{x^2} = x$. Then

$$\sqrt{x^n + x^2} = \sqrt{x^2\left(x^{n-2} + 1\right)} = x\sqrt{1 + x^{n-2}},$$

so the integrand is

$$\frac{1}{x\sqrt{1 + x^{n-2}}},$$

which has the form in $(*)$ with $m = n - 2$ and $\mathrm{f}(u) = \sqrt{1+u}$.

Ask yourself What happens when $n = 2$?

Then $m = 0$ and the general result does not apply. But the integral is easy directly: the integrand is $\dfrac{1}{\sqrt{2x^2}} = \dfrac{1}{\sqrt{2}\,x}$, so

$$\int \frac{1}{\sqrt{x^2 + x^2}}\,\mathrm{d}x = \frac{1}{\sqrt{2}}\ln x + c \qquad (n = 2).$$

For $n \neq 2$ we need $\mathrm{F}(u) = \displaystyle\int \frac{1}{u\sqrt{1+u}}\,\mathrm{d}u$, where $u = x^{n-2} > 0$. Substitute $t = \sqrt{1+u}$, so $u = t^2 - 1$, $\mathrm{d}u = 2t\,\mathrm{d}t$ and $t > 1$:

$$\int \frac{1}{u\sqrt{1+u}}\,\mathrm{d}u = \int \frac{2t}{(t^2-1)\,t}\,\mathrm{d}t = \int\left(\frac{1}{t-1} - \frac{1}{t+1}\right)\mathrm{d}t = \ln\left(\frac{t-1}{t+1}\right),$$

with no modulus needed since $t > 1$. This tidies up nicely: multiplying top and bottom by $t-1$ and using $t^2 - 1 = u$,

$$\frac{t-1}{t+1} = \frac{(t-1)^2}{t^2-1} = \frac{\left(\sqrt{1+u}-1\right)^2}{u}, \qquad \text{so} \qquad \mathrm{F}(u) = 2\ln\left(\sqrt{1+u}-1\right) - \ln u.$$

Now put $u = x^{n-2}$ into $(*)$ with $m = n-2$. The term $-\ln u$ becomes $-(n-2)\ln x$, and dividing by $m = n-2$ turns it into $-\ln x$:

$$\boxed{\int \frac{1}{\sqrt{x^n + x^2}}\,\mathrm{d}x = \frac{2}{n-2}\ln\left(\sqrt{1+x^{n-2}}-1\right) - \ln x + c, \qquad n \neq 2,\ x > 0.}$$

Equivalently, the answer is $\dfrac{1}{n-2}\ln\left(\dfrac{\sqrt{1+x^{n-2}}-1}{\sqrt{1+x^{n-2}}+1}\right) + c$; either form is fine.

Checks

- Differentiate the answers back. For (i) with $n = 2$ the formula gives $\ln|x-1| - \ln|x|$, whose derivative is $\frac{1}{x-1} - \frac{1}{x} = \frac{1}{x^2-x}$, as it should be.
- For (i) with $n = 3$, the formula gives $\displaystyle\int_2^3 \frac{\mathrm{d}x}{x^3 - x} = \tfrac{1}{2}\ln\tfrac{32}{27} \approx 0.084950$, matching Simpson's rule.
- For (ii) with $n = 3$, the formula gives $\displaystyle\int_1^3 \frac{\mathrm{d}x}{\sqrt{x^3 + x^2}} = \ln\frac{3+2\sqrt{2}}{3} \approx 0.664135$, again matching numerical integration. We also tested both formulae for many values of n, including negative and non-integer ones.

Common slips

- Forgetting the factor $\frac{1}{m}$: the general result has m in the numerator, and our integrands don't, so we must divide by m.
- Writing $\sqrt{x^2} = x$ without comment. It is true only for $x \geq 0$; for $x < 0$ it is $-x$ (or $|x|$).
- Missing the exceptional values $n = 1$ in (i) and $n = 2$ in (ii), where $m = 0$.

Beyond STEP

The opening result is the substitution rule seen from the other side: substitution is just the chain rule run backwards, and that is how it is proved in a first analysis course, with careful hypotheses on the functions involved. Integrals such as (ii), involving square roots of polynomials, lead to the question of which integrals can be done in elementary functions at all; $\int \mathrm{d}x/\sqrt{1+x^4}$, for instance, cannot. For a careful treatment

of the fundamental theorem of calculus and substitution, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 2005, Question 8

Pure • Complex Numbers and Circles

STEP III 2005, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q8

The key idea

Every length in the Argand diagram can be turned into algebra with $|z|^2 = zz^*$. Once the first identity is in hand, a right angle becomes Pythagoras and a tangent becomes a right angle between the tangent and the radius. So "OA touches the circle" turns into a single equation, which fixes the radius squared as $cc^* - aa^*$. After that, both "P on the circle" and "P' on the circle" reduce to the same equation.

Solution

The identity and the right angle

Since $|z|^2 = zz^*$ and conjugation respects subtraction,

$$\begin{aligned} |a-c|^2 &= (a-c)(a-c)^* = (a-c)(a^*-c^*) \\ &= aa^* + cc^* - ac^* - ca^*, \end{aligned}$$

as required.

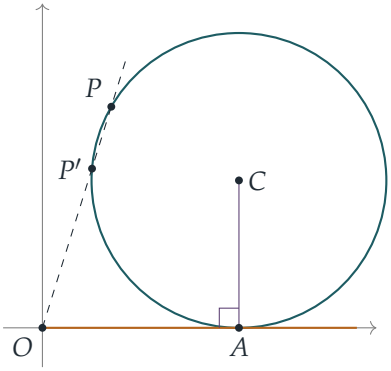
The sides of triangle OAC have lengths $OA = |a|$, $OC = |c|$ and $AC = |a-c|$, all non-zero because a, c are non-zero and distinct. By Pythagoras' theorem and its converse, the angle at A is a right angle if and only if $OC^2 = OA^2 + AC^2$, that is,

$$cc^* = aa^* + \left(aa^* + cc^* - ac^* - ca^*\right) \iff 2aa^* = ac^* + ca^*.$$

Every step here is reversible, so the "if and only if" holds.

(The converse of Pythagoras also rules out a degenerate "triangle". If O, A, C were collinear then one of the three lengths would equal the sum of the other two, and that can't happen alongside $OC^2 = OA^2 + AC^2$ when all three lengths are positive.)

The tangent circle



A tangent is perpendicular to the radius at the point of contact. So OA being tangent at A means that angle OAC is a right angle, and by the first part

$$2aa^* = ac^* + ca^*. \qquad (*)$$

The radius r satisfies $r^2 = |a-c|^2 = aa^* + cc^* - (ac^* + ca^*) = cc^* - aa^*$, using $(*)$. (That's just Pythagoras again: $r^2 = OC^2 - OA^2$.)

Write $k = bb^* > 0$ and $s = abc^* + a^*b^*c$. Then

$$|ab-c|^2 = (ab-c)(a^*b^* - c^*) = aa^*k - s + cc^*,$$
$$\left|\frac{a}{b^*} - c\right|^2 = \left(\frac{a}{b^*} - c\right)\left(\frac{a^*}{b} - c^*\right) = \frac{aa^*}{k} - \frac{ac^*}{b^*} - \frac{a^*c}{b} + cc^*.$$

Multiplying the second by $k = bb^*$ gives

$$k\left|\frac{a}{b^*} - c\right|^2 = aa^* - s + k\,cc^*. \qquad (**)$$

P on the circle. $|ab-c|^2 = cc^* - aa^*$ holds if and only if

$$aa^*k - s + cc^* = cc^* - aa^* \iff aa^*(k+1) = s.$$

P' on the circle. $|a/b^* - c|^2 = cc^* - aa^*$ holds if and only if (multiplying by $k \neq 0$ and using $(**)$)

$$aa^* - s + k\,cc^* = k\,cc^* - k\,aa^* \iff aa^*(k+1) = s.$$

The two conditions are identical, so P lies on the circle if and only if P' does.

The converse

Now drop the tangency assumption. Suppose the circle has centre C and passes through A, so $r^2 = |a-c|^2 = aa^* + cc^* - ac^* - ca^*$. Suppose also that both ab and a/b^* lie on it, where $k = bb^* \neq 1$. Using the same expansions,

$$aa^*k - s + cc^* = r^2,$$
$$aa^* - s + k\,cc^* = kr^2 \qquad \text{(from } (**)\text{)}.$$

Subtracting the second equation from the first eliminates s:

$$(k-1)\,aa^* - (k-1)\,cc^* = (1-k)\,r^2.$$

Ask yourself Why does the question insist on $bb^* \neq 1$?

If $bb^* = 1$ then $b^* = 1/b$, so $a/b^* = ab$ and the two points coincide. "Both points on the circle" is then a single condition, which certainly doesn't force tangency. In our algebra this is exactly the step where we divide by $k - 1 \neq 0$:

$$r^2 = cc^* - aa^*.$$

Comparing with $r^2 = aa^* + cc^* - ac^* - ca^*$ gives $2aa^* = ac^* + ca^*$. By the first part, angle OAC is a right angle. So OA is perpendicular to the radius CA at the point A of the circle, which means OA is a tangent to the circle.

Checks

- There's a neat geometric reason behind the result. Since $\dfrac{a}{b^*} = \dfrac{ab}{bb^*}$, the point P' is P scaled by the positive real $1/|b|^2$. So O, P, P' are collinear, with $OP \cdot OP' = |a|^2 = OA^2$, which is exactly the tangent–secant theorem (see the diagram).
- Numerically, take $a = 2$, $c = 2 + 1.5\mathrm{i}$ (so $r^2 = 2.25 = |c|^2 - |a|^2$), and $P = 0.701 + 2.25\mathrm{i}$ on the circle. Then $b = P/2$ and $P' = a/b^* \approx 0.505 + 1.620\mathrm{i}$, with $|P' - c|^2 \approx 2.250$.
- A computer checked the identity and the tangent case on random a, b, c. It also took 1000 random non-tangent circles with $|b| \neq 1$: in every case, P on the circle forced P' off it.

Common slips

- Writing $(a-c)^* = a^* - c^*$ correctly but then expanding $|a-c|^2$ as $(a-c)^2$. The modulus squared needs the conjugate.
- In the converse, assuming tangency (and so $r^2 = cc^* - aa^*$) at the start. That is what you have to prove. Start from $r = |a - c|$ only.
- Dividing by $1 - bb^*$ without saying why it's non-zero.

Beyond STEP

Since $|a|^2/(ab)^* = aa^*/(a^*b^*) = a/b^*$, the point P' is the image of P under $z \mapsto |a|^2/z^*$. This map is *inversion* in the circle $|z| = |a|$: it sends each point to the point on the same ray from O at distance $|a|^2/OP$. This question shows that circles orthogonal to the circle of inversion are mapped to themselves. Inversion and the Möbius transformations it

generates are central to first-year complex analysis and geometry. Needham's *Visual Complex Analysis* has a beautiful chapter-length treatment.

WORKED SOLUTION

STEP III 2005, Question 9

Mechanics • Collisions and Restitution

STEP III 2005, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

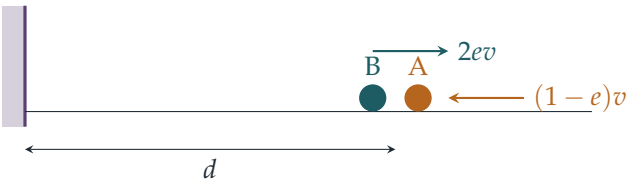
step.maths.org/questions/05-s3-q9

The key idea

The masses look strange, but they have been chosen so that the first collision gives very tidy answers. Once you have the second collision, compare the velocities just before it with the velocities just before the first one: they are the same, but each multiplied by e. Collision laws are linear in the velocities, so the whole story then repeats on a smaller scale, and that gives every later collision at once.

Solution

We model A and B as particles on a smooth horizontal line, with Newton's experimental law at each impact. Take the direction *towards the wall* as positive, and measure distances from the wall.



(The picture shows the instant just before the first impact, with the two particles drawn slightly apart.)

The first collision

Let A and B have velocities v_A and v_B (towards the wall) afterwards. Before, A has velocity $(1-e)v$ and B has velocity $-2ev$.

Conservation of momentum gives

$$\begin{aligned} 4em\,v_A + (1-e)^2 m\,v_B &= 4em(1-e)v - (1-e)^2 m\cdot 2ev \\ &= 2e(1-e)v\big[2-(1-e)\big]m = 2e(1-e)(1+e)v\,m. \end{aligned}$$

The speed of approach is $(1-e)v + 2ev = (1+e)v$, so Newton's law gives

$$v_B - v_A = e(1+e)v.$$

Substituting $v_B = v_A + e(1+e)v$ into the momentum equation and dividing by m:

$$\big[4e + (1-e)^2\big]v_A = 2e(1-e)(1+e)v - (1-e)^2 e(1+e)v = e(1-e)(1+e)v\big[2-(1-e)\big].$$

Since $4e + (1-e)^2 = (1+e)^2$, the right-hand side is $e(1-e)(1+e)^2 v$, and so

$$v_A = e(1-e)v, \qquad v_B = e(1-e)v + e(1+e)v = 2ev.$$

Both are positive: after the collision **A moves towards the wall with speed** $e(1-e)v$ **and B moves towards the wall with speed** $2ev$.

The second collision

B is faster, so it pulls ahead of A and reaches the wall after time $t_1 = d/(2ev)$. In that time A covers $e(1-e)v \cdot \frac{d}{2ev} = \frac{1}{2}(1-e)d$, so A is at distance

$$d - \tfrac{1}{2}(1-e)d = \tfrac{1}{2}(1+e)d$$

from the wall. B rebounds with speed $e \cdot 2ev = 2e^2 v$, now moving away from the wall, while A is still moving towards it with speed $e(1-e)v$. They are approaching each other, with closing speed $e(1-e)v + 2e^2 v = e(1+e)v > 0$, so they must meet again (before A could reach the wall, since B lies between them). This takes a time

$$t_2 = \frac{\frac{1}{2}(1+e)d}{e(1+e)v} = \frac{d}{2ev},$$

during which B travels $2e^2 v \cdot \frac{d}{2ev} = ed$ from the wall. So **a second collision takes place, at distance** de **from the wall**, as required.

Further collisions

> **Ask yourself** How can we deduce that there are more collisions without doing all the work again?

Look at the state just before the second collision: A moves towards the wall with speed $e(1-e)v$ and B moves away from it with speed $2e^2 v = 2e(ev)$. This is exactly the initial state with v replaced by ev and d replaced by de (the masses and e are unchanged). Everything we did above holds for any $v > 0$ and $d > 0$, so the same argument applies with (ev, de) in place of (v, d): there is a third collision, at distance $(de)e = de^2$, and just before it the state is the initial one with $(e^2 v, de^2)$. By induction there is an $(n+1)$th collision for every n, so collisions go on for ever.

Just before the nth collision the state is the initial one with v and d replaced by $e^{n-1}v$ and de^{n-1}, so

$$\text{the } n\text{th collision takes place at distance } de^{n-1} \text{ from the wall.}$$

From the second part, the time from one collision to the next is $t_1 + t_2 = d/(ev)$. Starting from the nth collision, this becomes

$$\frac{de^{n-1}}{e \cdot e^{n-1}v} = \frac{d}{ev},$$

which does not depend on n. So the times between successive collisions are all equal, to $d/(ev)$. The distance and the speeds both shrink by the factor e each cycle, and these two effects cancel in the time.

Checks

- Momentum in the first collision: with $e = \frac{1}{2}$, $m = v = 1$, the masses are 2 and $\frac{1}{4}$; before, $2 \cdot \frac{1}{2} - \frac{1}{4} \cdot 1 = \frac{3}{4}$; after, $2 \cdot \frac{1}{4} + \frac{1}{4} \cdot 1 = \frac{3}{4}$. The relative speed goes from $\frac{3}{2}$ to $\frac{3}{4}$, a factor e.
- We simulated the motion exactly (with fractions) for 200 random choices of e, v, d, m and eight collisions each: every collision was at de^{n-1} and the gaps were all $d/(ev)$. For $e = \frac{1}{2}$, $v = d = 1$ the collisions are at times $0, 2, 4, 6$ and distances $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$.

Common slips

- Mixing up the signs: B starts out moving *away* from the wall, so its initial velocity is negative in our convention.
- Forgetting that the wall also has coefficient e, so B leaves the wall with speed $2e^2v$, not $2ev$.
- Redoing the second collision by brute force instead of noticing the scaling; it works, but it's much longer and the pattern is easy to miss.

Beyond STEP

The deduction here uses linearity and *self-similarity*: the equations of the collisions are unchanged if every velocity is multiplied by the same constant, so a state that reproduces itself up to scale repeats for ever. The same idea underlies the analysis of a bouncing ball with infinitely many bounces, and dimensional analysis more generally. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and momentum with this kind of care.

WORKED SOLUTION

STEP III 2005, Question 10

Mechanics • Elastic Band and Friction

STEP III 2005, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q10

The key idea

Everything depends on one variable: the distance d between the centres. The band's length is fixed by d, so its tension and stored energy are simple functions of d. From there, energy (elastic energy against work done by friction) tells us where the discs stop, and comparing the band's pull with the limiting friction tells us whether they stay stopped.

Solution

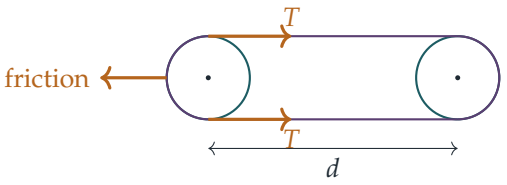
We model the discs as identical, the band as light and elastic with smooth contact, and the friction as having magnitude μmg whenever a disc slides. By symmetry the two discs always move equal distances in opposite directions, so the straight sections of the band stay parallel.

The band as a function of d. When the centres are a distance d apart, the band runs half-way round each disc and along two straight sections of length d. So its length is $2\pi r + 2d$ and its extension is $x = 2d$. By Hooke's law, with modulus $\lambda = \frac{\pi mg}{12}$ and natural length $2\pi r$,

$$T = \frac{\lambda x}{2\pi r} = \frac{\pi mg}{12} \cdot \frac{2d}{2\pi r} = \frac{mg\,d}{12r}, \qquad E = \frac{\lambda x^2}{2 \cdot 2\pi r} = \frac{mg\,d^2}{12r}.$$

Because the contact is smooth, the tension is the same all along the band. Each disc is pulled by the two straight sections, both parallel to the line of centres and pointing towards the other disc. So the resultant pull on each disc is

$$P = 2T = \frac{mg\,d}{6r}.$$



The figure, seen from above, shows the horizontal forces on the left-hand disc while it slides towards the other disc. Its weight and the normal reaction are vertical and cancel.

At release $d = 6r$, so $E_0 = 3mgr$ and $P = mg$. At contact $d = 2r$, so $E = \frac{1}{3}mgr$.

Energy while the discs approach. While the discs move inwards from $d = 6r$, each moves a distance $\frac{1}{2}(6r - d)$ against friction μmg. So the total work done against friction is $\mu mg(6r - d)$. The total kinetic energy is therefore

$$\begin{aligned} K(d) &= 3mgr - \frac{mg\,d^2}{12r} - \mu mg(6r - d) \\ &= mg\,(6r - d)\left(\frac{6r + d}{12r} - \mu\right), \end{aligned}$$

using $3r - \frac{d^2}{12r} = \frac{(6r-d)(6r+d)}{12r}$.

Part (i)

Starting. At release the pull on each disc is $P = mg$ and friction can supply at most μmg. So the discs start sliding exactly when $mg > \mu mg$, that is, when $\mu < 1$.

Stopping. Suppose $\mu < 1$. For $d < 6r$, $K(d)$ vanishes only where $\frac{6r+d}{12r} = \mu$, that is, at

$$d^* = 12\mu r - 6r.$$

As d decreases from $6r$, the second factor $\frac{6r+d}{12r} - \mu$ starts at $1 - \mu > 0$ and decreases, so $K > 0$ until d reaches d^*. The discs come to rest before colliding exactly when d^* is reached before contact, that is, when $2r < d^* < 6r$:

$$8r < 12\mu r < 12r \iff \tfrac{2}{3} < \mu < 1.$$

If $\mu \le \frac{2}{3}$, then $K(2r) \ge 0$ and the discs reach each other first. (When $\mu = \frac{2}{3}$ they arrive just touching with zero speed, which is not "before".)

> **Ask yourself** Once the discs stop at d^*, might the band start them moving again?

At d^* the pull is $P = \frac{mg\,d^*}{6r} = (2\mu - 1)mg$. This is less than the limiting friction μmg because $\mu < 1$, so static friction holds the discs and they stay at rest. Combining the two conditions, the discs start sliding but come to rest before colliding if and only if $\frac{2}{3} < \mu < 1$. ■

Part (ii)

If the discs collide at least once, they must reach $d = 2r$ for the first time moving inwards, with the approach formula still valid. So the total kinetic energy just before the first collision is

$$K(2r) = mg\,(4r)\left(\frac{8r}{12r} - \mu\right) = 4mgr\left(\tfrac{2}{3} - \mu\right) = \tfrac{4}{3}mgr(2 - 3\mu),$$

as required. (This is positive precisely when $\mu < \frac{2}{3}$, consistent with part (i).)

Part (iii)

First, $\mu^2 < \frac{4}{9}$ gives $\mu < \frac{2}{3}$, so by parts (i) and (ii) the discs do collide. Half the kinetic energy is lost, so they rebound from $d = 2r$ with total kinetic energy

$$K_1 = \tfrac{2}{3}mgr(2 - 3\mu).$$

Now the discs move apart. The band and friction both act towards the other disc, so each disc decelerates. While the separation increases from $2r$ to d, the elastic energy rises by $\frac{mg(d^2-4r^2)}{12r}$ and the work done against friction is $\mu mg(d - 2r)$, so

$$K(d) = K_1 - \frac{mg(d^2 - 4r^2)}{12r} - \mu mg(d - 2r).$$

This is strictly decreasing in d, so it vanishes exactly once, at the separation d_1 where the discs come to rest. Writing $d_1 = 2r + u$ and multiplying by $\frac{12r}{mg}$:

$$u^2 + 4r(1 + 3\mu)\,u - 8r^2(2 - 3\mu) = 0.$$

Taking the positive root and simplifying the discriminant, $(1 + 3\mu)^2 + 2(2 - 3\mu) = 9\mu^2 + 5$, gives

$$u = 2r\left(\sqrt{9\mu^2 + 5} - (1 + 3\mu)\right), \qquad d_1 = 2r\left(\sqrt{9\mu^2 + 5} - 3\mu\right).$$

(One can check that $2r < d_1 \le 6r$, so the band stays taut and the formula for its energy still applies.)

> **Ask yourself** The discs are momentarily at rest at d_1. What decides whether they stay there?

They stay at rest if the band's pull does not exceed the limiting friction:

$$\frac{mg\,d_1}{6r} \le \mu mg \iff \sqrt{9\mu^2 + 5} - 3\mu \le 3\mu \iff 9\mu^2 + 5 \le 36\mu^2 \iff \mu^2 \ge \tfrac{5}{27},$$

where squaring is valid because both sides are positive. So if $\frac{5}{27} < \mu^2 < \frac{4}{9}$, the discs collide, rebound, decelerate monotonically to rest at d_1, and are then held there by static friction for good. They come to rest exactly once after the first collision. ■

Checks

- At $\mu = \frac{2}{3}$ both part (i)'s stopping point $d^* = 12\mu r - 6r$ and part (ii)'s kinetic energy give "arrive at $d = 2r$ with zero speed", so the two parts agree at the boundary.
- With $\mu = 0.5$ (inside the range of part (iii)): $K(2r) = \frac{2}{3}mgr$, $K_1 = \frac{1}{3}mgr$ and $d_1 = 2r(\sqrt{7.25} - 1.5) \approx 2.385r$. The pull there is $\approx 0.397mg < 0.5mg$, so the discs stay put. A step-by-step numerical simulation of the motion reproduced all of this and also the stopping point $d^* = 12\mu r - 6r$ for several $\mu \in (\frac{2}{3}, 1)$. For $\mu^2 < \frac{5}{27}$ (for example $\mu = 0.42$) it showed the discs moving again after stopping, as predicted.

Common slips

- Taking the extension as d instead of $2d$: both straight sections stretch.
- Using only one tension on each disc: the band pulls each disc with *two* strands, giving $2T$.
- Forgetting to check that the discs stay at rest once stopped. "Speed zero" and "at rest for good" are different conditions.

Beyond STEP

Friction makes this a piecewise system. The equation of motion changes whenever the velocity changes sign, and "stuck" states appear where the applied force is below the friction threshold. Systems like this (stick–slip motion) are studied with phase-plane methods in first-year dynamics. For small μ, repeated collisions with energy loss produce infinitely many bounces in a finite time, which is worth thinking about. For more on energy methods and friction, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 2005, Question 11

Mechanics • Rigid Body Equilibrium

STEP III 2005, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q11

The key idea

The spindle and rods form a rigid body turning about a fixed horizontal axis, so one angle θ describes everything and energy is the natural tool. The potential energy is a sum of three cosines whose angles differ by $\frac{2}{3}\pi$; combining them gives a single expression $A\cos\theta + B\sin\theta$, which is one cosine wave in disguise. Once it's written in that form, the equilibria, their stability and the energy needed to get over the top can all be read off.

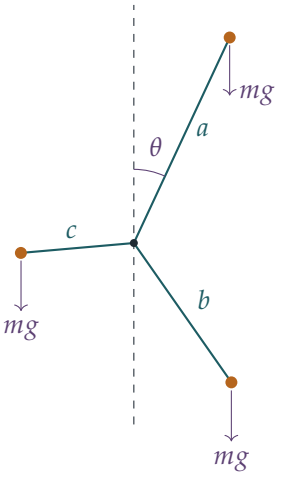
Solution

The energy

We model the particles as point masses on light rigid rods, with a smooth bearing so no energy is lost. Measure every angle clockwise from the upward vertical, as in the diagram. The rods of lengths a, b, c then make angles θ, $\theta + \frac{2}{3}\pi$ and $\theta + \frac{4}{3}\pi$ with the upward vertical, so the particles are at heights

$$a\cos\theta, \qquad b\cos\left(\theta + \tfrac{2}{3}\pi\right), \qquad c\cos\left(\theta + \tfrac{4}{3}\pi\right)$$

above the spindle.



Using $\cos\frac{2}{3}\pi = \cos\frac{4}{3}\pi = -\frac{1}{2}$, $\sin\frac{2}{3}\pi = \frac{\sqrt{3}}{2}$ and $\sin\frac{4}{3}\pi = -\frac{\sqrt{3}}{2}$,

$$\cos\left(\theta + \tfrac{2}{3}\pi\right) = -\tfrac{1}{2}\cos\theta - \tfrac{\sqrt{3}}{2}\sin\theta, \qquad \cos\left(\theta + \tfrac{4}{3}\pi\right) = -\tfrac{1}{2}\cos\theta + \tfrac{\sqrt{3}}{2}\sin\theta.$$

Taking the spindle as the zero of potential energy,

$$\begin{aligned} V &= mg\left[a\cos\theta + b\cos\left(\theta + \tfrac{2}{3}\pi\right) + c\cos\left(\theta + \tfrac{4}{3}\pi\right)\right] \\ &= mg\left[\tfrac{1}{2}(2a - b - c)\cos\theta - \tfrac{\sqrt{3}}{2}(b - c)\sin\theta\right]. \end{aligned}$$

Each particle moves on a circle about the spindle with angular velocity $\dot{\theta}$, so the speeds are $a\dot{\theta}$, $b\dot{\theta}$, $c\dot{\theta}$. The rods are light, so the total energy is

$$E = \tfrac{1}{2}m\left(a^2 + b^2 + c^2\right)\dot{\theta}^2 + \tfrac{1}{2}mg\left[(2a - b - c)\cos\theta - \sqrt{3}\,(b - c)\sin\theta\right].$$

Equilibrium

The system is in equilibrium where $\mathrm{d}V/\mathrm{d}\theta = 0$, that is,

$$-\tfrac{1}{2}mg\left[(2a - b - c)\sin\theta + \sqrt{3}\,(b - c)\cos\theta\right] = 0.$$

Since $a > b$ and $a > c$, we have $2a - b - c > 0$. So $\cos\theta \neq 0$ at an equilibrium: if $\cos\theta = 0$ then $\sin\theta = \pm 1$ and the bracket would be $\pm(2a - b - c) \neq 0$. Dividing by $\cos\theta$,

$$\tan\theta = -\frac{(b - c)\sqrt{3}}{2a - b - c},$$

as required. The argument also runs backwards: any θ satisfying this equation makes $\mathrm{d}V/\mathrm{d}\theta = 0$. So the equilibria are exactly the solutions of this equation.

Exactly two equilibria, and which is stable

The tangent function has period π and takes each real value exactly once on each interval of length π where it is defined. So the equation has exactly two solutions in $0 \le \theta < 2\pi$, which differ by π. These are the two equilibrium positions.

To decide stability, write V as a single cosine. Put

$$A = \tfrac{1}{2}(2a - b - c), \qquad B = -\tfrac{\sqrt{3}}{2}(b - c),$$

so $V = mg(A\cos\theta + B\sin\theta)$. Then

$$\begin{aligned} 4(A^2 + B^2) &= (2a - b - c)^2 + 3(b - c)^2 \\ &= 4a^2 + 4b^2 + 4c^2 - 4ab - 4bc - 4ca \\ &= 2\left[(a - b)^2 + (b - c)^2 + (c - a)^2\right] = 4R^2, \end{aligned}$$

so $A^2 + B^2 = R^2$, with $R > 0$ since the lengths are distinct. Choose α with $\cos\alpha = A/R$ and $\sin\alpha = B/R$. Since $A > 0$ and $B < 0$, $-\frac{1}{2}\pi < \alpha < 0$. Then

$$V = mgR\cos(\theta - \alpha).$$

The two equilibria are $\theta = \alpha$, where $V = mgR$ is a maximum, and $\theta = \alpha + \pi$, where $V = -mgR$ is a minimum.

? **Ask yourself** Why does a minimum of V mean stable?

Here $V''(\theta) = -mgR\cos(\theta - \alpha)$. At $\theta = \alpha + \pi$ this is $mgR > 0$, so a small displacement raises the potential energy. With total energy fixed, the system can't wander far, and it oscillates about this position. At $\theta = \alpha$ we have $V'' = -mgR < 0$: a small displacement lowers V, the kinetic energy grows, and the system moves away. So the stable equilibrium is $\theta = \alpha + \pi$, the one with $\frac{1}{2}\pi < \theta < \pi$, where $\cos\theta < 0$ and $\sin\theta > 0$. In that position the longest rod a points below the horizontal, at an angle $|\alpha|$ to the downward vertical. The other equilibrium, with $-\frac{1}{2}\pi < \theta < 0$ and rod a above the horizontal, is unstable.

Complete revolutions

Energy is conserved, since the bearing is smooth and gravity is conservative. Write $I = m(a^2 + b^2 + c^2)$, so that

$$\tfrac{1}{2}I\dot{\theta}^2 + mgR\cos(\theta - \alpha) = E.$$

To make complete revolutions the system must get past every angle, and in particular past $\theta = \alpha$, where V takes its largest value mgR. Since $\frac{1}{2}I\dot{\theta}^2 \ge 0$ everywhere, this needs $E \ge mgR$. Let ω be the angular velocity when the system passes through the stable position $\theta = \alpha + \pi$, where $V = -mgR$. Then $E = \frac{1}{2}I\omega^2 - mgR$, and the condition becomes

$$\tfrac{1}{2}I\omega^2 - mgR \ge mgR \quad \Longleftrightarrow \quad \omega^2 \ge \frac{4mgR}{I} = \frac{4gR}{a^2 + b^2 + c^2}.$$

So the system must pass through its stable equilibrium with angular velocity at least $\sqrt{4gR/(a^2+b^2+c^2)}$, as required.

(In the borderline case of equality, the system creeps towards the unstable equilibrium and only reaches it in infinite time. Strictly, a full revolution needs ω strictly greater than this value; the question asks only for the necessary bound.)

Checks

- If $a = b = c$ then $R = 0$ and $V \equiv 0$: the three equal masses balance in every position (their centre of mass is on the axis), and any spin at all goes round. This matches the bound tending to zero.
- If $b = c$, the tangent condition gives $\tan\theta = 0$. By symmetry, the long rod points straight up (unstable) or straight down (stable).
- A numerical check confirms the form of V, the identity $A^2 + B^2 = R^2$, and that there are exactly two equilibria with the stated stability, for random lengths. For $a = 3$, $b = 2$, $c = 0.5$ and $g = 9.81$, the critical speed is 2.5406 rad s^{-1}. Stepping the equation of motion shows that 2% below this the system swings back, while 2% above it goes right round.

Common slips

- Putting the rods at $\theta - \frac{2}{3}\pi$ and $\theta + \frac{2}{3}\pi$ the wrong way round relative to the diagram flips the sign of the $\sin\theta$ term, and with it the sign in the tangent condition.
- Concluding "two solutions" without noting that $\cos\theta = 0$ can't be an equilibrium, so that dividing by $\cos\theta$ loses nothing.
- Using kinetic energy $\frac{1}{2}m(a+b+c)^2\dot{\theta}^2$ or similar. Each particle has its own speed $r\dot{\theta}$, so the moment of inertia is $m(a^2+b^2+c^2)$.

Beyond STEP

The system is a compound pendulum, a rigid body about a fixed axis. Its energy depends only on the moment of inertia and on the position of the centre of mass, which lies at distance $R/3$ from the axis here. The picture of $\frac{1}{2}I\dot{\theta}^2 + V(\theta) = E$ as a curve in the $(\theta, \dot{\theta})$ phase plane, with oscillations inside a separatrix and rotations outside it, is a central idea in first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation and energy methods for this kind of problem.

WORKED SOLUTION

STEP III 2005, Question 12

Probability and Statistics • Combining Unbiased Estimators

STEP III 2005, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q12

The key idea

Everything rests on two facts about linear combinations of independent random variables: expectations add, and variances add with the squares of the coefficients. Requiring an expectation to come out right *whatever* the true lap times means matching coefficients, which turns each condition into an equation for the constants. Then X is a sensible estimate of the race time, and Y is a way of estimating the unknown σ from the disagreement between the two methods of timing.

Solution

Let the true lap times be $\mu_1, \ldots, \mu_4$ and the true race time $\mu = \mu_1 + \mu_2 + \mu_3 + \mu_4$. Write

$$S = T_1 + T_2 + T_3 + T_4.$$

Then $\mathrm{E}(S) = \mu_1 + \cdots + \mu_4 = \mu$ and, since $T_1, \ldots, T_4$ are independent (the timers are independent), $\mathrm{Var}(S) = 4\sigma^2$. Also $\mathrm{E}(T) = \mu$, $\mathrm{Var}(T) = \sigma^2$, and T is independent of S, because S depends only on the other four timers.

Finding X

For $X = aT + bS$,

$$\mathrm{E}(X) = a\mu + b\mu = (a+b)\mu.$$

This must equal μ whatever the true lap times, so in particular for some $\mu \neq 0$; hence

$$a + b = 1.$$

By independence of T and S,

$$\mathrm{Var}(X) = a^2\,\mathrm{Var}(T) + b^2\,\mathrm{Var}(S) = \sigma^2\left(a^2 + 4b^2\right) = \sigma^2\left(a^2 + 4(1-a)^2\right).$$

The quadratic $a^2 + 4(1-a)^2 = 5a^2 - 8a + 4$ has positive leading coefficient, so it is minimised where its derivative $10a - 8$ vanishes, at $a = \frac{4}{5}$. (Completing the square:

$5a^2 - 8a + 4 = 5\left(a - \frac{4}{5}\right)^2 + \frac{4}{5}$.) Hence $a = \frac{4}{5}$, $b = \frac{1}{5}$, and

$$X = \tfrac{4}{5}T + \tfrac{1}{5}(T_1 + T_2 + T_3 + T_4), \qquad \mathrm{Var}(X) = \tfrac{4}{5}\sigma^2.$$

This makes sense: T is four times as reliable as S (variance σ^2 against $4\sigma^2$), so it gets four times the weight.

Finding Y

For $Y = cT + dS$ we use $\mathrm{E}(Y^2) = \mathrm{Var}(Y) + \big(\mathrm{E}(Y)\big)^2$:

$$\mathrm{E}(Y^2) = \sigma^2(c^2 + 4d^2) + (c + d)^2\mu^2.$$

Ask yourself How can this equal σ^2 whatever the true lap times are?

The constants c, d may not depend on μ, but μ can be any positive number. The expression is $\sigma^2(c^2 + 4d^2)$ plus a multiple of μ^2, so for it to stay equal to σ^2 as μ varies the coefficient of μ^2 must vanish: $(c + d)^2 = 0$, so $d = -c$. Then we need $\sigma^2(c^2 + 4c^2) = \sigma^2$, that is, $c^2 = \frac{1}{5}$. Taking $c = 1/\sqrt{5}$,

$$Y = \frac{T - (T_1 + T_2 + T_3 + T_4)}{\sqrt{5}}$$

(the other sign gives $-Y$, which works equally well since only Y^2 matters). So Y measures the disagreement between the two timings; its expectation is 0, and its square is an unbiased estimate of σ^2.

The particular race

With $T = 220$ and $S = 220.5$,

$$X = \tfrac{4}{5}(220) + \tfrac{1}{5}(220.5) = 176 + 44.1 = 220.1,$$
$$Y = \frac{220 - 220.5}{\sqrt{5}} = -\frac{0.5}{\sqrt{5}}, \qquad Y^2 = \frac{0.25}{5} = 0.05.$$

So we estimate $\sigma^2 \approx 0.05$, and hence

$$\mathrm{Var}(X) = \tfrac{4}{5}\sigma^2 \approx \tfrac{4}{5}(0.05) = 0.04, \qquad \text{standard deviation of } X \approx 0.2 \text{ s}.$$

Our estimate of the true race time is therefore 220.1 ± 0.2 seconds: the true time lies roughly in the interval

$$[219.9,\ 220.3] \text{ seconds},$$

taking one standard deviation either side. If we wanted more confidence we could go two standard deviations either side, giving $[219.7,\ 220.5]$ seconds; for normally distributed errors this would be about a 95% interval. Bear in mind that σ^2 has been estimated from a single value of Y^2, so the width of the interval is itself only a rough estimate.

Checks

- With $a = \frac{4}{5}$, $\mathrm{Var}(X) = \frac{4}{5}\sigma^2$ is smaller than both $\mathrm{Var}(T) = \sigma^2$ and $\mathrm{Var}(S) = 4\sigma^2$, as combining two independent measurements should achieve.
- $X = 220.1$ lies between the two measurements, much closer to $T = 220$, the more reliable one.
- A simulation with 200 000 races ($\sigma = 0.7$) gave a mean of X within 0.001 of the true time, $\mathrm{Var}(X) \approx 0.393$ against $\frac{4}{5}(0.49) = 0.392$, and a mean of Y^2 of 0.491 against $\sigma^2 = 0.49$.

Common slips

- Taking $\mathrm{Var}(S) = \sigma^2$ or $16\sigma^2$: it is $4\sigma^2$, the sum of four independent variances.
- Forgetting the $(\mathrm{E}\,Y)^2$ term in $\mathrm{E}(Y^2)$, which is what forces $d = -c$.
- Using $Y^2 = 0.05$ as the variance of X, rather than scaling by $\frac{4}{5}$.

Beyond STEP

X is the minimum-variance unbiased linear combination of two estimates: the general rule is to weight each independent estimate by the reciprocal of its variance, which is the basis of inverse-variance weighting in meta-analysis and of the Gauss–Markov theorem for least squares. Y^2 is an unbiased estimator of σ^2, like the sample variance with its divisor $n - 1$. Grimmett and Welsh, *Probability: An Introduction*, covers expectation and variance of sums carefully, and a first statistics course takes up estimators and confidence intervals properly.

WORKED SOLUTION

STEP III 2005, Question 13

Probability and Statistics • Geometric Distribution

STEP III 2005, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q13

The key idea

The cards from $w+1$ to n do nothing at all: they pay nothing and they don't end the game. So in part (i) we can simply ignore them and watch only the draws that show $0, 1, \ldots, w$; each of those is a fair pick from $w+1$ equally likely numbers, and the total win is then a geometric random variable. In part (ii), where the draws are no longer independent, we split the total win into a sum of indicators, one for each paying card, and use linearity of expectation.

Solution

Call a draw *relevant* if it shows one of $0, 1, \ldots, w$, and *irrelevant* if it shows one of $w+1, \ldots, n$.

Part (i)

Why the two events have the same probability. An irrelevant draw changes nothing: the player receives nothing and the game carries on. So the player's total depends only on the sequence of relevant draws: each non-zero relevant draw earns £1, and the first relevant draw showing 0 ends the game. Hence the player wins exactly £3 precisely when, among the relevant draws, the first three show non-zero numbers and the fourth shows 0. These are the same event, so they have the same probability.

(Each draw is relevant with probability $\frac{w+1}{n+1} > 0$, independently, so with probability 1 there are infinitely many relevant draws and in particular a first, second, third and fourth one; the game ends with probability 1.)

Ask yourself Why is each relevant draw equally likely to show any of $0, 1, \ldots, w$, regardless of what came before?

Because the card is replaced each time, the draws are independent, each uniform on $\{0, 1, \ldots, n\}$. Given that a particular draw is relevant, it is uniform on $\{0, 1, \ldots, w\}$, since

those $w+1$ outcomes were equally likely to begin with. Independence of the draws means the values of successive relevant draws are also independent. So each relevant draw is 0 with probability $\frac{1}{w+1}$ and non-zero with probability $\frac{w}{w+1}$, independently of the others.

Therefore

$$P(\text{win exactly £3}) = \left(\frac{w}{w+1}\right)^3 \cdot \frac{1}{w+1} = \frac{w^3}{(w+1)^4},$$

as required.

General r. Exactly the same argument with r in place of 3 (the first r relevant draws non-zero, the $(r+1)$th zero) gives

$$P(\text{win exactly £}r) = \frac{w^r}{(w+1)^{r+1}}, \qquad r = 0, 1, 2, \ldots$$

Writing $q = \frac{w}{w+1}$, this is $(1-q)q^r$, a geometric distribution; these probabilities sum to 1 as they should.

Expected total win. Let X be the total win in pounds. Differentiating $\sum_{r\geq 0} q^r = \frac{1}{1-q}$ for $|q| < 1$ and multiplying by q gives $\sum_{r\geq 0} rq^r = \frac{q}{(1-q)^2}$. So

$$E(X) = \sum_{r=0}^{\infty} r(1-q)q^r = (1-q)\cdot\frac{q}{(1-q)^2} = \frac{q}{1-q} = \frac{w/(w+1)}{1/(w+1)} = w.$$

The expected total win is £w.

Part (ii)

Now the cards are not replaced, so the game is the same as shuffling the pack into a uniformly random order and turning cards over until 0 appears. The player receives £1 for each of the cards $1, \ldots, w$ that comes before card 0 in this order (every card before 0 is drawn, and none after it).

For $k = 1, \ldots, w$ let $I_k = 1$ if card k comes before card 0, and $I_k = 0$ otherwise. The total win is $Y = I_1 + I_2 + \cdots + I_w$.

> **Ask yourself** The I_k are not independent. Does that matter?

No: the expectation of a sum is the sum of the expectations whatever the dependence. So we need only $P(I_k = 1)$. In a uniformly random order, swapping the positions of cards k and 0 is a bijection between orders with k before 0 and orders with 0 before k, so the two are equally likely and $P(I_k = 1) = \frac{1}{2}$. Hence

$$E(Y) = \sum_{k=1}^{w} E(I_k) = \sum_{k=1}^{w} \tfrac{1}{2} = \frac{w}{2},$$

which is half of the expected win w in part (i), as required.

Checks

- With $w = n$ (every non-zero card pays) and no replacement, the position of card 0 is uniform on $1, \ldots, n+1$, so the mean number of cards before it is $\frac{n}{2}$, matching $\frac{w}{2}$.
- With replacement and $w = 1$, each relevant draw is a fair coin between 0 and 1, so $P(X = r) = 2^{-(r+1)}$ and $E(X) = 1$, matching $w^r/(w+1)^{r+1}$ and $E(X) = w$.
- A simulation of 2×10^5 games with $n = 5$, $w = 2$ gave $P(X = 3) \approx 0.0995$ against $\frac{8}{81} \approx 0.0988$, and a mean win of 1.995 against 2. Exact enumeration of all orders for small packs (for example $n = 5$, $w = 3$) gives mean $\frac{3}{2} = \frac{w}{2}$ without replacement.

Common slips

- Using $\frac{1}{n+1}$ for the chance that a draw is 0. That is the chance per draw, but irrelevant draws can occur any number of times in between; conditioning on relevant draws is what gives the clean $\frac{1}{w+1}$.
- Forgetting $r = 0$: the first relevant draw may be 0, giving probability $\frac{1}{w+1}$ of winning nothing.
- Trying to find the full distribution of the win in part (ii). It isn't needed; indicators and linearity do the job in two lines.

Beyond STEP

Ignoring the irrelevant draws is an example of *thinning*, or looking at a process only at the times something interesting happens; it reappears with Markov chains (the "jump chain") and Poisson processes. Writing a count as a sum of indicators and using linearity of expectation, even when the indicators are dependent, is one of the most useful tricks in first-year probability and combinatorics. Both ideas are developed in Blitzstein and Hwang, *Introduction to Probability*, and in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 2005, Question 14

Probability and Statistics • Continuous Random Variables

STEP III 2005, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q14

The key idea

Everything here runs on the one integral we are allowed to quote, so the work is in spotting which m and n each integral needs. The substitution $x = k^2/u$ swaps the interval $[0, k]$ with $[k, \infty)$ while leaving the integrand's shape unchanged, and that symmetry pins down the median at once. For the time $T = s/V$, the same reciprocal trick shows that T has a density of exactly the same family as V, with k replaced by s/k. After that, the medians and means can be read straight off the results for V.

Solution

The constant C

Take $m = a$ and $n = 2a$ in the given result. This is allowed because a is a positive integer and $2a \geq a$. Then $n + 2 = 2a + 2$ and $n - m + 1 = a + 1$, so

$$\int_0^\infty \frac{x^a}{(x+k)^{2a+2}}\,\mathrm{d}x = \frac{a!\,a!}{(2a+1)!\,k^{a+1}}.$$

A density integrates to 1 over its range, so

$$1 = Ck^{a+1} \cdot \frac{a!\,a!}{(2a+1)!\,k^{a+1}} \qquad \Longrightarrow \qquad C = \frac{(2a+1)!}{a!\,a!},$$

as required.

The substitution and the median

In the left-hand integral, put $x = k^2/u$, so that $\mathrm{d}x = -\dfrac{k^2}{u^2}\,\mathrm{d}u$. As x runs from 0 to v, u runs from ∞ down to k^2/v. Also

$$x + k = \frac{k^2}{u} + k = \frac{k(u+k)}{u},$$

so

$$\frac{x^a}{(x+k)^{2a+2}} = \frac{k^{2a}/u^a}{k^{2a+2}(u+k)^{2a+2}/u^{2a+2}} = \frac{u^{a+2}}{k^2\,(u+k)^{2a+2}}.$$

Multiplying by $\mathrm{d}x$ cancels the k^2/u^2, and the minus sign reverses the limits:

$$\int_0^v \frac{x^a}{(x+k)^{2a+2}}\,\mathrm{d}x = \int_\infty^{k^2/v} \frac{u^a}{(u+k)^{2a+2}}\,(-\mathrm{d}u) = \int_{k^2/v}^\infty \frac{u^a}{(u+k)^{2a+2}}\,\mathrm{d}u,$$

which is the required identity.

Ask yourself Which value of v makes this identity say something about the median?

Choose $v = k$, so that $k^2/v = k$ as well. After multiplying both sides by Ck^{a+1}, the identity becomes

$$\begin{aligned}\mathrm{P}(V < k) &= \int_0^k \mathrm{f}(x)\,\mathrm{d}x \\ &= \int_k^\infty \mathrm{f}(u)\,\mathrm{d}u = \mathrm{P}(V > k).\end{aligned}$$

V is continuous, so $\mathrm{P}(V = k) = 0$ and these two probabilities add to 1. Each is therefore $\frac{1}{2}$, and the median of V is k.

The expected value of V

$$\mathrm{E}(V) = Ck^{a+1}\int_0^\infty \frac{x^{a+1}}{(x+k)^{2a+2}}\,\mathrm{d}x.$$

Now take $m = a+1$ and $n = 2a$. This needs $2a \geq a+1$, which holds because $a \geq 1$. Then $n - m + 1 = a$, and

$$\mathrm{E}(V) = \frac{(2a+1)!}{a!\,a!}\,k^{a+1}\cdot\frac{(a+1)!\,(a-1)!}{(2a+1)!\,k^a} = k\cdot\frac{(a+1)!}{a!}\cdot\frac{(a-1)!}{a!} = \frac{(a+1)k}{a}.$$

The distribution of T

Speeds are positive, so for $t > 0$ we can multiply through by $V/t > 0$ without reversing the inequality:

$$T < t \iff \frac{s}{V} < t \iff V > \frac{s}{t}.$$

Hence

$$\mathrm{P}(T < t) = \mathrm{P}\Big(V > \frac{s}{t}\Big) = \int_{s/t}^\infty \frac{Ck^{a+1}x^a}{(x+k)^{2a+2}}\,\mathrm{d}x, \qquad (*)$$

as required.

Now substitute $u = s/x$ in $(*)$, so that $x = s/u$ and $\mathrm{d}x = -\dfrac{s}{u^2}\,\mathrm{d}u$. As x runs from s/t to ∞, u runs from t down to 0. Since $x + k = (s+ku)/u$,

$$\mathrm{P}(T < t) = \int_0^t Ck^{a+1}\,\frac{s^a/u^a}{(s+ku)^{2a+2}/u^{2a+2}}\cdot\frac{s}{u^2}\,\mathrm{d}u = \int_0^t \frac{Ck^{a+1}s^{a+1}\,u^a}{(s+ku)^{2a+2}}\,\mathrm{d}u.$$

This holds for every $t > 0$, and $\mathrm{P}(T < t) = 0$ for $t \le 0$. So differentiating with respect to t gives the density of T:

$$\mathrm{g}(t) = \frac{Ck^{a+1}s^{a+1}\,t^a}{(s+kt)^{2a+2}} \qquad (t > 0).$$

To compare this with f, take out k^{2a+2} from the denominator and write $K = s/k$:

$$\mathrm{g}(t) = \frac{C\,(s/k)^{a+1}\,t^a}{(t+s/k)^{2a+2}} = \frac{CK^{a+1}t^a}{(t+K)^{2a+2}}.$$

This is exactly the density f with k replaced by $K = s/k$. So T belongs to the same family of distributions as V.

Medians and means

Because T has the same form of distribution with parameter $K = s/k$, the earlier results apply directly:

$$\text{median of } T = \frac{s}{k}, \qquad \mathrm{E}(T) = \frac{(a+1)}{a} \cdot \frac{s}{k}.$$

The product of the median time and the median speed is therefore

$$\frac{s}{k} \cdot k = s,$$

which is the distance, as claimed. The product of the expected time and the expected speed is

$$\mathrm{E}(T)\,\mathrm{E}(V) = \frac{(a+1)s}{ak} \cdot \frac{(a+1)k}{a} = s\Big(1 + \frac{1}{a}\Big)^2 > s,$$

since a is a positive integer, so $1 + 1/a > 1$. This completes the question.

Checks

- For $a = 1$, $k = 1$ we get $C = 3!/1 = 6$, and numerical integration of $6x/(x+1)^4$ over $[0, \infty)$ gives 1.000000. Over $[0, 1]$ it gives 0.500000, confirming the median.
- Numerical quadrature for several pairs, such as $a = 2$, $k = 0.7$, gives $\mathrm{E}(V) = 1.05000 = \frac{3}{2}(0.7)$. With $s = 3.2$ it gives $\mathrm{E}(T)\mathrm{E}(V) = 7.20000 = 3.2 \times \frac{9}{4}$.
- A simulation of 2×10^5 speeds with $a = 2$, $k = s = 1$ gave median speed $\times$ median time ≈ 1.000001, but mean speed ≈ 1.504 (theory: 1.5).
- The median product equalling s can be seen without any formulae. $T = s/V$ is a decreasing function of V, so the median of T is s divided by the median of V.

Common slips

- Forgetting to check $n \ge m$ when quoting the given integral. For $\mathrm{E}(V)$ it needs $a \ge 1$, which is exactly why a must be a positive integer. For larger moments the formula would eventually fail, and indeed $\mathrm{E}(V^{a+1})$ is infinite.

- Losing the factor s/t^2 (the chain rule) when differentiating $(*)$ to find the density of T.
- Assuming $\mathrm{E}(s/V) = s/\mathrm{E}(V)$. The whole point of the last part is that this is false.

Beyond STEP

The final inequality is an instance of Jensen's inequality: $1/v$ is convex for $v > 0$, so $\mathrm{E}(1/V) \geq 1/\mathrm{E}(V)$, with equality only if V is constant. Medians, by contrast, pass through any monotonic function. The substitution $V/(V+k)$ turns V into a $\mathrm{Beta}(a+1, a+1)$ variable, which is symmetric about $\frac{1}{2}$, and this is another way to see why the median is k. Blitzstein and Hwang, *Introduction to Probability*, covers Jensen's inequality, change of variables and the Beta family very readably.

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